

Chapter 1

Introduction to Statics

Engineering Statics is the gateway into *engineering mechanics*, which is the application of Newtonian physics to design and analyze objects, systems, and structures with respect to motion, deformation, and failure. In addition to learning the subject itself, you will also develop skills in the art and practice of problem solving and mathematical modeling, skills that will benefit you throughout your engineering career.

The subject is called “statics” because it is concerned with particles and rigid bodies that are in equilibrium, and these will usually be stationary, i.e. static.

The chapters in this book are:

[Introduction to Statics](#)— an overview of statics and an introduction to units and problem solving.

[Forces and Other Vectors](#)— basic principles and mathematical operations on force and position vectors.

[Equilibrium of Particles](#)— an introduction to equilibrium and problem solving.

[Moments and Static Equivalence](#)— the rotational tendency of forces, and simplification of force systems.

[Rigid Body Equilibrium](#)— balance of forces and moments for single rigid bodies.

[Equilibrium of Structures](#)— balance of forces and moments on interconnected systems of rigid bodies.

[Centroids and Centers of Gravity](#)— an important geometric property of shapes and rigid bodies.

[Internal Forces](#)— forces and moments within beams and other rigid bodies.

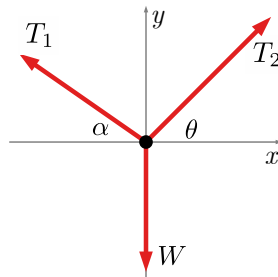
[Friction](#)— equilibrium of bodies subject to friction.

Moments of Inertia— an important property of geometric shapes used in many applications.

Your statics course may not cover all of these topics, or may move through them in a different order.

Below are two examples of the types of problems you'll learn to solve in statics. Notice that each can be described with a picture and problem statement, a free-body diagram, and equations of equilibrium.

Equilibrium of a particle: A 140 lb person walks across a slackline stretched between two trees. If angles α and θ are known, find the tension in each end of the slackline.



Person's point of contact to slackline:

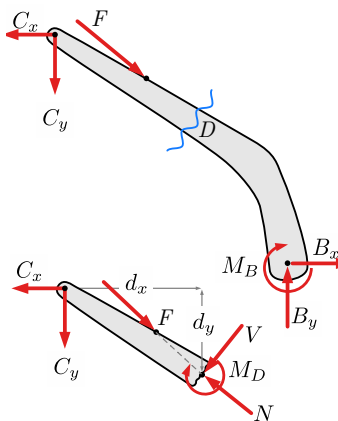
$$\Sigma F_x = 0$$

$$-T_1 \cos \alpha + T_2 \cos \theta = 0$$

$$\Sigma F_y = 0$$

$$T_1 \sin \alpha + T_2 \sin \theta - W = 0$$

Equilibrium of a rigid body: Given the interaction forces at point C on the upper arm of the excavator, find the internal axial force, shear force, and bending moment at point D .



Section cut FBD:

$$\Sigma F_x = 0$$

$$-C_x + F_x - V_x - N_x = 0$$

$$\Sigma F_y = 0$$

$$-C_y - F_y - V_y + N_y = 0$$

$$\Sigma M_D = 0$$

$$+(d_y)C_x + (d_x)C_y - M_D = 0$$

The knowledge and skills gained in Statics will be used in your other engineering courses, in particular in Dynamics, Mechanics of Solids (also called Strength or Mechanics of Materials), and in Fluid Mechanics. Statics will be a foundation of your engineering career.

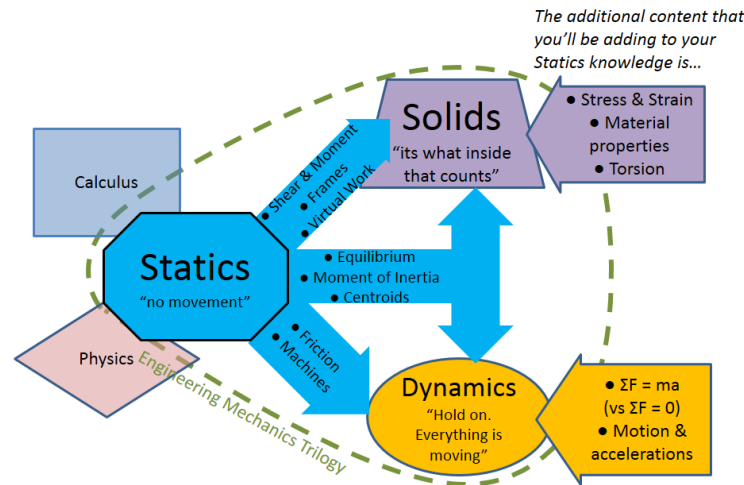


Figure 1.0.1 Map of how Statics builds upon the prerequisites of Calculus and Physics and then informs the later courses of Mechanics of Solids and Dynamics.

1.1 Newton's Laws of Motion

Key Questions

- What are the two types of motion?
- What three relationships do Newton's laws of motion define?
- What are physical examples for each of Newton's three laws of motion?

The English scientist Sir Isaac Newton laid the foundation of mechanics in 1687 with his three laws of motion, which describe the relationship between forces, objects, and motion. Motion can be categorized into two types:

Translation— where a body changes position without changing its orientation in space, and

Rotation— where a body spins about an axis fixed in space, without changing its average position.

Some moving bodies purely translate, others purely rotate, and many do both. Conveniently, we can separate translation and rotation and analyze them individually using independent equations.

Newton's three laws and their implications for translation and rotation are described below.

1.1.1 Newton's 1st Law

Newton's first law states that

an object will remain at rest or in uniform motion in a straight line unless acted upon by an external force.

This law, also sometimes called the “law of inertia,” tells us that bodies maintain their current velocity unless a net force is applied to change it. In other words, an object at rest it will remain at rest and a moving object will hold its current speed and direction unless an unbalanced force causes a velocity change. Remember that velocity is a vector quantity that includes both speed and direction, so an unbalanced force may cause an object to speed up, slow down, or change direction.



Figure 1.1.1 This rock, likely a glacial erratic dropped here in the last ice age, is at rest with zero velocity and will remain at rest until an unbalanced force (like a crane or another ice age) causes it to move.

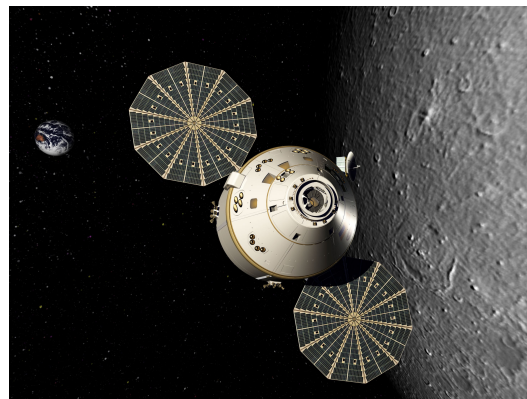


Figure 1.1.2 In deep space, where friction and gravitational forces are negligible, an object moves with constant velocity; near a celestial body gravitational attraction continuously changes its velocity.

Newton's first law also applies to angular velocities, however instead of force, the relevant quantity which causes an object to rotate is called a **torque** by physicists, but usually called a **moment** by engineers. A moment, as you will learn in [Chapter 4](#), is the rotational tendency of a force. Just as a force will cause a change in linear velocity, a moment will cause a change in angular velocity. This can be seen in things like tops, flywheels, stationary bikes, and other objects that spin on an axis when a moment is applied, but eventually stop because of the opposite moment produced by friction.



In the absence of friction this top would spin forever, but the small frictional moment exerted at the point of contact with the table will eventually bring it to a stop.

Figure 1.1.3 A spinning top demonstrates rotary motion.

1.1.2 Newton's 2nd Law

Newton's second law is usually succinctly stated with the familiar equation

$$\mathbf{F} = m\mathbf{a} \quad (1.1.1)$$

where $\mathbf{F}$ is net force, m is mass, and $\mathbf{a}$ is acceleration.

You will notice that the force and the acceleration are in bold face. This means these are vector quantities, having both a magnitude and a direction. Mass on the other hand is a scalar quantity, which has only a magnitude. This equation indicates that a force will cause an object to accelerate in the direction of the net force, and the magnitude of the acceleration will be proportional to the net force but inversely proportional to the mass of the object.

When studying Statics we are only concerned with bodies which are *not* accelerating which simplifies things considerably. When an object is not accelerating $a = 0$, which implies that it is either at rest or moving with a constant velocity. With this restriction Newton's Second Law for translation simplifies to

$$\Sigma \mathbf{F} = 0 \quad (1.1.2)$$

where $\Sigma \mathbf{F}$ is read as “the sum of the forces” and used to indicate the *net* force acting on the object.

Newton's second law for rotational motions is similar

$$\mathbf{M} = I\boldsymbol{\alpha}. \quad (1.1.3)$$

This equation states that a net moment $\mathbf{M}$ acting on an object will cause an angular acceleration $\boldsymbol{\alpha}$ proportional to the net moment and inversely proportional to I , a quantity known as the **mass moment of inertia**. Mass moment of inertia for rotational acceleration is analogous to ordinary mass for linear acceleration. We will have more to say about the moment of inertia in [Chapter 10](#).

Again, we see that the net moment and angular acceleration are vectors, quantities with magnitude and direction. The mass moment of inertia, on the

other hand, is a scalar quantity and has only a magnitude. Also, since Statics deals only with objects which are *not* accelerating $\alpha = 0$, they will always be at rest or rotating with constant angular velocity. With this restriction Newton's second law implies that the net moment on all static objects is zero.

$$\Sigma \mathbf{M} = 0 \quad (1.1.4)$$

1.1.3 Newton's 3rd Law

Newton's Third Law states

For every action, there is an equal and opposite reaction.

The actions and reactions Newton is referring to are **forces**. Forces occur whenever one object interacts with another, either directly like a push or pull, or indirectly like magnetic or gravitational attraction. Any force acting on one body is always paired with another equal-and-opposite force acting on some other body.

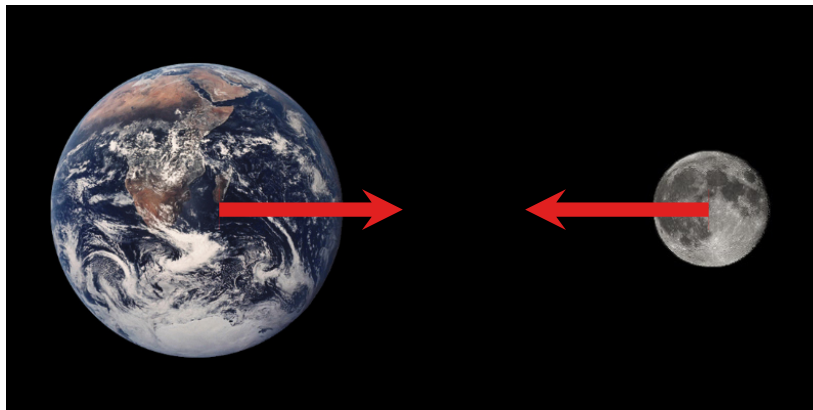


Figure 1.1.4 The earth exerts a gravitational force on the moon, and the moon exerts an equal and opposite force on the earth.

These equal-and-opposite pairs can be confusing, particularly when there are multiple interacting bodies. To clarify, we always begin solving statics problems by drawing a **free-body diagram**—a sketch where we isolate a body or system of interest and identify the forces acting *on* it, while ignoring any forces exerted *by* it on interacting bodies.

Consider the situation in [Figure 1.1.5](#). Diagram (a) shows a book resting on a table supported by the floor. The weights of the book and table are placed at their centers of gravity. To solve for the forces on the legs of the table, we use the free-body diagram in (b) which treats the book and the table as a single system and replaces the floor with the forces of the floor *on* the table. In diagram (c) the book and table are treated as independent objects. By separating them, the

equal-and-opposite interaction forces of the book *on* the table and the table *on* the book are exposed.

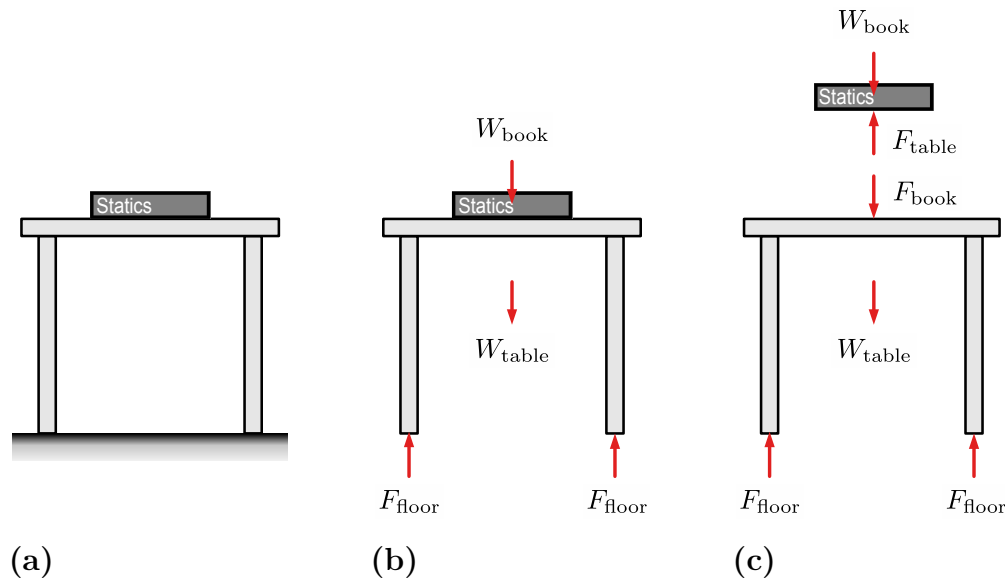


Figure 1.1.5 Free-body diagrams are used to isolate objects and identify relevant forces and moments.

This will be discussed further in [Chapter 3](#) and [Chapter 5](#).

1.2 Units

Key Questions

- What are the similarities and differences between the SI, British Gravitational, and English Engineering unit systems?
- How do you convert a value into different units?
- When a Statics problem give a value in pounds, is this referring to pounds-force or pounds-mass?

Engineering quantities consist of a numeric value and an associated unit (like 150 kg, 9.81 m/s^2 , or 17 ft). The values by themselves (150, 9.81, or 17) are just numbers, but the units give the numbers context. When discussing a quantity, you must *always* include the associated unit. The exceptions to this rule are unitless quantities, which are typically ratios where the units all cancel out.

Related units are defined as a coherent **unit system**. All unit systems are based on seven base units, the important ones for Statics being mass, length, and time. These base units combine to form all other measurement units. For example:

- Acceleration is defined as length $[L]$ divided by time $[t]$ squared, so has

units $a = [L/t^2]$

- Force is the product of mass and acceleration, as defined by [Newton's second law](#), so the units of force are $F = [mL/t^2]$.

1.2.1 Unit Systems

Multiple unit systems are generally used in engineering practice in the United States, including the SI and British Gravitational systems. A third, the English Engineering system, is commonly only used in Physics applications.

The SI system, abbreviated from the French *Système International (d'unités)* is the modern form of the metric system. The SI system is used for most international engineering and increasingly used in the U.S. in fields like science, medicine, electronics, and the military.

In the SI system, the unit of force is the **newton**, abbreviated N , and the unit of mass is the **kilogram**, abbreviated kg . The base unit of time, used by all systems, is the **second**, abbreviated s . Prefixes are added to unit names are used to specify the base-10 multiple of the original unit. One newton is equal to $1 \text{ kg} \cdot \text{m}/\text{s}^2$ because 1 N of force applied to 1 kg of mass causes the mass to accelerate at a rate of $1 \text{ m}/\text{s}^2$.

The other unit systems, British Gravitational and English Engineering, fall under the general name of “Imperial units,” given their broad use in the British Empire. The British Gravitational system uses the **foot**, abbreviated ft , as the base unit of distance, the **second** for time, and the **slug** for mass. Force is expressed in the unit of **pound-force**, abbreviated lbf , or lb for short. One pound-force will accelerate a mass of one slug at $1 \text{ ft}/\text{s}^2$, so $1 \text{ lbf} = 1 \text{ slug} \cdot \text{ft}/\text{s}^2$. On earth, a 1 slug mass weighs 32.174 lbf .

The English Engineering system also uses **foot** as the base unit of distance and **second** for time, but, unlike the British Gravitational system, uses **pound-mass** as the base unit of mass, where

$$32.174 \text{ lbf} = 1 \text{ slug} = 14.6 \text{ kg}.$$

While gravitational acceleration remains the same between the British Gravitational and English Engineering systems, a conversion factor is required to maintain unit consistency.

$$1 = \left[\frac{1 \text{ lbf} \cdot \text{s}^2}{32.174 \text{ ft} \cdot \text{lbf}} \right] = \left[\frac{1 \text{ slug}}{32.174 \text{ lbf}} \right] \quad (1.2.1)$$

The perceived advantage of the English Engineering system is that on Earth, a mass of 1 lbf weighs 1 lbf . However, this numerical equivalence often causes confusion, given the required gravitational conversion factor mentioned in the previous paragraph and the simple fact that mass and weight are different. Mass describes how much matter an object contains, while weight is a force — the effect of gravity on a mass.

Therefore, the U.S. engineering community primarily embraces the British Gravitational system but avoids the mass unit of slugs, instead stating material amounts as weight in pounds-force, or for brevity, pounds.

Table 1.2.1 shows the standard units of weight, mass, length, time, and gravitational acceleration in three unit systems.

Table 1.2.1 Fundamental Units

Unit System	Force	Mass	Length	Time	g (Earth)
SI	N	kg	m	s	9.81 m/s^2
British Gravitational	lbf	slug	ft	s	32.2 ft/s^2
English Engineering	lbf	lbm	ft	s	1 lbf/1 lbm

Force or Mass? The confusion caused by U.S. engineering students learning about the pounds-mass + pounds-force system in Physics, closely followed by adopting a slugs + pounds-force system in Statics, often leads students to ask “Is this pound a mass or force?” to which Statics faculty commonly respond “All pounds are forces in Statics.”

Dr. Dan Baker created a clever song to help students remember that in Statics pounds always means pounds-force. Here’s a link to the song [Pounds are a Force](#) at Suno.com.

Our world is engineering, let’s get it right,
Pounds are a force, not mass in our sight.
We measure with care, we never confuse,
Pounds-mass is a unit that we never use.

[chorus]

Pounds are force, feel their weight,
That is what Engineers calculate.
SI uses Kilograms for mass, Newtons for force,
But in the U.S. we are different of course.

Gravity pulls, and we feel the strain,
Newtons and pounds, they make it plain.
Kilograms and slugs, they are the masses,
The amount of material across our classes.

[chorus]

Pounds are force, feel their weight,
That is what Engineers calculate.
SI uses Kilograms for mass, Newtons for force,
But in the U.S. we are different of course.

From meters to feet, we can convert,
 9.81 meters is simply 32.2 feet, that's certain.
 So don't ask: do we multiply or divide?
 Just write $W = mg$, and then decide.

[chorus]
 Pounds are force, feel their weight,
 That is what Engineers calculate.
 SI uses Kilograms for mass, Newtons for force,
 But in the U.S. we are different of course.

So when you measure, know your course,
 In engineering, pounds are force.
 Leave pounds mass back in the dust,
 In the US Customary System we trust.

By Dan Baker, with lyrics assistance from ChatGPT and music composed by Suno.com.

1.2.2 Units within Newton's 2nd Law

Given that Newton's 2nd Law, $\Sigma \mathbf{F} = m\mathbf{a}$, is one of the foundational equations in engineering mechanics and the most basic equation relating mass and weight, let's examine its use in more detail. First we need to relate $\Sigma \mathbf{F} = m\mathbf{a}$ to weight, mass, and gravity. Consider an object in free fall. The only vertical force on the body is its weight W ; thus, the object's mass m is accelerated by gravity g . Therefore, we can transform Newton's 2nd Law $\Sigma \mathbf{F} = m\mathbf{a}$ to $\mathbf{W} = m\mathbf{g}$.

The scalar version, $W = mg$, also serves as the conversion equation among weight, mass, and gravity. You likely know this relationship for problems using the SI system. Hence, if you are given a 100 kg object here on earth and are asked to find the weight, you compute:

$$\begin{aligned} W &= mg \\ W &= 100 \text{ kg } (9.81 \text{ m/s}^2) \\ W &= 981 \text{ N} \end{aligned}$$

The confusion often starts when students in the United States are asked to find the mass of a known-weight object. Let's say you are asked to find the mass of a 100 lb object. Given the U.S. engineering convention that all pounds are force, the *lb* unit listed in the problem is pounds-force. Here's the root of the confusion: ***SI problems give an object's mass, while U.S. problems give an object's weight.*** Life would be easier if this was not the case, but never fear — we can use $W = mg$ to solve the problem. This time we will solve for m instead of W .

$$W = mg$$

$$\begin{aligned}
 100 \text{ lb} &= m \, 32.2 \text{ ft/s}^2 \\
 m &= \frac{100 \text{ lb}}{32.2 \text{ ft/s}^2} \\
 m &= 3.1 \text{ slug}
 \end{aligned}$$

Next we will investigate the values of the gravitational acceleration constant in SI and the British Gravitational system. As both measure the acceleration of an object toward the Earth, it should make sense that they are equivalent:

$$g = 9.81 \text{ m/s}^2 \left(\frac{3.28 \text{ ft}}{1 \text{ m}} \right) \approx 32.2 \text{ ft/s}^2$$

As shown in this unit conversion, *always make sure to multiply terms by the correct ratios* to get your desired units. In this case we multiplied by (3.28 ft/1 m) with meters in the bottom so it cancels out with the meters in the top of 9.81 m/s².

Here are two examples to test your understanding of the relationships among weight, mass, and gravity.

Example 1.2.2

How much does a 5 kg bag of flour weigh on Earth?

Hint. A value in kg is a mass. Weight is a force.

Solution.

$$\begin{aligned}
 W &= mg \\
 &= 5 \text{ kg}(9.81 \text{ m/s}^2) \\
 &= 49.05 \text{ N}
 \end{aligned}$$

Example 1.2.3

How much does a 5 lb bag of sugar weigh?

Hint. When a problem says *pounds*, it is referring to *pounds-force*.

Solution. The weight was given:

$$W = 5 \text{ lb} = 5 \text{ lbf.}$$

If you needed the *mass in slugs*, the mass is

$$m = \frac{5 \text{ lbf}}{32.2 \text{ ft/s}^2} = 0.155 \text{ slug.}$$

1.2.3 Avoiding Common Unit Errors

Awareness of units will help you prevent errors in your engineering calculations. You should always:

- Pay attention to the units of every quantity in the problem. Forces should have force units, distances should have distance units, etc.
- Use the unit system given in the problem statement and ensure any constants used agree with the units of the other terms. For example, $g = 9.81 \text{ m/s}^2$ should only be used when the rest of the problem is in SI units, matching the units of m and s.
- Avoid unit conversions when possible. If you must, convert given values to a consistent set of units and stick with them.
- Ensure unit consistency in your calculations. You can only add or subtract quantities with the same units. When multiplying or dividing quantities with units, apply the same operation to the units. Finally, the units on both sides of an equation must match.
- Develop a sense of the magnitudes of the units and consider your answers for reasonableness. On Earth, the four commonly used masses and weights, from smallest to largest, are roughly equivalent to:

Newton = 1 apple,
pound-force = box of butter,
kilogram = a pineapple, and
slug = a medium dog (like a Corgi).

- Assume every answer in Statics, and engineering in general, has units. A few answers will be dimensionless, but not many.

Understanding the relationship between weight, mass, and gravitational acceleration is essential for solving Statics problems. Always identify whether weight or mass is given and use $W = mg$ with the correct gravitational acceleration. This intentional process will help you confidently work across both SI and U.S. unit systems.

1.3 Forces

Key Questions

- What are some of the fundamental types of forces used in statics?
- Why do we often simplify distributed forces with equivalent point forces?

A force is simply a linear push or pull, yet forces come from a variety of sources and occur in many different situations. As such we need a specialized vocabulary to talk about them. We are also interested in forces that cause rotation, and we have special terms to describe these too.

Some terms used to describe forces are given below; others will be defined as needed later in the book.

Point Forces, also called **concentrated forces**, are forces that act at a single point. Examples are the push you give to open a door, the thrust of a rocket engine, or the pull at the chain suspending a wrecking ball. Point forces are actually an idealization, because real forces always act over an area and not at a mathematical point. However, point forces are the easiest type to deal with computationally so we will usually represent other types of forces as equivalent concentrated forces.

Distributed forces are forces that are spread out over a line, area or volume. Water pressure on the side of a swimming pool and the weight of snow on a roof are examples of forces distributed over an area. Distributed forces are represented graphically by an array of force vectors.

Body forces are distributed forces acting over the volume of a body. The most common body force is the body's weight, but there are others including buoyancy and forces caused by electric and magnetic fields. Weight and buoyancy will be the only body forces we consider in this book.

In many situations, body forces are small in comparison to the other forces acting on the object, and as such may be neglected. In practice, the decision to neglect forces must be made on the basis of sound engineering judgment; however, in this course you should consider the weight in your analysis if the problem statement provides enough information to determine it, otherwise you may ignore it.

Loads are the forces which an object must support in order to perform its function. Loads can be either static or dynamic, however only static loads will be considered here. Forces which hold a loaded object in equilibrium or hold parts of an object together are not considered loads.

Reaction forces or simply **reactions** are the forces and moments which hold or constrain an object or mechanical system in equilibrium. They are called the reactions because they react when other forces on the system change. If the load on a system increases, the reaction forces will automatically increase in response to maintain equilibrium. Reaction forces are introduced in [Chapter 3](#)

and reaction moments are introduced in [Chapter 5](#).

Interaction forces are forces that exist between rigid bodies. They are defined by Newton's 3rd Law as being equal and opposite. Interaction forces become important in [Chapter 6](#) when working with multi-body structures.

Internal forces are forces and moments that exist within the material of a rigid body that resist the external loads to keep the rigid body from breaking. We model these internal loads as an axial force, a shear force, and a bending moment. Internal forces will be discussed in [Chapter 8](#).

The figure below illustrates multiple of these force types using a heavy crate being pulled by a rope across a rough surface.

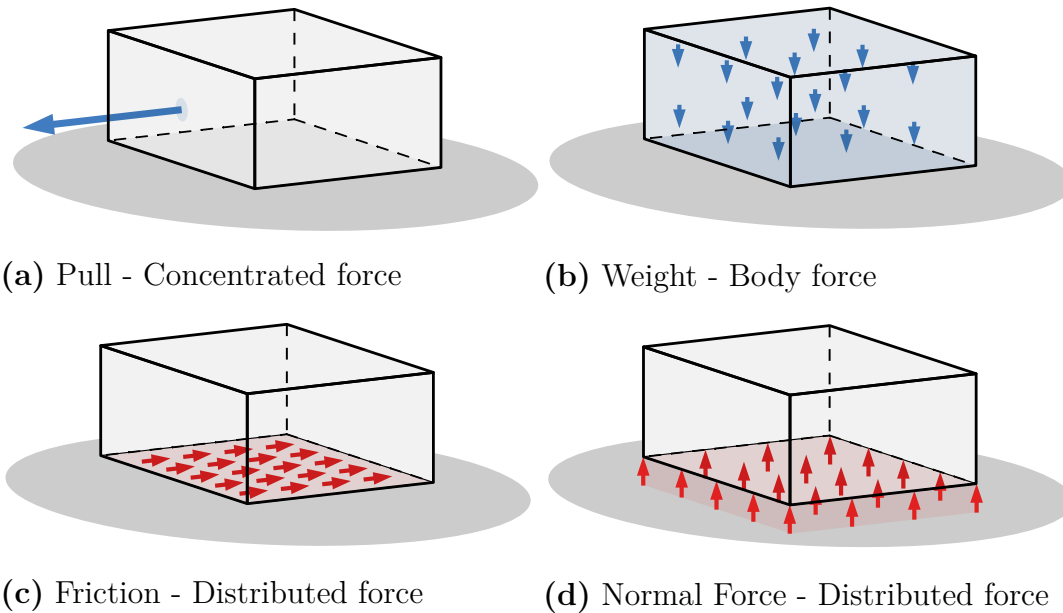


Figure 1.3.1 Forces on a crate being pulled across a rough surface.

The pull of the rope and the weight of the crate are *loads*. The rope applies a force at a single point, so is a *concentrated force*. The force of the ground holding the crate in equilibrium is a *reaction force*. This force can be divided into two components: a **tangential** friction component which acts parallel to the ground and resists the pull of the cable, and a **normal** component which acts perpendicular to the bottom surface and supports the crate's weight. The normal and tangential components are *distributed forces* since they act over the bottom surface area. The weight is also a distributed force, but one that acts over the entire crate so it's considered a *body force*. For computational simplicity we usually model all these distributed forces as equivalent concentrated forces. This process is discussed in [Chapter 7](#).

1.4 Problem Solving

Key Questions

- What are some strategies to practice selecting a tool from your problem-solving toolbox?
- What is the basic problem-solving process for equilibrium?

Statics may be the first course you take where you are required to decide on your own how to approach a problem. Unlike your previous Physics courses, you can't just memorize a formula and plug-and-chug to get an answer; there are often multiple ways to solve a problem, not all of them equally easy, so before you begin you need a plan or strategy. This seems to cause a lot of students difficulty.

The ways to think about forces, moments and equilibrium, and the mathematics used to manipulate them are like tools in your toolbox. Solving statics problems requires acquiring, choosing, and using these tools. Some problems can be solved with a single tool, while others require multiple tools. Sometimes one tool is a better choice, sometimes another. You need familiarity and practice to get skilled using your tools. As your skills and understanding improve, it gets easier to recognize the most efficient way to get a job done.

Struggling statics students often say things like:

"I don't know where to start the problem."

"It looks so easy when you do it."

"If I only knew which equation to apply, I could solve the problem."

These statements indicate that the students think they know how to use their tools, but are skipping the planning step. They jump right to writing equations and solving for things without making much progress towards the answer, or they start solving the problem using a reasonable approach but abandon it in mid-stream to try something else. They get lost, confused and give up.

Choosing a strategy gets easier with experience. Unfortunately, the way you get that experience is to solve problems. If you think that this is a chicken and egg problem, you are correct. Here are some suggestions which will help you become a better problem-solver.

- Hone your math skills, in the areas of algebra, trigonometry, and calculus. These are complimentary tools to solve engineering problems, but only help after you get the problem set up.
- Build your understanding of the terminology. If you do not understand the words in the problem statement, then you will have difficulty setting up your equations to solve the problem.

- Every engineering equation comes with assumptions. Learn these assumptions so you can decide whether a specific equation is valid within the context of a problem.
- Work through, or at least set up, lots of problems. Start with simple ones to build your skills.
- Start by viewing examples or videos, but progress to working problems on your own. Seeing someone else's solution will make you overconfident that you can do the same thing. The only way you'll know is to try on your own.
- Solve problems using multiple approaches. Confirm that alternate approaches produce the same results and try to understand why one method was easier than the other.
- Draw neat, clear, labeled diagrams.
- When you get frustrated, you have just found something new to learn. Don't get stuck for more than 15-20 minutes. Work to identify what is confusing you and find informative resources or ask questions.

The majority of the topics in this book focus on equilibrium. The remaining topics are either preparing you for solving equilibrium problems or setting you up with skills that you will use in later classes. For equilibrium problems, the problem-solving steps are:

1. Read and understand the problem.
2. Identify what you are asked to find and what is given.
3. Stop, think, and decide on an strategy.
4. Draw a free-body diagram with all the required elements:
 - (a) An isolated body or group of bodies
 - (b) All applied loads
 - (c) All reactions from supports
 - (d) An axis system
 - (e) Labeled vectors
5. Write equations of equilibrium based on the free-body diagram.
6. Check if the number of equations equals the number of unknowns. If it doesn't, you are missing something. You may need additional free-body diagrams or other relationships.
7. Solve for unknowns, then evaluate the reasonableness of your answer(s) magnitude, units, and sign/direction.

Using these steps does not guarantee that you will get the right solution, but it will help you be critical and conscious of your chosen strategies. This reflection will help you learn more quickly and increase the odds that you choose the right tool for the job.

Chapter 2

Forces and Other Vectors

To progress to the later topics of static equilibrium, structures, and friction, you will need to first build a robust set of basic vector analysis skills. These skills include:

- Understanding the basic terminology around vectors.
- Computing vector components from the various ways that vector magnitude and direction can be defined.
- Extracting vector direction in the form of a unit vector.
- Adding and subtracting vectors.
- Finding the parallel projections of vectors with dot products.
- Computing cross products to find the rotational tendency of forces.

This chapter will help you build this skillset. You may notice that many of the tools in this chapter are also covered in your Calculus III course. Use this overlap to your advantage to dial in your computational skills in calculus, while learning about the application of the math tools here in statics.

2.1 Vectors

Key Questions

- What differentiates a vector from a scalar?
- What are the standard notations for vectors and scalars in this textbook?
- How do you identify the tip, tail, line of action, direction, and magnitude of a vector?

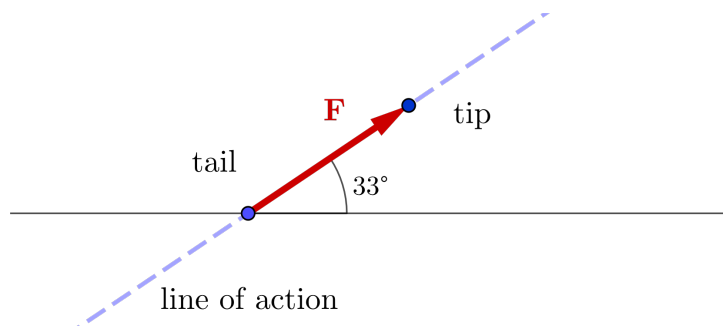
- What is the difference between the sense and orientation of a vector?

Before you can solve statics problems, you will need to understand the basic physical quantities used in Statics: scalars and vectors.

Scalars are physical quantities that have no associated direction and can be described by a positive or negative number, or even zero. Scalar quantities follow the usual laws of algebra, and most scalar quantities have units. Mass, time, temperature, and length are all scalars.

Vectors represent physical quantities that have magnitude and direction. Vectors are identified by a symbolic name which will be typeset in bold like **r** or **F** to indicate its vector nature. The primary vector quantity you will encounter in statics will be **force**, but **moment** and **position** are also important vectors. Computations involving vectors must always consider the directionality of each term and follow the rules of vector algebra as described in this chapter.

You can visualize a vector as an **arrow** pointing in a particular direction. The **tip** is the pointed end and the **tail** the trailing end. The tip and tail of a vector define a **line of action**. A line of action can be thought of as an invisible string along which a vector can slide. Sliding a vector along its line of action does not change its magnitude or its direction. Sliding a vector can be a handy way to simplify vector problems.



[Live Link](#)

Figure 2.1.1 Vector Definitions

The standard notation for a vector uses the vector's name in bold font, or an arrow or bar above the vector's name. All three of these notations mean the same thing.

$$\mathbf{F} = \vec{F} = \bar{F} = \text{a vector named } F$$

Most printed works including this book will use the bold symbol to indicate vectors, but for handwritten work, you and your instructor will use the bar or arrow notation.

Force vectors acting on physical objects have a **point of application**, which

is the point where the force is applied. Other vectors, such as moment vectors, are **free vectors**, which means that the point of application is not significant. Free vectors can be moved freely to any location as long as the magnitude and direction are maintained.

The vector's **magnitude** is a *positive real number* including units which describes the 'strength' or 'intensity' of the vector. Graphically a vector's magnitude is represented by the length of its vector arrow, and symbolically by enclosing the vector's symbol with vertical bars. This is the same notation as for the absolute value of a number. The absolute value of a number and the magnitude of a vector can both be thought of as a distance from the origin, so the notation is appropriate. By convention the magnitude of a vector is also indicated by the same letter as the vector, but in non-bold font.

$$F = |\mathbf{F}| = \text{the magnitude of vector } \mathbf{F}$$

By itself, a vector's magnitude is a scalar quantity, but it makes no sense to speak of a vector with a negative magnitude so vector magnitudes are always positive or zero. Multiplying a vector by -1 produces a vector with the same magnitude but pointing in the opposite direction.

Vector directions are described with respect to a **coordinate system**. A coordinate system is an arbitrary reference system used to establish the **origin** and the primary directions. Distances are usually measured from the origin, and directions from a primary or **reference direction**. You are probably familiar with the Cartesian coordinate system with mutually perpendicular x , y and z axes and the origin at their intersection point.

Another way of describing a vector's direction is to specify its **orientation** and **sense**. Orientation is the angle the vector's line of action makes with a specified reference direction, and sense defines the direction the vector points along its line of action. A vector with a positive sense points towards the positive end of the reference axis and vice-versa. A vector representing an object's weight has a vertical reference direction and a downward sense or negative sense, for example.

A third way to represent a vector is with a **unit vector** multiplied by a **scalar component**. Unit vectors (which we'll learn about in [Section 2.5](#)) are vectors with a magnitude of one (unitless), and scalar components are signed values with units. Together, they fully define a vector quantity; the unit vector specifies the direction of its line of action, and the scalar component specifies its magnitude and sense. The scalar component "scales" the unit vector.

Be careful not to confuse scalar components, which can be positive or negative, with vector magnitudes, which are always positive.

Vectors are either constant or vary as a function of time, position, or something else. For example, if a force varies with time according to the function $F(t) = (10 \text{ N/s})t$, where t is the time in seconds, then the force will be 0 N at $t = 0$ s and increase by 10 N each second thereafter.

2.2 One-Dimensional Vectors

Key Questions

- Why is one-dimensional vector math just like scalar math?
- Given two one-dimensional vectors, how do you compute and then draw the resultant?
- What happens when you multiply a vector by a scalar?

The simplest vector calculations involve one-dimensional vectors. With one-dimensional vectors, vector arithmetic is the same as *ordinary* arithmetic.

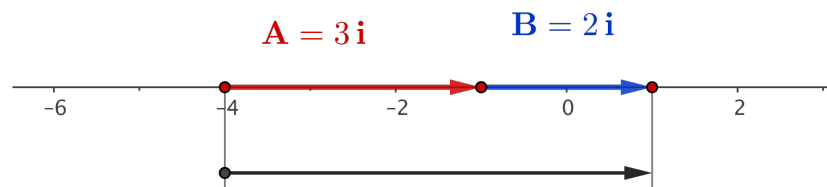
By definition, one-dimensional vectors all act along a single line. The line they share is called the **line of action**, but note that vectors that act along a line may point towards either end.

If we place a coordinate axis along the line of action and give vectors that point towards the positive end *positive* values equal to their magnitudes, and give vectors that point the other way *negative* values, then the vectors add, subtract and multiply by constants just like ordinary numbers along the number line. The resulting sign indicates the direction of the result. Note that, by convention, the labeled end of an axis is the positive end.

2.2.1 Vector Addition

Adding multiple vectors together finds the **resultant** vector. Resultant vectors can be thought of as the *sum* of or *combination* of two or more vectors.

To find the resultant vector **R** of two one-dimensional vectors **A** and **B** you can use the tip-to-tail technique in Figure 2.1.1 below. In the tip-to-tail technique, you slide vector **B** until its tail is at the tip of **A**, and the vector from the tail of **A** to the tip of **B** is the resultant **R**. Note that vector addition is commutative: the resultant **R** is the same whether you add **A** to **B** or **B** to **A**.



[Live Link](#)

Figure 2.2.1 One Dimensional Vector Addition

2.2.2 Vector Subtraction

The easiest way to handle vector subtraction is to add the negative of the vector you are subtracting to the other vector. In this way, you can still use the tip-to-tail technique after flipping the vector you are subtracting.

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}) \quad (2.2.1)$$

Example 2.2.2 Vector subtraction.

Find $\mathbf{A} - \mathbf{B}$ where $\mathbf{A} = 2 \mathbf{i}$ and $\mathbf{B} = 3 \mathbf{i}$.

Solution. You can simulate this in [Figure 2.2.1](#).

1. Set $\mathbf{A}$ to a value of $2 \mathbf{i}$ and $\mathbf{B}$ to a value of $-3 \mathbf{i}$, the negative of its actual value.
2. Move the vectors until they are tip-to-tail. The order does not matter because vector addition is commutative.

$$\mathbf{R} = -1 \mathbf{i}.$$

2.2.3 Vector Multiplication by a Scalar

Multiplying or dividing a vector by a scalar changes the vector's magnitude but maintains its original line of action. In other words, multiplying a vector by a scalar "scales" the vector.

One common transformation is to find the negative of a vector. To find the negative of vector $\mathbf{A}$, we multiply it by -1; in equation form

$$-\mathbf{A} = (-1)\mathbf{A}$$

Spatially, the effect of multiplying a vector by -1 , is to reverse it end for end. The magnitude and line of action stay the same, but the sense reverses, so now the arrowhead points in the opposite direction. When multiplying vector components by -1 , simply flip the $+/-$ sign of each term to flip the direction of the vector.

2.3 2D Coordinate Systems & Vectors

Key Questions

- Why are orthogonal coordinate systems useful?
- How do you transform between polar and Cartesian coordinates?

- How is the notation for point coordinates different than that of vector coordinates?

A coordinate system provides a reference frame to describe a system we want to analyze. Although there are many different coordinate systems that can be used, selecting an appropriate one will make a problem easier to solve.

In statics, we use **orthogonal** coordinate systems, where orthogonal means “perpendicular.” The advantage of orthogonal coordinate systems is that the coordinate directions are all perpendicular and, therefore, independent. The intersection of the coordinate axes is called the **origin**, and measurements are made from there. Both points and vectors are described with a set of numbers called the **coordinates**. For points in space, the coordinates specify the distance you must travel in each of the coordinate directions to get from the origin to the point in question. Together, the coordinates can be thought of as specifying a **position vector**, a vector from the origin directly to the point. The position vector gives the magnitude and direction needed to travel directly from the origin to the point.

In the case of force vectors, the coordinates are the **scalar components** of the force in each of the coordinate directions. These components locate the tip of the vector and they can be interpreted as the fraction of the total force which acts in each of the coordinate directions.

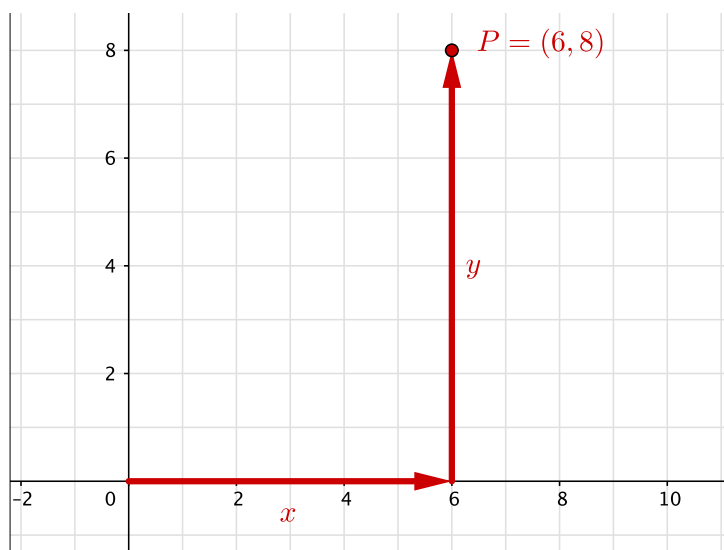
Three coordinate directions are needed to map our real three-dimensional world but in this section we will start with two, simpler, two-dimensional orthogonal systems: **rectangular** and **polar** coordinates, and the tools to convert from one to the other.

2.3.1 Rectangular Coordinates

The most important coordinate system is the **Cartesian** system, which was named after the French mathematician René Descartes. In two dimensions the coordinate axes are straight lines rotated 90° apart named x , and y .

In most cases, the x axis is horizontal and points to the right, and the y axis points vertically upward, however, we are free to rotate or translate this entire coordinate system if we like. It is usually mathematically advantageous to establish the origin at a convenient point to make measurements from, and to align one of the coordinate axes with a major feature of the problem.

Points are specified as an ordered pair of coordinate values separated by a *comma* and enclosed in parentheses, $P = (x, y)$.



[Live Link](#)

Figure 2.3.1 Cartesian Coordinate System

Similarly, forces and other vectors will be specified with an ordered pair of scalar components enclosed by angle brackets,

$$\mathbf{F} = \langle F_x, F_y \rangle.$$

2.3.2 Polar Coordinates

The **polar** coordinate system is an alternate orthogonal system which is useful in some situations. In this system, a point is specified by its distance r from the origin, and an angle θ measured counter-clockwise from a reference direction – usually the positive x axis. Polar coordinates are useful if we know more about distances and angles from a central point. Examples of systems that would use polar coordinates would be tower cranes to build skyscrapers, robotic arm motion, and radar systems.

In this text, points in polar coordinates will be specified as an ordered pair of values separated by a *semicolon* and enclosed in parentheses

$$P = (r; \theta).$$

Angles can be measured in either radians or degrees, so be sure to include a degree sign on angle θ if that is what you intend.

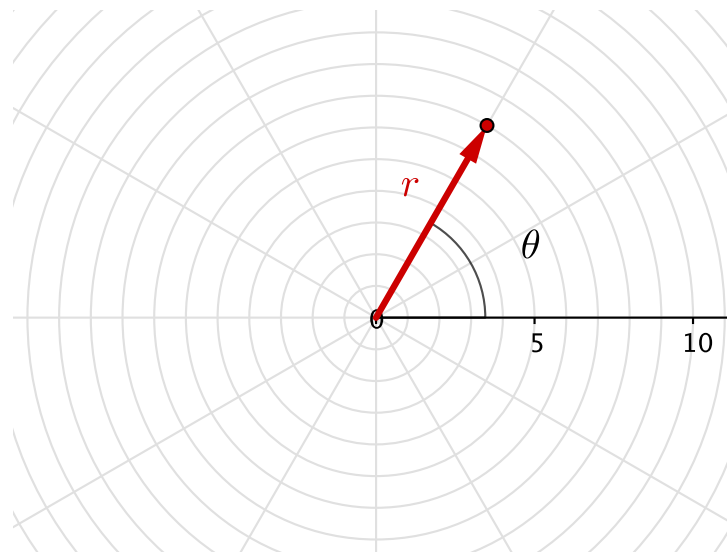
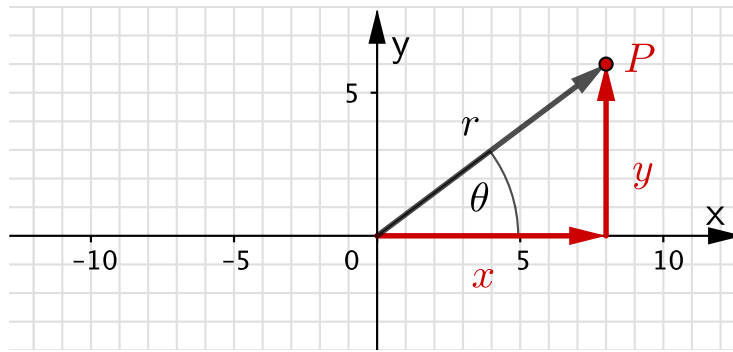
[Live Link](#)

Figure 2.3.2 Polar Coordinate System

2.3.3 Coordinate Transformation

You should be able to translate points from one coordinate system to the other whenever necessary. The relation between (x, y) coordinates and $(r; \theta)$ coordinates are illustrated in the diagram and right-triangle trigonometry is all that is needed to convert from one representation to the other.

Luckily, it is quite easy to translate points between the rectangular (x, y) and polar $(r; \theta)$ coordinate systems using right-triangle trigonometry. As you move point P around the figure below, you can see how the two coordinate types are related.



[Live Link](#)

Figure 2.3.3 Coordinate Transformation

Rectangular To Polar for points (Given: x and, y).

$$r = \sqrt{x^2 + y^2} \quad (2.3.1)$$

$$\theta = \tan^{-1} \left(\frac{y}{x} \right) \quad (2.3.2)$$

$$P = (r ; \theta) \quad (2.3.3)$$

Note 2.3.4

Take care when using the inverse tangent function on your calculator. Calculator angles are always in the first or fourth quadrant, and you may need to add or subtract 180° to the calculator angle to locate the point in the correct quadrant.

Polar to Rectangular for points (Given: r and, θ).

$$x = r \cos \theta \quad (2.3.4)$$

$$y = r \sin \theta \quad (2.3.5)$$

$$P = (x, y) \quad (2.3.6)$$

Rectangular To Polar for forces (Given: rectangular components). If you are working with forces rather than distances, the process is exactly the same but triangle is labeled differently. The hypotenuse of the triangle is the magnitude of the vector, and sides of the right triangle are the scalar components of the force, so for vector **A**

$$A = \sqrt{A_x^2 + A_y^2} \quad (2.3.7)$$

$$\theta = \tan^{-1} \left(\frac{A_y}{A_x} \right) \quad (2.3.8)$$

$$\mathbf{A} = (A; \theta) \quad (2.3.9)$$

Polar to Rectangular for forces (Given: magnitude and direction).

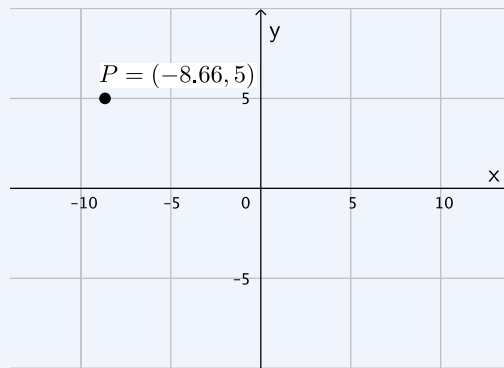
$$A_x = A \cos \theta \quad (2.3.10)$$

$$A_y = A \sin \theta \quad (2.3.11)$$

$$\mathbf{A} = \langle A_x, A_y \rangle = A \langle \cos \theta, \sin \theta \rangle \quad (2.3.12)$$

Example 2.3.5 Rectangular to Polar Representation.

Express point $P = (-8.66, 5)$ in polar coordinates.



Solution 1. Given: $x = -8.66$, $y = 5$

$$\begin{aligned} r &= \sqrt{x^2 + y^2} & \theta &= \tan^{-1} \left(\frac{y}{x} \right) \\ &= \sqrt{(-8.66)^2 + (5)^2} & &= \tan^{-1} \left(\frac{5}{-8.66} \right) \\ &= 10 & &= \tan^{-1}(-0.577) \\ & & &= -30^\circ \end{aligned}$$

You must be careful here and use some common sense. The -30° angle your calculator gives you in this problem is incorrect because point P is in the second quadrant, but your calculator doesn't know this. It can't tell whether the argument of $\tan^{-1}(-0.577)$ is negative because the x was negative or because the y was negative, so it must make an assumption and in this case it is wrong.

The arctan function on calculators will always return values in the first and fourth quadrant. If, by inspection of the x and the y coordinates, you see that the point is in the second or third quadrant, you must add or subtract 180° to the calculator's answer.

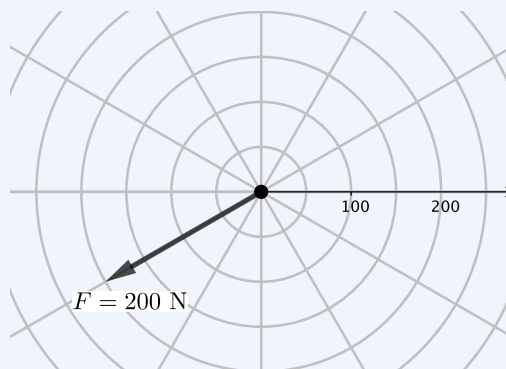
So in this problem, θ is really $-30^\circ + 180^\circ$. After making this adjustment, the location of P in polar coordinates is:

$$P = (10; 150^\circ)$$

Solution 2. Most scientific calculators include handy polar-to-rectangular and rectangular-to-polar functions that can save you time and help you avoid errors. Perhaps you should [google your calculator model](#) to find out if yours does and learn how to use it?

Example 2.3.6 Polar to Rectangular Representation.

Express 200 N force $\mathbf{F}$ as a pair of scalar components.



Solution 1. Given: The magnitude of force $\mathbf{F} = 200$ N, and from the diagram we see that the direction of $\mathbf{F}$ is 30° counter-clockwise from the negative x axis.

Letting $\theta = 30^\circ$ we can find the components of $\mathbf{F}$ with right triangle trigonometry.

$$\begin{aligned} F_x &= F \cos \theta & F_y &= F \sin \theta \\ &= 200 \text{ N} \cos 30^\circ & &= 200 \text{ N} \sin 30^\circ \\ &= 173.2 \text{ N} & &= 100 \text{ N} \end{aligned}$$

Since the force points down and to the left into the third quadrant, these values are actually negative, and the signs must be applied manually. After making this adjustment, the location of $\mathbf{F}$ expressed in rectangular coordinates is:

$$\mathbf{F} = \langle -173.2 \text{ N}, -100 \text{ N} \rangle$$

Solution 2. If you would prefer not to apply the negative signs by hand, you can convert the 30° to an angle measured from the positive x axis and let your calculator takes care of the signs. You may use either $\theta = 30^\circ \pm 180^\circ$.

For $\theta = -150^\circ$

$$\begin{aligned}
 F_x &= F \cos \theta & F_y &= F \sin \theta \\
 &= 200 \text{ N} \cos(-150^\circ) & &= 200 \text{ N} \sin(-150^\circ) \\
 &= -173.2 \text{ N} & &= -100 \text{ N}
 \end{aligned}$$

$$\mathbf{F} = \langle -173.2 \text{ N}, -100 \text{ N} \rangle$$

Although this approach is mathematically correct, experience has shown that it can lead to errors and we recommend that when you work with right triangles, use angles between zero and 90° , and apply signs manually as required by the physical situation.

2.4 3D Coordinate Systems & Vectors

Key Questions

- How can we check if a rectangular coordinate system follows the right-hand rule?
- What are direction cosine angles and why are they always less than 180° ?
- How are spherical coordinates different than cylindrical coordinates?

In this section we will discuss four methods to specify points and vectors in three-dimensional space.

The most commonly used method is an extension of two-dimensional **rectangular coordinates** to three-dimensions. Alternately, points and vectors in three dimensions can be specified in terms of **direction cosines**, or using **spherical** or **cylindrical** coordinate systems. These will be discussed in the following sections.

You will often need to convert from one representation to another. Good visualization skills are helpful here.

2.4.1 Rectangular Coordinates

We can extend the two-dimensional Cartesian coordinate system into three dimensions easily by adding a z axis perpendicular to the two-dimensional Cartesian plane. The notation is similar to the notation used for two-dimensional vectors. Points and forces are expressed as ordered triples of rectangular coordinates following the same notation used previously.

$$P = (x, y, z)$$

$$\mathbf{F} = \langle F_x, F_y, F_z \rangle$$

For nearly all three-dimensional problems, you will need the rectangular x , y , and z locations of points in space and components of vectors before proceeding with the computations. If you are given the components upfront, then you are set to move forward, but otherwise, you will need to transform one coordinate system into rectangular coordinates.

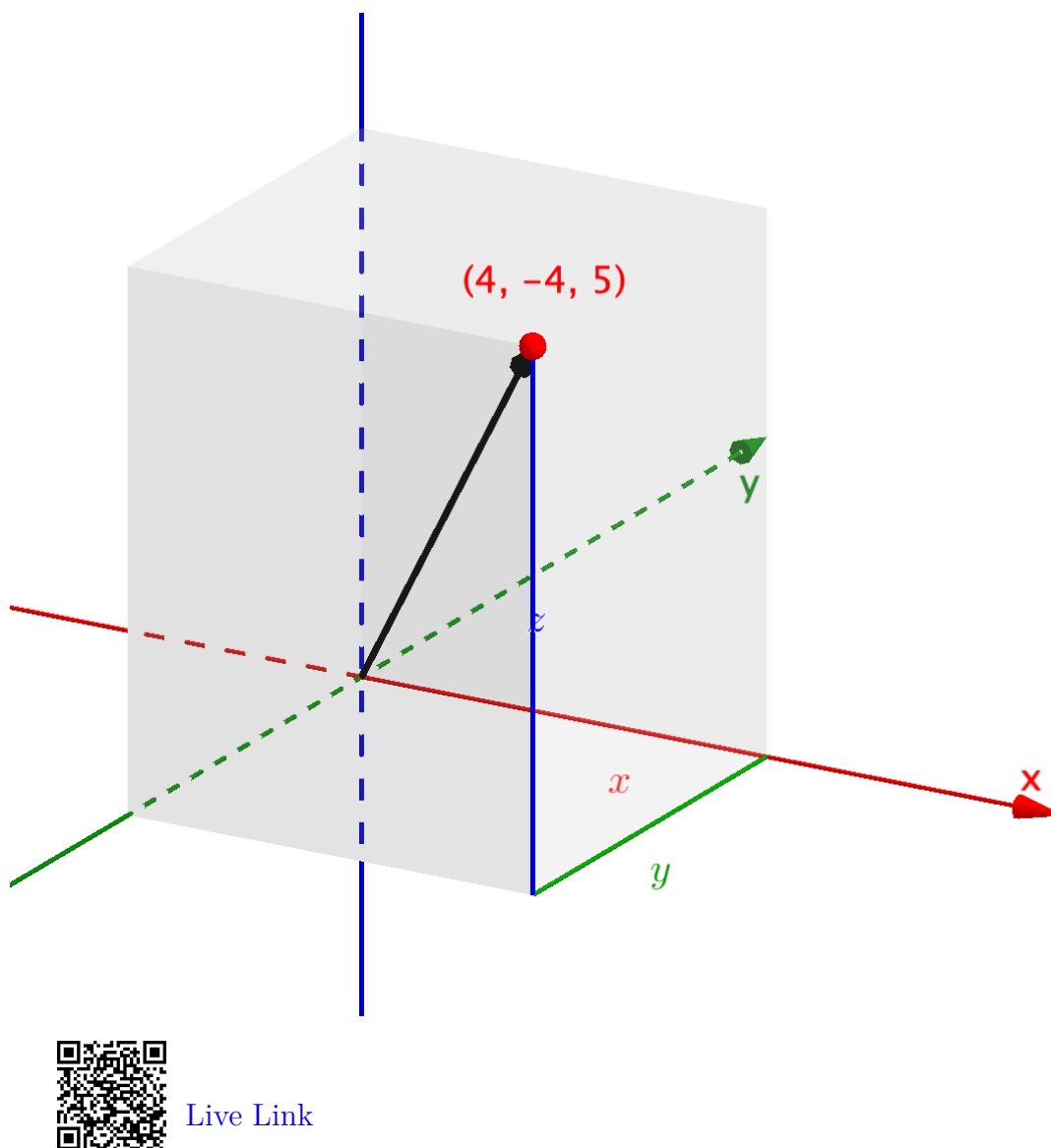


Figure 2.4.1 Three-Dimensional Rectangular Coordinates

Thinking Deeper 2.4.2 Right-Handed Coordinate Systems.

Does it matter which way the axes are oriented? Is it OK to make the x axis point left or the y axis point down?

The answers to the questions above, like so many in life, are "It depends." If you are simply summing the forces of a system, you typically can select any perpendicular axes you wish, but for systems involving any cross-product relationships (like the rotational moments you'll learn about in [Chapter 4](#)), we need to use a standard orientation. In mathematics and engineering, the default is a **right-handed coordinate system**, where the coordinate axes are oriented according to the **right-hand rule** shown in the figure.

To apply the right-hand rule, orient your thumb and first two fingers at right angles to each other and align them with three coordinate axes. Starting with your thumb, name the axes in alphabetical order x - y - z . In mathematical terms, when we cross the x -axis into the y -axis, we need to get the direction of the z -axis.

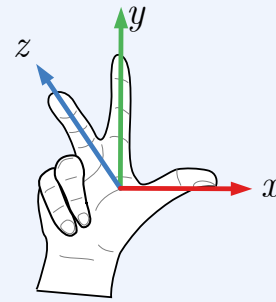
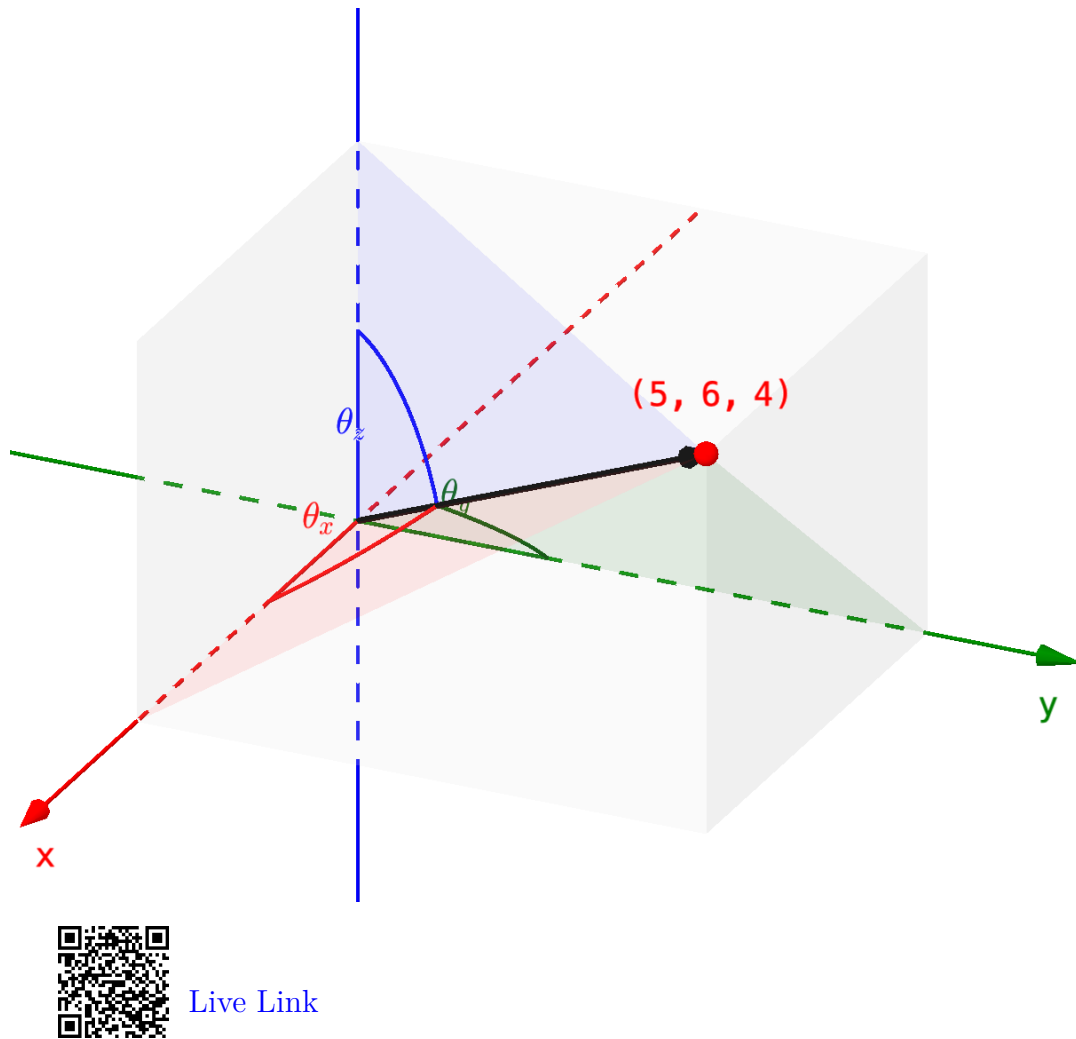


Figure 2.4.3 Right-handed coordinate system.

2.4.2 Direction Cosine Angles

The direction of a vector in two-dimensional systems could be expressed clearly with a single angle measured from a reference axis, but adding an additional dimension means that one angle is no longer enough.

One way to define the direction of a three-dimensional vector is by using **direction cosine angles**, also commonly known as **coordinate direction angles**. The direction cosine angles are the angles between the positive x , y , and z axes to a given vector and are traditionally named θ_x , θ_y , and θ_z . Three-dimensional vectors, components, and angles are often difficult to visualize because they do not commonly lie in the Cartesian planes.

**Figure 2.4.4** Direction Cosine Angles

We can relate the components of a vector to its direction cosine angles using the following equations.

$$\cos \theta_x = \frac{A_x}{|A|} \quad \cos \theta_y = \frac{A_y}{|A|} \quad \cos \theta_z = \frac{A_z}{|A|} \quad (2.4.1)$$

These cosine relationships come from the red, green, and blue right triangles shown in Figure 2.4.4, where the side adjacent to each angle is the component and the hypotenuse is the full vector. Note that the component in the numerator of each direction cosine equation is positive or negative as defined by the coordinate system, and the vector magnitude in the denominator is always positive.

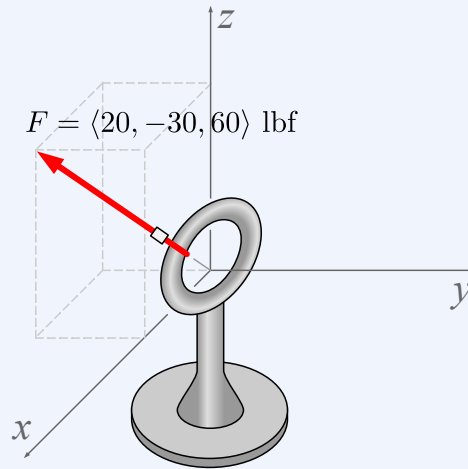
From these equations, we can conclude that:

- Direction cosines are signed value between -1 and 1.
- Direction cosine angles must always be between 0° and 180° or

$$0^\circ \leq \theta_n \leq 180^\circ.$$

- Any direction cosine angle greater than 90° indicates a negative component along that respective axis. Spatially this is because all direction cosine angles are measured from the positive side of each axis. Mathematically this is because the cosine of any angle between 90 and 180 degrees is numerically negative.

Example 2.4.5 Direction Cosine Angles.



A rope pulls on an anchor ring centered at the origin with force $\mathbf{F} = \langle 20, -30, 60 \rangle$ lbf.

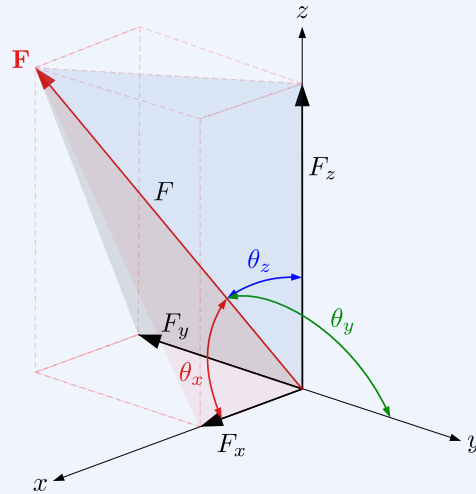
Find the magnitude of $\mathbf{F}$ and the direction cosine angles, θ_x , θ_y , and θ_z components.

Solution. Since the three components of F are perpendicular, we can apply the Pythagorean Theorem to find the magnitude of F .

$$\begin{aligned} F = |\mathbf{F}| &= \sqrt{F_x^2 + F_y^2 + F_z^2} \\ &= \sqrt{20^2 + (-30)^2 + 60^2} \text{ lbf} \\ &= 70 \text{ lbf} \end{aligned}$$

Direction cosine angles are equal to the inverse cosine of each Cartesian force component divided by the force magnitude.

$$\begin{aligned} \theta_x &= \cos^{-1} \left(\frac{F_x}{|\mathbf{F}|} \right) = \cos^{-1} \left(\frac{20}{70} \right) = 73.4^\circ \\ \theta_y &= \cos^{-1} \left(\frac{F_y}{|\mathbf{F}|} \right) = \cos^{-1} \left(\frac{-30}{70} \right) = 115.38^\circ \\ \theta_z &= \cos^{-1} \left(\frac{F_z}{|\mathbf{F}|} \right) = \cos^{-1} \left(\frac{60}{70} \right) = 31.0^\circ \end{aligned}$$



Since the direction cosine angles are measured from the positive x , y , and z axes, the negative component of F_y means that $\theta_y > 90^\circ$, while both θ_x and θ_z are less than 90° as their components are positive.

2.4.3 Spherical Coordinates

In spherical coordinates, points are specified with these three coordinates

- r , the **radial distance** from the origin to the point in space. This can also be thought of as the magnitude of a position vector $\mathbf{r}$ pointing from the origin to the point in space.
- θ , the **azimuthal angle**, measured counter-clockwise from the positive x axis to the projection of the vector onto the xy plane, and
- ϕ , the **polar angle** from the z axis to the vector.

Other vectors are specified similarly by using the magnitude of the vector to define the radius of the sphere in appropriate units.

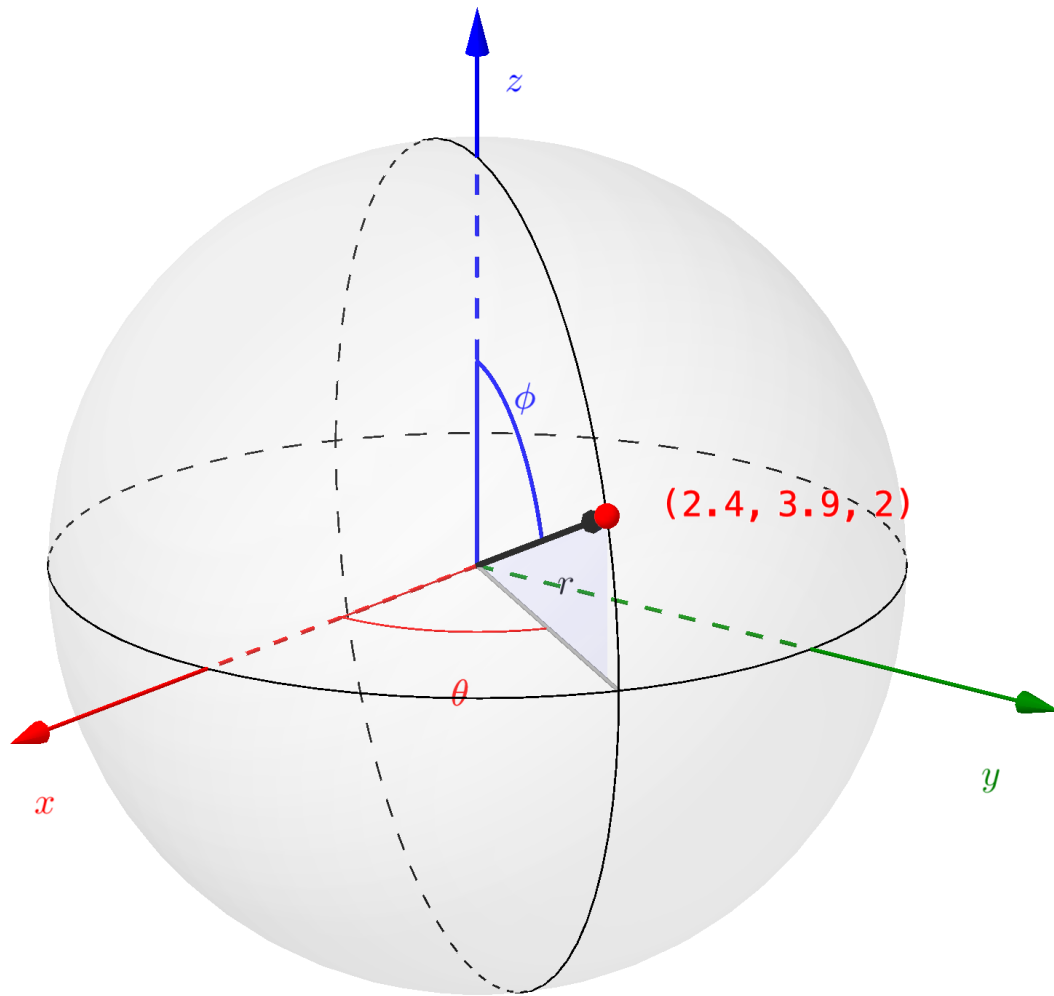
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Figure 2.4.6 Spherical Coordinate System

Question 2.4.7

What are the differences between polar coordinates and terrestrial latitude/longitude locations?

Answer. In terrestrial measurements

- Coordinate r is not needed since all points are on the surface of the globe.
- Longitude is measured 0° to 180° East or West of the prime meridian, rather than 0° to 360° counter-clockwise from the x axis.

- Latitude is measured 0° to 90° North or South of the equator, whereas polar angle ϕ is 0° to 180° measured from the “North Pole”.

When the two given spherical angles are defined in the manner shown here, the rectangular components of the vector $\mathbf{A} = (A ; \theta ; \phi)$ are found thus:

$$A' = A \sin \phi \quad (2.4.2)$$

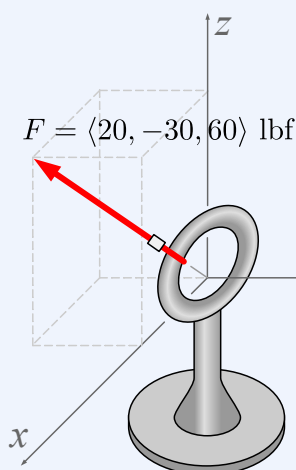
$$A_z = A \cos \phi \quad (2.4.3)$$

$$A_x = A' \cos \theta = A \sin \phi \cos \theta \quad (2.4.4)$$

$$A_y = A' \sin \theta = A \sin \phi \sin \theta \quad (2.4.5)$$

Reflect on the equations above. Can you think through the process of how they were derived? The generalized steps are as follows. First, draw an accurate sketch of the given information and define the right triangles related to both θ and ϕ . Then use trig identities on the right triangle involving the vector, the z axis and angle ϕ to find A_z , and A' , the projection of $\mathbf{A}$ onto the xy plane. Finally, use trig identities on the right triangle involving vector $\mathbf{A}'$ and θ to find the remaining components of $\mathbf{A}$.

Example 2.4.8 Spherical Coordinates.



A rope pulls on an anchor ring centered at the origin with force $\mathbf{F} = \langle 20, -30, 60 \rangle$ lbf.

Find the *spherical* coordinates of $\mathbf{F}$.

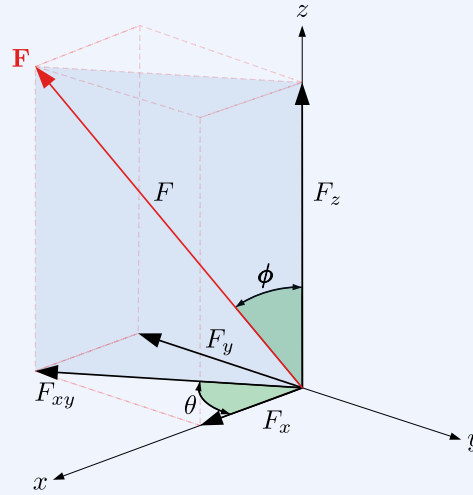
Solution. To represent $\mathbf{F}$ in spherical coordinates, we must find the radial distance r , the azimuthal angle θ , and the polar angle ϕ . Coordinate r is simply the magnitude of force $\mathbf{F}$. Since the three components of $\mathbf{F}$ are perpendicular, we can apply the Pythagorean Theorem to find it.

$$\begin{aligned} F = |\mathbf{F}| = r &= \sqrt{F_x^2 + F_y^2 + F_z^2} \\ &= \sqrt{20^2 + (-30)^2 + 60^2} \text{ lbf} \end{aligned}$$

$$= 70 \text{ lbf}$$

Azimuthal angle θ measures the angle between the x axis and the projection of $\mathbf{F}$ onto the xy plane, F_{xy} .

Using a right triangle with sides F_x , F_y , and F_{xy} , we can find θ using the inverse tangent of the ratio of the opposite to adjacent sides.



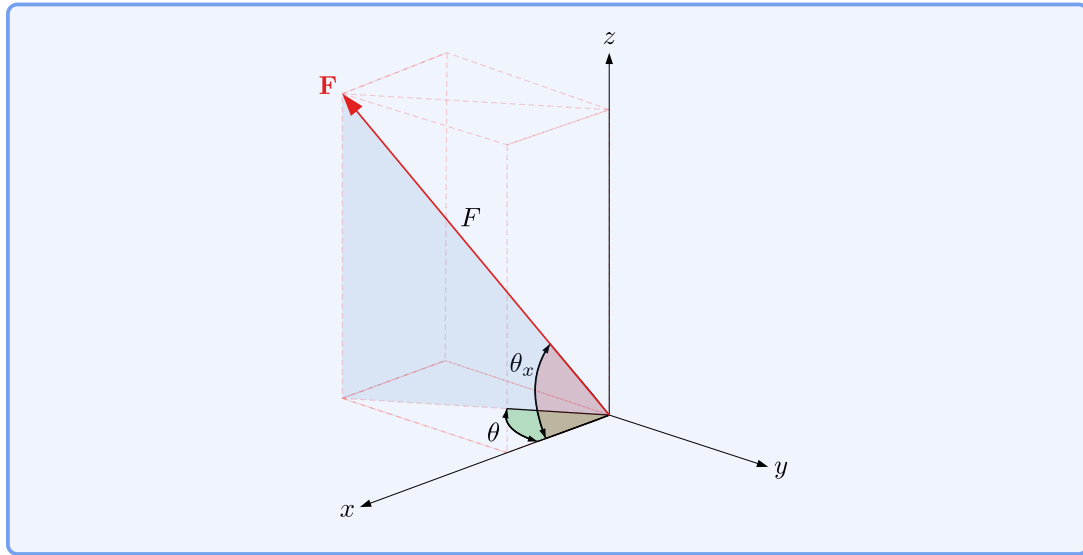
$$\theta = \tan^{-1} \frac{F_y}{F_x} = \tan^{-1} \left(\frac{-30}{20} \right) = -56.31^\circ$$

This angle is negative because it is measured *clockwise* from the positive x axis, opposite the standard CCW direction.

The polar angle ϕ is measured down from the $+z$ axis to the vector $\mathbf{F}$. We can find it using a right triangle with sides F , F_z , and F_{xy} . Note that ϕ is the same as the direction cosine angle θ_z .

$$\phi = \theta_z = \cos^{-1} \frac{F_z}{|\mathbf{F}|} = \cos^{-1} \left(\frac{60}{70} \right) = 31.0^\circ$$

Also notice that the azimuthal angle θ is smaller than the direction cosine angle θ_x , since θ is in the xy plane, but θ_x is a 3D angle from the x axis to the vector $\mathbf{F}$.



2.4.4 Cylindrical Coordinates

The cylindrical coordinate system is seldom used in statics, however, it is useful in certain geometries. Cylindrical coordinates extend two-dimensional polar coordinates by adding a z coordinate indicating the distance above or below the xy plane.

Points and vectors are specified with these three cylindrical coordinates.

- r , the radius of the cylinder. This is the distance from the origin to the projection of the tip of the vector onto the xy plane, or equivalently the magnitude of the component of the vector in the xy plane.
- θ , the azimuthal angle, measured counter-clockwise from the positive x axis to the projection of the vector onto the xy plane
- z , the vertical height of the vector tip, or equivalently the magnitude of the z component.

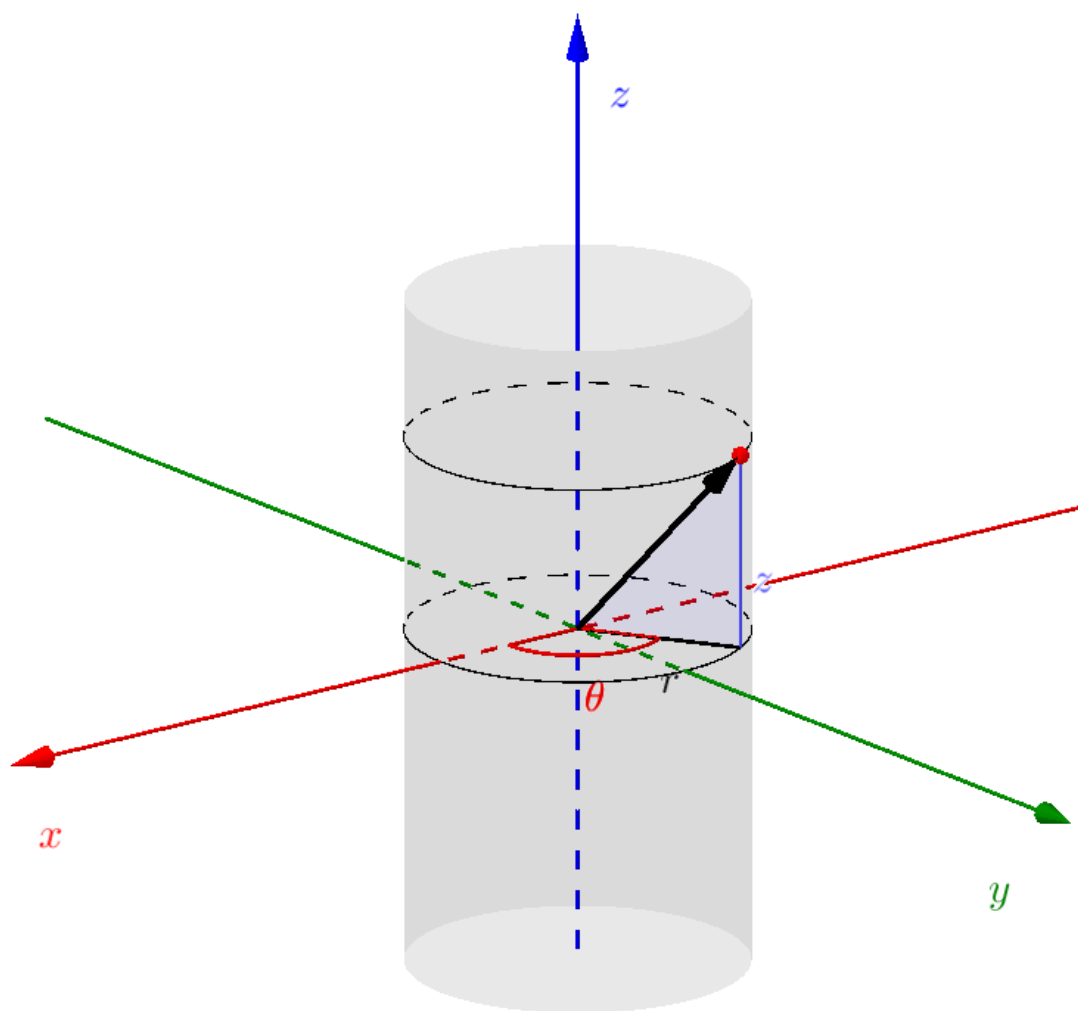
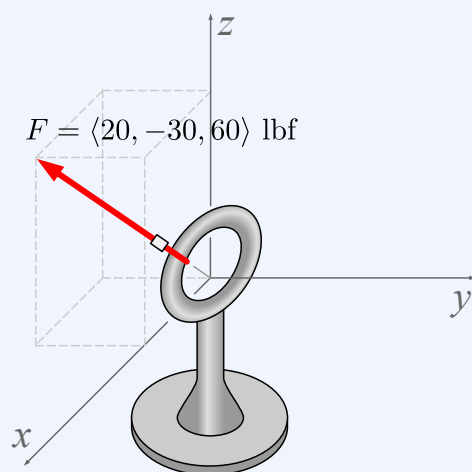
[Live Link](#)

Figure 2.4.9 Cylindrical Coordinate System

Unfortunately, not all problems give the angles θ and ϕ as defined here; so you will need to find them from the given angles in other situations.

You can use the interactive diagram in this section to practice visualizing and finding the components of a vector in all of these coordinate systems. You should be able to find the x , y , and z coordinates given direction angles, spherical or cylindrical coordinates, and vice-versa.

Example 2.4.10 Cylindrical Coordinates.



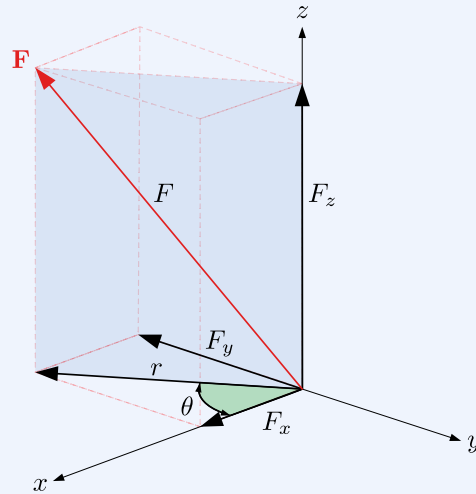
A rope pulls on an anchor ring centered at the origin with force $\mathbf{F} = \langle 20, -30, 60 \rangle$ lbf.

Find the *cylindrical* coordinates of $\mathbf{F}$.

Solution. To represent $\mathbf{F}$ in cylindrical coordinates, we must find the radial distance, r , the azimuthal angle, θ , and the axial coordinate, z . In cylindrical coordinates, r is the radius of the cylinder rather than the radius of the enclosing sphere. r is the projection of $\mathbf{F}$ onto the xy plane, F_{xy} , and can be found by applying the Pythagorean Theorem to the x and y components of $\mathbf{F}$.

$$\begin{aligned} r = F_{xy} &= \sqrt{F_x^2 + F_y^2} \\ &= \sqrt{20^2 + (-30)^2} \text{ lbf} \\ &= 36.06 \text{ lbf} \end{aligned}$$

The azimuthal angle θ is the same in both cylindrical and spherical coordinates. It measures the angle between the x axis and the projection of $\mathbf{F}$ onto the xy plane. θ can be found using a right triangle in the xy plane with sides F_x and F_y .



$$\theta = \tan^{-1} \frac{F_y}{F_x} = \tan^{-1} \left(\frac{-30}{20} \right) = -56.31^\circ$$

Finally, the z component is the vertical component of the force, F_z , which was given.

$$F_z = 60.0 \text{ lbf}$$

2.5 Unit Vectors

Key Questions

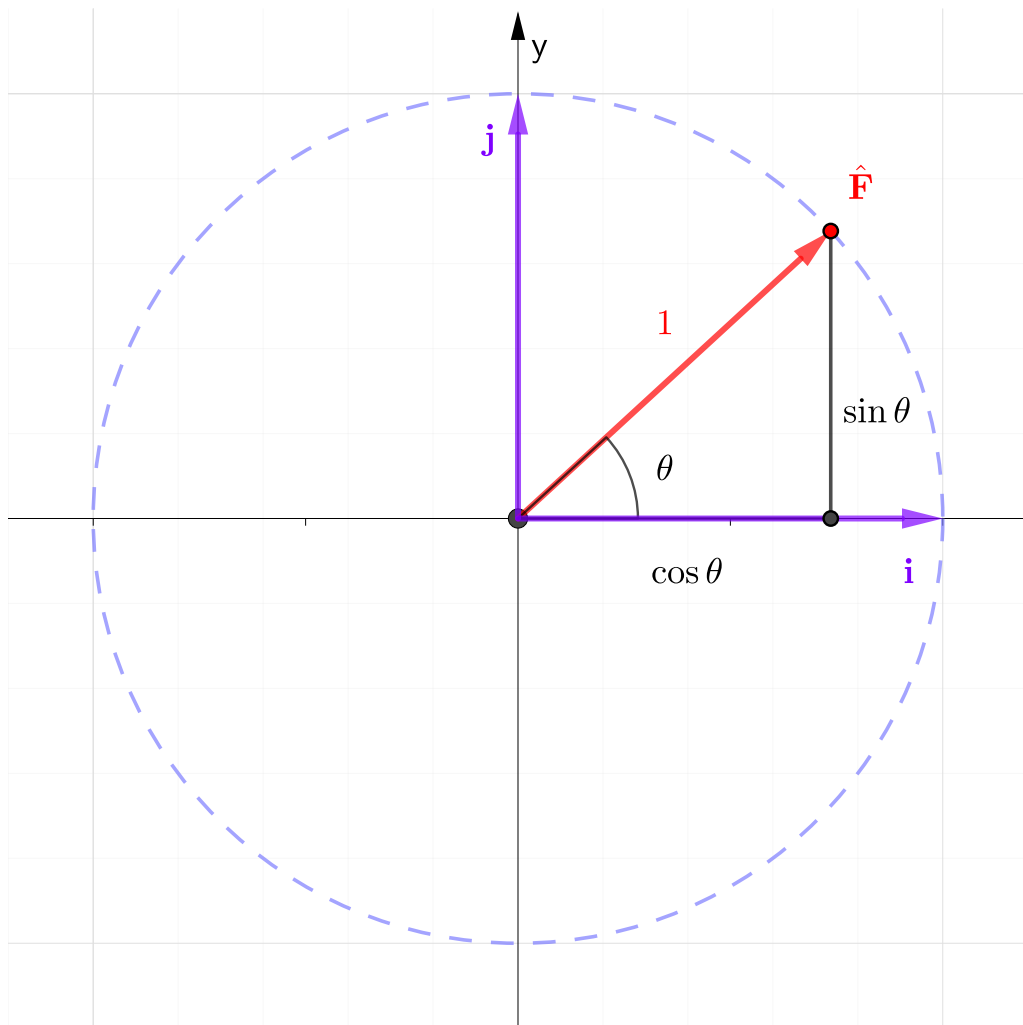
- Why are unit vectors useful?
- What are the unit vectors along the Cartesian x , y , and z axes?
- How do you find the force vector components of known force magnitude along a geometric line?
- How can you find unit vector components from direction cosine angles?

A unit vector is a vector with a magnitude of one and no units. As such, a unit vector represents a pure direction. By convention, a unit vector is indicated by a *hat* over a vector symbol. This may sound like a new concept, but it's a simple one, directly related to the unit circle, the Pythagorean Theorem, and the definitions of sine and cosine.

2.5.1 Cartesian Unit Vectors

A unit vector can point in any direction, but because they occur so frequently the unit vectors in each of the three Cartesian coordinate directions are given their own symbols, which are:

- $\mathbf{i}$, for the unit vector pointing in the x direction,
- $\mathbf{j}$, for the unit vector pointing in the y direction, and
- $\mathbf{k}$, for the unit vector pointing in the z direction..



[Live Link](#)

Figure 2.5.1 Unit Vector Interactive

Applying the Pythagorean Theorem to the triangle gives the equation for a unit circle

$$\cos^2 \theta + \sin^2 \theta = 1^2$$

No matter what angle a unit vector makes with the x axis, $\cos \theta$ and $\sin \theta$ are its scalar components. This relation assumes that the angle θ is measured from the x axis, if it is measured from the y axis the sine and cosine functions reverse, with $\sin \theta$ defining the horizontal component and the $\cos \theta$ defining the vertical component.

The x and y components of a point on the unit circle are also the scalar components of $\hat{\mathbf{F}}$, so

$$\left. \begin{array}{l} F_x = \cos \theta \\ F_y = \sin \theta \end{array} \right\} \implies \hat{\mathbf{F}} = \langle \cos \theta, \sin \theta \rangle.$$

2.5.2 Relation between Vectors and Unit Vectors

When a purely-directional unit vector is multiplied by a scalar value it is *scaled* by that amount. For example, when a unit vector pointing to the right is multiplied by 100 N the result is a 100 N vector pointing to the right. When a unit vector pointing up is multiplied by -50 N, the negative magnitude flips the direction of the unit vector and the result is a 50 N vector pointing down.

In general,

$$\mathbf{F} = |\mathbf{F}| \hat{\mathbf{F}}, \quad (2.5.1)$$

where $|\mathbf{F}|$ is the magnitude of vector $\mathbf{F}$, and $\hat{\mathbf{F}}$ is the unit vector pointing in the direction of $\mathbf{F}$.

Solving equation (2.5.1) for $\hat{\mathbf{F}}$ gives the approach to find the unit vector of known vector $\mathbf{F}$.

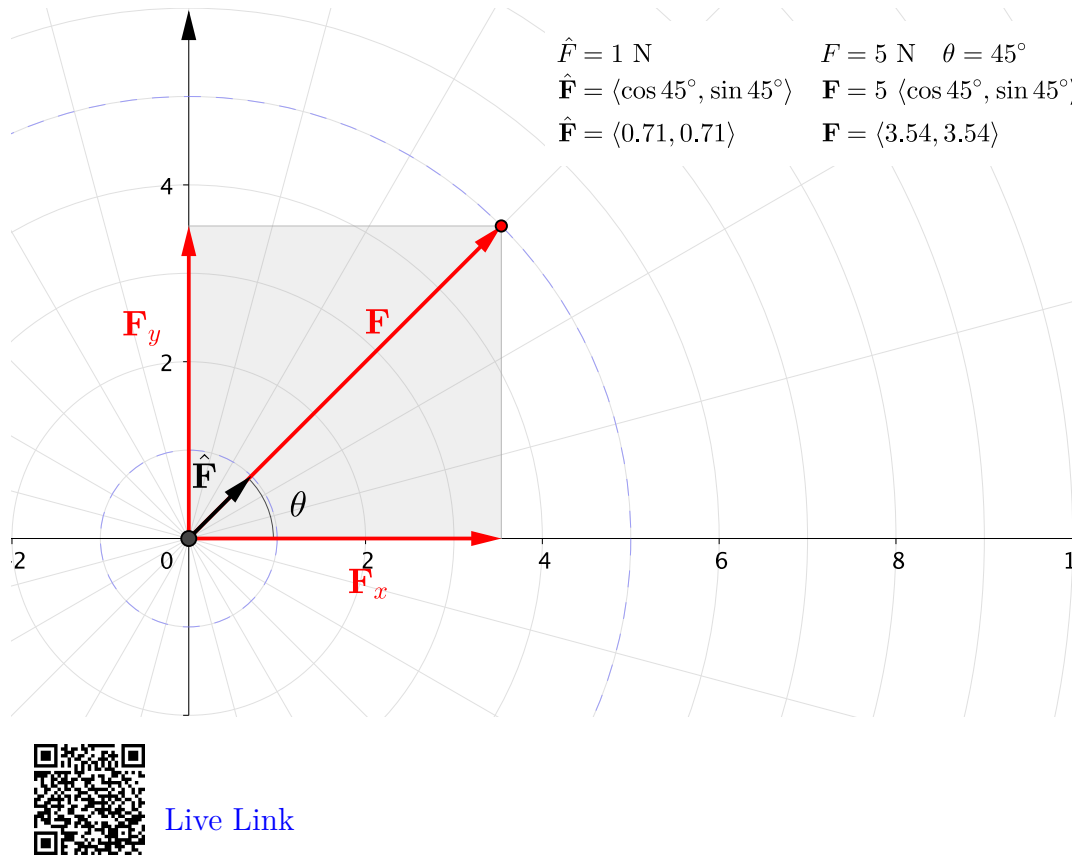
The process is straightforward— divide the vector by its magnitude.

$$\hat{\mathbf{F}} = \frac{\mathbf{F}}{|\mathbf{F}|} \quad (2.5.2)$$

To emphasize that unit vectors are pure direction, recall that vectors consist of both a magnitude and direction, so when we divide a vector by its own magnitude we are just left with direction.

$$\text{unit vector} = \frac{\mathbf{F}}{|\mathbf{F}|} = \frac{[\text{vector}]}{[\text{magnitude}]} = \frac{[\cancel{\text{magnitude}}] \cdot [\text{direction}]}{[\cancel{\text{magnitude}}]} = [\text{direction}]$$

This interactive shows vector $\mathbf{F}$, its associated unit vector $\hat{\mathbf{F}}$, and expressions for $\mathbf{F}$ in terms of its unit vector $\hat{\mathbf{F}}$.

**Figure 2.5.2** Unit Vectors**Example 2.5.3** Find unit vector of a force.

Find the unit vector corresponding to a 100 N force at 60° above the positive x -axis.

Solution. In polar coordinates, the unit vector is a vector of magnitude 1, pointing in the same direction as the force, so, by inspection

$$\mathbf{F} = (100 \text{ N}; 60^\circ)$$

$$\hat{\mathbf{F}} = (1; 60^\circ)$$

In rectangular coordinates, first express $\mathbf{F}$ in terms of its x and y components.

$$\left. \begin{aligned} F_x &= F \cos 60^\circ \\ F_y &= F \sin 60^\circ \end{aligned} \right\} \implies \mathbf{F} = \langle F \cos 60^\circ, F \sin 60^\circ \rangle.$$

$$\hat{\mathbf{F}} = \frac{\mathbf{F}}{F} = \frac{\langle 100 \text{ N} \cos 60^\circ, 100 \text{ N} \sin 60^\circ \rangle}{100 \text{ N}} = \langle \cos 60^\circ, \sin 60^\circ \rangle$$

2.5.3 Force Vectors from Position Vectors

Unit vectors are generally the best approach when working with forces and distances in three dimensions.

For example, when the locations of two points on the line of action of a force are known, the unit vector of the line of action can be found and used to determine the components of the force acting along that line. This can be accomplished as follows, where A and B are points on the line of action.

1. Use the problem geometry to find $\mathbf{AB}$, the displacement vector from point A to point B .

You can either subtract the coordinates of the starting point A from the coordinates of the destination point B ,

$$\begin{aligned} A &= (A_x, A_y, A_z) \\ B &= (B_x, B_y, B_z) \\ \mathbf{AB} &= (B_x - A_x)\mathbf{i} + (B_y - A_y)\mathbf{j} + (B_z - A_z)\mathbf{k}, \text{ or} \end{aligned}$$

or, write the displacements directly by noting the change in the x , y , and z coordinates when moving from A to B .

$$\begin{aligned} AB_x &= \Delta x = B_x - A_x \\ AB_y &= \Delta y = B_y - A_y \\ AB_z &= \Delta z = B_z - A_z \\ \mathbf{AB} &= AB_x \mathbf{i} + AB_y \mathbf{j} + AB_z \mathbf{k} \end{aligned}$$

The result is the same with either method.

2. Find the distance between point A and point B using the Pythagorean Theorem. This distance is also the magnitude of $\mathbf{AB}$ or $|\mathbf{AB}|$.

$$|\mathbf{AB}| = \sqrt{(AB_x)^2 + (AB_y)^2 + (AB_z)^2}$$

3. Find $\widehat{\mathbf{AB}}$, the unit vector from A to B , by dividing vector $\mathbf{AB}$ by its magnitude. This is a unitless vector with a magnitude of 1 which points from A to B .

$$\widehat{\mathbf{AB}} = \left\langle \frac{AB_x}{|\mathbf{AB}|}, \frac{AB_y}{|\mathbf{AB}|}, \frac{AB_z}{|\mathbf{AB}|} \right\rangle$$

4. Finally, multiply the magnitude of the force by the unit vector $\widehat{\mathbf{AB}}$ to get force $\mathbf{F}_{AB}$.

$$\begin{aligned} \mathbf{F}_{AB} &= F_{AB} \widehat{\mathbf{AB}} \\ &= F_{AB} \left\langle \frac{AB_x}{|\mathbf{AB}|}, \frac{AB_y}{|\mathbf{AB}|}, \frac{AB_z}{|\mathbf{AB}|} \right\rangle \end{aligned}$$

The interactive below can be used to visualize the displacement vector and its unit vector, and practice this procedure.

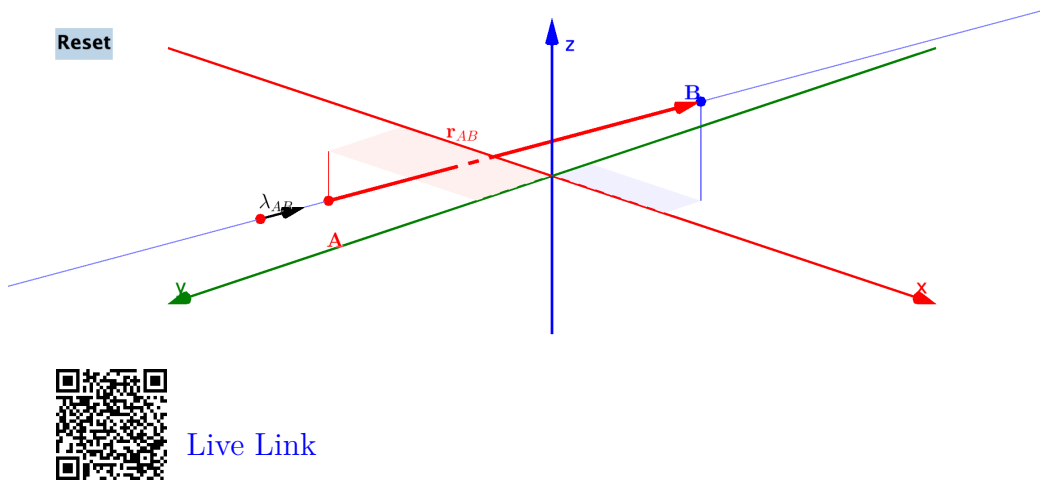


Figure 2.5.4 Unit Vectors in Space

Example 2.5.5 Component in a Specified Direction.

Determine the components of a 5 kN force $\mathbf{F}$ acting at point A , in the direction of a line from A to B . Given: $A = (2, 3, -2.1)$ m and $B = (-2.5, 1.5, 2.2)$ m

We will take the solution one step at a time.

- (a) Draw a good diagram.

Hint. The interactive in [Figure 2.5.4](#) may be useful for this problem.

- (b) Find the displacement vector from A to B .

Answer.

$$\mathbf{AB} = \langle -4.5, -1.5, 4.3 \rangle \text{ m}$$

Solution.

$$\begin{aligned} \mathbf{AB} &= (B_x - A_x)\mathbf{i} + (B_y - A_y)\mathbf{j} + (B_z - A_z)\mathbf{k} \\ &= [(-2.5 - 2)\mathbf{i} + (1.5 - 3)\mathbf{j} + (2.2 - (-2.1))\mathbf{k}] \text{ m} \\ &= (-4.5\mathbf{i} - 1.5\mathbf{j} + 4.3\mathbf{k}) \text{ m} \\ &= \langle -4.5, -1.5, 4.3 \rangle \text{ m} \end{aligned}$$

- (c) Find the magnitude of the displacement vector.

Answer.

$$|\mathbf{AB}| = 6.402 \text{ m}$$

Solution.

$$\begin{aligned} |\mathbf{AB}| &= \sqrt{(\Delta_x)^2 + (\Delta_y)^2 + (\Delta_z)^2} \\ &= \sqrt{(-4.5)^2 + (-1.5)^2 + 4.3^2} \text{ m} \\ &= \sqrt{40.99} \text{ m} \\ &= 6.402 \text{ m} \end{aligned}$$

(d) Find the unit vector pointing from A to B .

Answer.

$$\widehat{\mathbf{AB}} = \langle -0.7, -0.23, 0.67 \rangle$$

Solution.

$$\begin{aligned} \widehat{\mathbf{AB}} &= \left\langle \frac{\Delta_x}{|\mathbf{AB}|}, \frac{\Delta_y}{|\mathbf{AB}|}, \frac{\Delta_z}{|\mathbf{AB}|} \right\rangle \\ &= \left\langle \frac{-4.5}{6.402}, \frac{-1.5}{6.402}, \frac{4.3}{6.402} \right\rangle \\ \widehat{\mathbf{AB}} &= \langle -0.7, -0.23, 0.67 \rangle \end{aligned}$$

(e) Find the force vector.

Answer.

$$\mathbf{F}_{AB} = \langle -3.51, -1.17, 3.36 \rangle \text{ kN}$$

Solution.

$$\begin{aligned} \mathbf{F}_{AB} &= F_{AB} \widehat{\mathbf{AB}} \\ &= 5 \text{ kN} \langle -0.7, -0.23, 0.67 \rangle \\ &= \langle -3.51, -1.17, 3.36 \rangle \text{ kN} \end{aligned}$$

Given the properties of unit vectors, there are some conceptual checks you can make after computing unit vector components which can prevent subsequent errors.

- The signs of unit vector components need to match the signs of the original position vector. A unit vector has the same line of action and sense as the

position vector but is scaled down to one unit in magnitude.

- Components of a unit vector must be between -1 and 1. If the magnitude of a unit vector is one, then it is impossible for it to have rectangular components larger than one.

2.5.4 Unit Vectors and Direction Cosines

If you look closely at the right side of equation (2.4.1), you will see that each equation consists of a component divided by the total vector magnitude. These are the same equations just used to find unit vectors. Thus, the cosine of each direction cosine angle collectively also computes the components of the unit vector; hence we can write an equation for $\hat{\mathbf{A}}$, i.e., the unit vector along $\mathbf{A}$.

$$\hat{\mathbf{A}} = \cos \theta_x \mathbf{i} + \cos \theta_y \mathbf{j} + \cos \theta_z \mathbf{k}$$

Combining the Pythagorean Theorem with our knowledge of unit vectors and direction cosine angles gives this result: if you know two of the three direction cosine angles you can manipulate the following equation to find the third.

$$\cos^2 \theta_x + \cos^2 \theta_y + \cos^2 \theta_z = 1 \quad (2.5.3)$$

2.6 Vector Addition

Key Questions

- How do you set up vectors for graphical addition using the Triangle Rule?
- Does it matter which vector you start with when using the Triangle Rule?
- Why can you separate a two-dimensional vector equation into two independent equations to solve for up to two unknowns?
- If you and another student define vectors using different direction coordinate systems, will you end up with the same resultant vector?
- What is the preferred technique to add vectors in three-dimensional systems?

In this section we will look at several different methods of vector addition. Vectors being added together are called the **components**, and the sum of the components is called the **resultant vector**.

These different methods are tools for your statics toolbox. They give you multiple different ways to think about vector addition and to approach a problem. Your goal should be to learn to use them all and to identify which approach will be the easiest to use in a given situation.

2.6.1 Triangle Rule of Vector Addition

All methods of vector addition are ultimately based on the tip-to-tail method discussed in a one-dimensional context in [Subsection 2.2.1](#). There are two ways to draw or visualize adding vectors in two or three dimensions, the **Triangle Rule** and **Parallelogram Rule**. Both are equivalent.

- *Triangle Rule.*

Place the tail of one vector at the tip of the other vector, then draw the resultant from the first vector's tail to the final vector's tip.

- *Parallelogram Rule.*

Place both vectors' tails at the origin, then complete a parallelogram with lines parallel to each vector through the tip of the other. The resultant is equal to the diagonal from the tails to the opposite corner.

The interactive below shows two forces **A** and **B** pulling on a particle at the origin, and the appropriate diagram for the triangle or parallelogram rule. Both approaches produce the same resultant force **R** as expected.

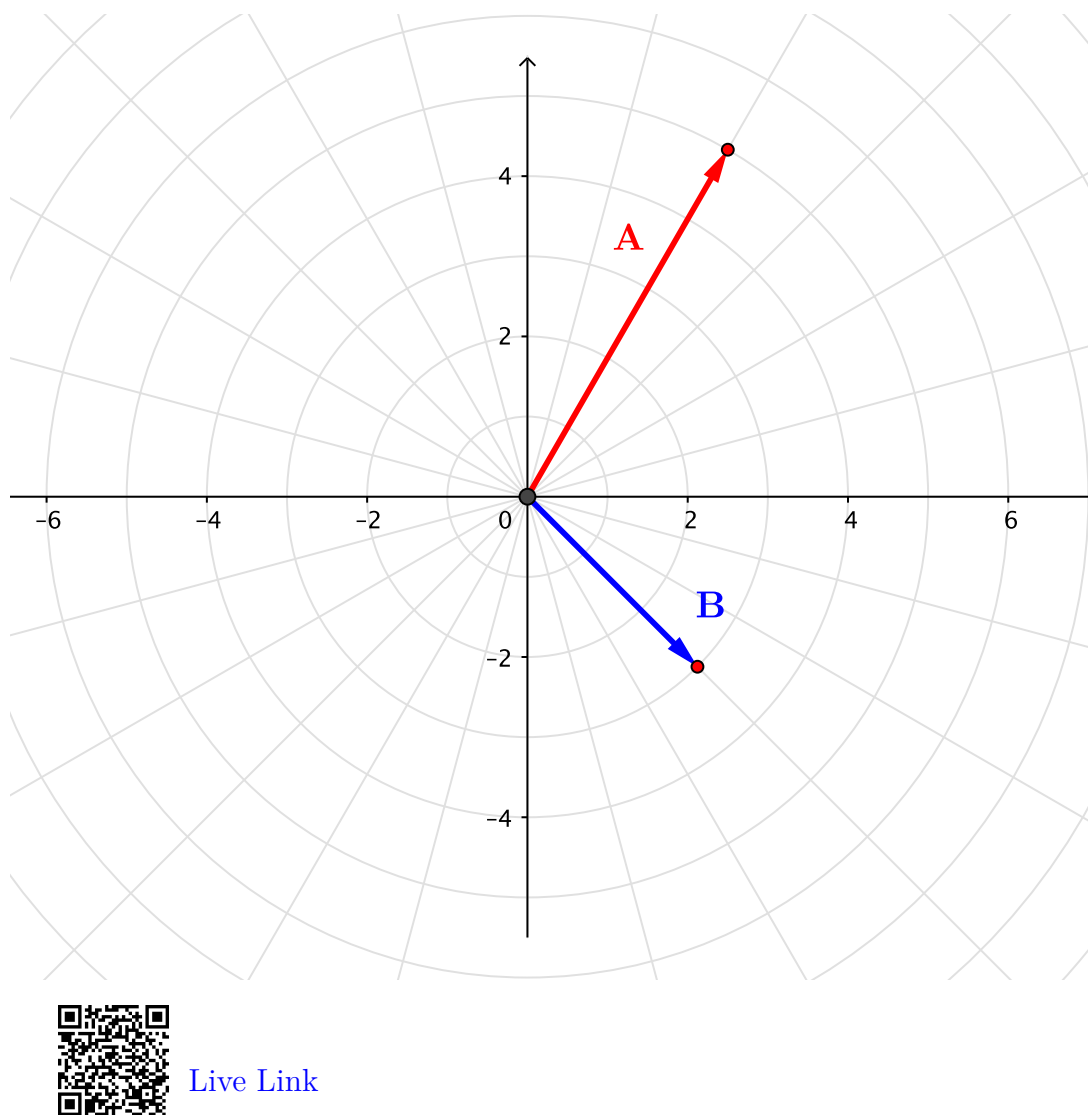


Figure 2.6.1 Vector Addition Methods

2.6.2 Graphical Vector Addition

Graphical vector addition involves drawing a scaled diagram using either the parallelogram or triangle rule, and then measuring the magnitudes and directions from the diagram. Graphical solutions work well enough for two-dimensional problems where all the vectors live in the same plane and can be drawn on a sheet of paper, but are not very useful for three-dimensional problems unless you use technology.

If you carefully draw the triangle accurately to scale and use a protractor and ruler you can measure the magnitude and direction of the resultant. However, your answer will only be as precise as your diagram and your ability to read your tools. If you use technology such as GeoGebra or a CAD program to make the diagram, your answer will be precise. The interactive in [Figure 2.6.1](#) may be

useful for this.

Even though the graphical approach has limitations, it is worth your attention because it provides a good way to visualize the effects of multiple forces, to quickly estimate ballpark answers, and to visualize the diagrams you need to use alternate methods to follow.

2.6.3 Trigonometric Vector Addition

You *can* get a precise answer from the triangle or parallelogram rule by

1. drawing a quick diagram using either rule,
2. identifying three known sides or angles,
3. using trigonometry to solve for the unknown sides and angles.

The trigonometric tools you will need are found in [Appendix B](#).

Using triangle-based geometry to solve vector problems is a quick and powerful tool, but includes the following limitations:

- There are only three sides in a triangle; thus vectors can only be added two at a time. If you need to add three or more vectors using this method, you must add the first two, then add the third to that sum and so on.
- If you fail to draw the correct vector triangle or identify the known sides and angles, you will not find the correct answer.
- The trigonometric functions produce scalar values. You can use them to find the magnitudes the angles between vectors, but the results are not, by themselves, vectors.

When you need to find the resultant of more than two vectors, it is generally best to use the algebraic methods described below.

2.6.4 Orthogonal Components

Any arbitrary two-dimensional vector $\mathbf{F}$ can be broken into two component vectors which are the sides of a parallelogram having $\mathbf{F}$ as its diagonal. The process of finding components of a vector in particular directions is called **vector resolution**. While a vector can be resolved into components in any two directions, it's generally most useful to resolve them into **rectangular** or **orthogonal** components, where the parallelogram is a rectangle and the components are perpendicular.

There are an infinite number of possible rectangles to choose from, so each vector has an infinite number of sets of orthogonal components. However, the most important set occurs when the sides of the rectangle are parallel to the x

and y axes. These orthogonal components are given x and y subscripts indicate that they're aligned with the coordinate axes. For vector $\mathbf{F}$,

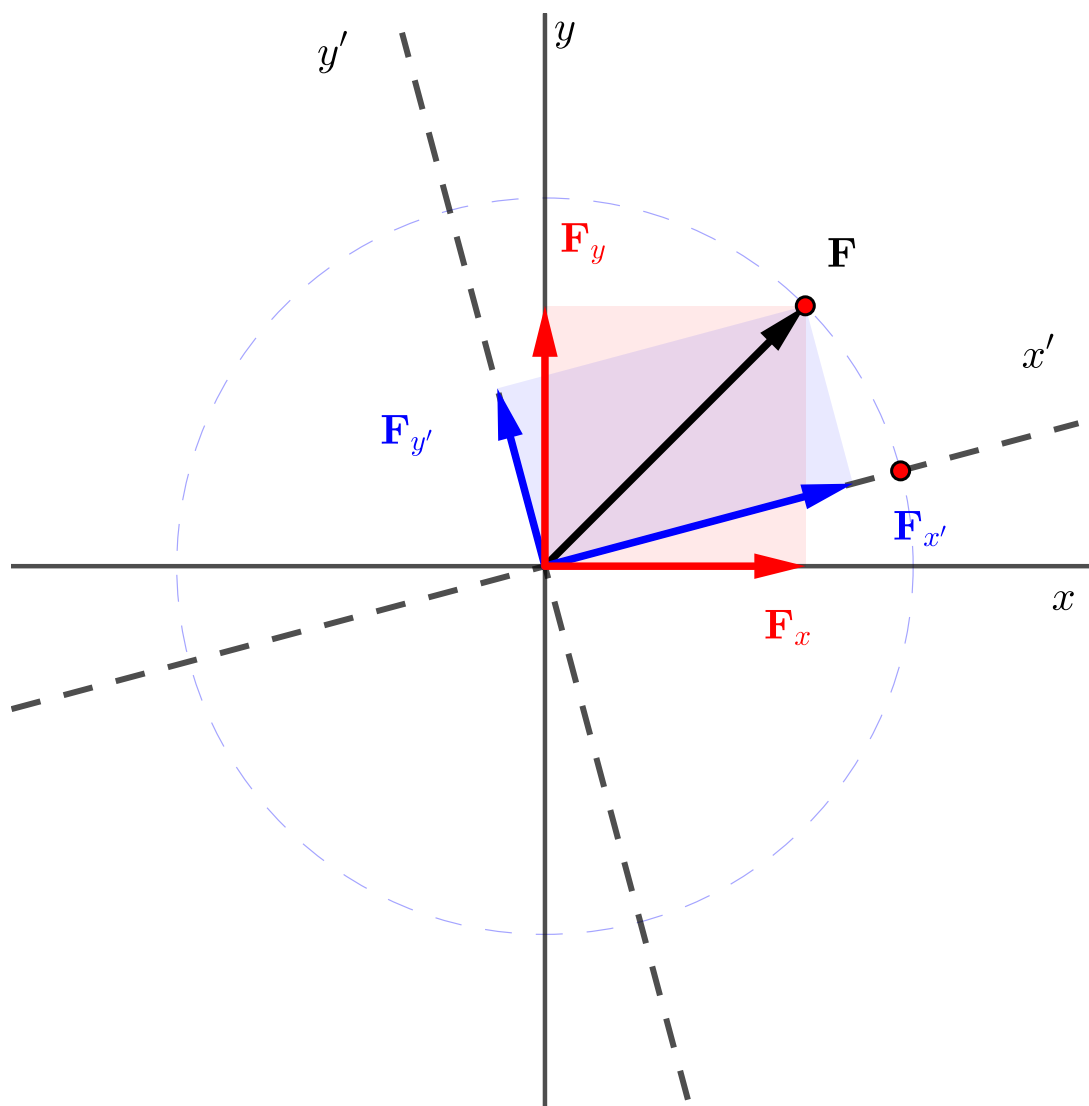
$$\mathbf{F} = \mathbf{F}_x + \mathbf{F}_y = F_x \mathbf{i} + F_y \mathbf{j}, \quad (2.6.1)$$

where F_x and F_y are the scalar components of $\mathbf{F}$. The advantage of this choice of components is that vector calculations can be replaced with ordinary algebraic calculation on scalar values for each orthogonal direction.

Alternately, you may rotate the coordinate system to any other convenient angle, and find the components in the directions of the rotated coordinate axes x' and y' . In either case, the vector is the sum of the rectangular components

$$\mathbf{F} = \mathbf{F}_x + \mathbf{F}_y = \mathbf{F}_{x'} + \mathbf{F}_{y'}. \quad (2.6.2)$$

The interactive below can help you visualize the relationship between a vector and its components in both the x - y and x' - y' directions.



[Live Link](#)

Figure 2.6.2 Orthogonal Components

2.6.5 Algebraic Addition of Components

While the parallelogram rule and the graphical and trigonometric methods are useful tools for visualizing and finding the sum of two vectors, they are not particularly suited for adding *more* than two vectors or working in three dimensions.

Consider vector $\mathbf{R}$ which is the sum of several vectors $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and perhaps more. Vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are the components of $\mathbf{R}$, and the $\mathbf{R}$ is the resultant of $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$.

It is easy enough to say that $\mathbf{R} = \mathbf{A} + \mathbf{B} + \mathbf{C}$, but how can we calculate $\mathbf{R}$?

You could draw the vectors arranged tip-to-tail and then use the triangle rule to add the first two components, then use it again to add the third component to that sum, and so forth until all the components have been added. The final sum is the resultant, $\mathbf{R}$. The process gets progressively more tedious the more components there are to sum.

This section introduces an alternate method to add multiple vectors which is straightforward, efficient and robust. This is called *algebraic method*, because the vector addition is replaced with a process of *algebraic* addition of *scalar* components. The algebraic technique works equally well for two and three-dimensional vectors, and for summing any number of vectors.

To find the sum of multiple vectors using the algebraic method:

1. Find the scalar components of each component vector in the x and y directions using the P to R procedure described in [Subsection 2.3.3](#).
2. Algebraically sum the scalar components in each coordinate direction. The scalar components will be positive if they point right or up, negative if they point left or down. These sums are the scalar components of the resultant.
3. Resolve the resultant's components to find the magnitude and direction of the resultant vector using the R to P procedure described in [Subsection 2.3.3](#).

The resultant $\mathbf{F}_R$ is simply the algebraic sum of the components in each coordinate direction.

$$\mathbf{F}_R = \Sigma F_x \mathbf{i} + \Sigma F_y \mathbf{j} + \Sigma F_z \mathbf{k}$$

or in bracket notation

$$\mathbf{F}_R = \langle \Sigma F_x, \Sigma F_y, \Sigma F_z \rangle \quad (2.6.3)$$

This process is illustrated in the following interactive diagram and in the next example.

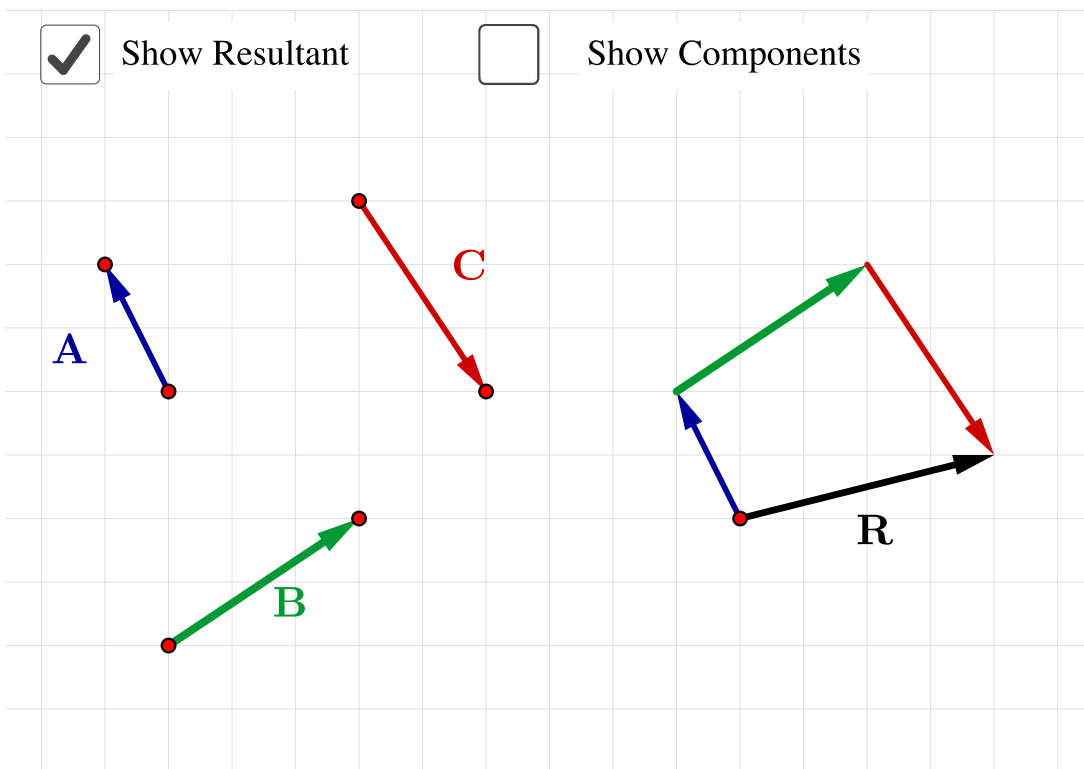

[Live Link](#)

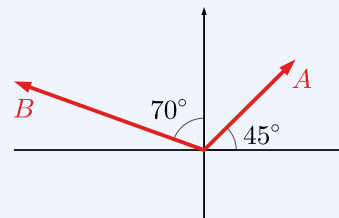
Figure 2.6.3 Vector addition by summing rectangular components.

Example 2.6.4 Vector Addition.

Vector $\mathbf{A} = 200 \text{ N} \angle 45^\circ$ counter-clockwise from the x axis, and vector $\mathbf{B} = 300 \text{ N} \angle 70^\circ$ counter-clockwise from the y axis. Find the resultant $\mathbf{R} = \mathbf{A} + \mathbf{B}$ by addition of scalar components.

Solution.

Use the given information to draw a sketch of the situation. By imagining or sketching the parallelogram rule, it should be apparent that the resultant vector points up and to the left.



$$\begin{aligned}
 A_x &= 200 \text{ N} \cos 45^\circ = 141.4 \text{ N} & B_x &= -300 \text{ N} \sin 70^\circ = -281.9 \text{ N} \\
 A_y &= 200 \text{ N} \sin 45^\circ = 141.4 \text{ N} & B_y &= 300 \text{ N} \cos 70^\circ = 102.6 \text{ N}
 \end{aligned}$$

$$R_x = A_x + B_x$$

$$= 141.4 \text{ N} - 281.9 \text{ N}$$

$$= -140.5 \text{ N}$$

$$R_y = A_y + B_y$$

$$= 141.4 \text{ N} + 102.6 \text{ N}$$

$$= 244.0 \text{ N}$$

$$R = \sqrt{R_x^2 + R_y^2} = 281.6 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{R_y}{R_x} \right) = -60.1^\circ$$

This answer indicates that the resultant points down and to the left, which is odd because the parallelogram rule shows that the resultant should point up and to the left.

This occurs because the calculator always returns angles in the first or fourth quadrant for $\tan^{-1}$. To get the actual direction of the resultant, add 180° to the calculator result.

$$\theta = -60.1^\circ + 180^\circ = 119.9^\circ$$

The final answer for the magnitude and direction of the resultant is

$$\mathbf{R} = 281.6 \text{ N} \angle 119.9^\circ$$

measured counter-clockwise from the x axis.

The process for adding three-dimensional vectors is exactly the same, except that the z component is included as well. This interactive allows you to input the three-dimensional vector components of forces $\mathbf{A}$ and $\mathbf{B}$ and view the resultant force $\mathbf{R}$ which is the sum of $\mathbf{A}$ and $\mathbf{B}$.

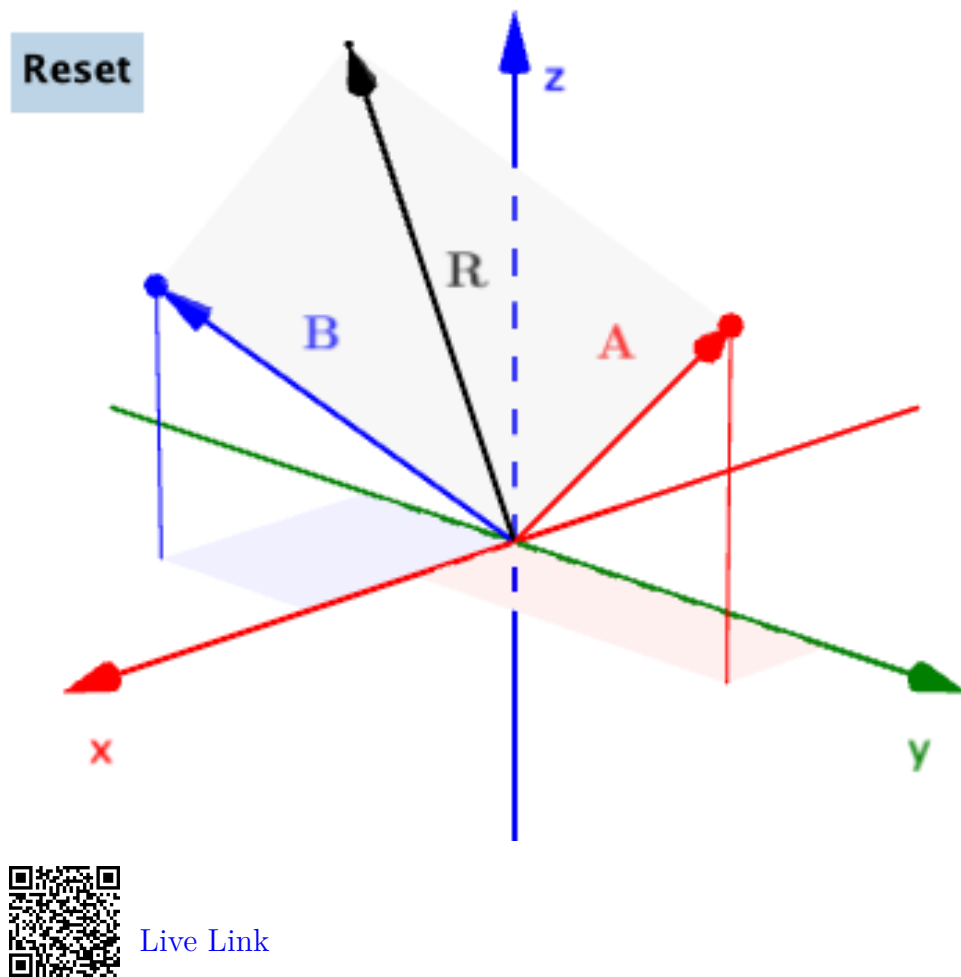


Figure 2.6.5 Vector Addition in Three Dimensions

2.6.6 Vector Subtraction

Like one-dimensional vector subtraction, the easiest way to handle two-dimensional vector subtraction is by taking the negative of a vector followed by vector addition. Multiplying a vector by -1 preserves its magnitude but flips its direction, which has the effect of changing the sign of the scalar components.

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$$

After negating the second vector you can choose any technique you prefer for vector addition.

2.7 Dot Products

Key Questions

- What are dot products used for?
- What does it mean when the dot product of two vectors is zero?
- How do you use a dot product to find the angle between two vectors?
- What does it mean when the scalar component of the projection $\|\text{proj}_{\mathbf{A}} \mathbf{B}\|$ is negative?

Unlike ordinary algebra where there is only one way to multiply numbers, there are two distinct vector multiplication operations: **dot product** and the **cross product**. Alternately, the first is referred to as the **scalar product** because its result is a scalar, and the second as the **vector product** because its result is a vector. The dot product and its applications will be discussed in this section and the cross product in the next.

2.7.1 Calculation of the Dot product

For two vectors $\mathbf{A} = \langle A_x, A_y, A_z \rangle$ and $\mathbf{B} = \langle B_x, B_y, B_z \rangle$, the dot product multiplication is computed by summing the products of the components.

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z \quad (2.7.1)$$

It can be shown that an alternate, equivalent method to compute the dot product is

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta = A B \cos \theta \quad (2.7.2)$$

where θ in the equation is the angle between the two vectors and $|\mathbf{A}|$ and $|\mathbf{B}|$ are the magnitudes of $\mathbf{A}$ and $\mathbf{B}$.

We can conclude from the second equation that the dot product of two perpendicular vectors is zero, because $\cos 90^\circ = 0$, and that the dot product of two parallel vectors equals the product of their magnitudes.

When dotting unit vectors that have a magnitude of one, the dot products of a unit vector with itself is one and the dot product two perpendicular unit vectors is zero, so for $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ we have

$$\begin{array}{lll} \mathbf{i} \cdot \mathbf{i} = 1 & \mathbf{j} \cdot \mathbf{i} = 0 & \mathbf{k} \cdot \mathbf{i} = 0 \\ \mathbf{i} \cdot \mathbf{j} = 0 & \mathbf{j} \cdot \mathbf{j} = 1 & \mathbf{k} \cdot \mathbf{j} = 0 \\ \mathbf{i} \cdot \mathbf{k} = 0 & \mathbf{j} \cdot \mathbf{k} = 0 & \mathbf{k} \cdot \mathbf{k} = 1 \end{array}$$

Dot products are commutative, associative and distributive:

1. **Commutative.** The order does not matter.

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A} \quad (2.7.3)$$

2. **Associative.** It does not matter whether you multiply a scalar value C by the final dot product, or either of the individual vectors, you will still get the same answer.

$$C(\mathbf{A} \cdot \mathbf{B}) = (C \mathbf{A}) \cdot \mathbf{B} = \mathbf{A} \cdot (C \mathbf{B}) \quad (2.7.4)$$

3. **Distributive.** If you are dotting one vector $\mathbf{A}$ with the sum of two more $(\mathbf{B} + \mathbf{C})$, you can either add $\mathbf{B} + \mathbf{C}$ first, or dot $\mathbf{A}$ by both and add the final value.

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \cdot \mathbf{B}) + (\mathbf{A} \cdot \mathbf{C}) \quad (2.7.5)$$

Dot products can be used to compute the magnitude of a vector, determine the angle between two vectors, and find the rectangular component or projection of a vector in a specified direction. These applications will be discussed in the following sections.

2.7.2 Magnitude of a Vector

Dot products can be used to find vector magnitudes. When a vector is dotted with itself using (2.7.1), the result is the square of the magnitude of the vector. (Recall that $|\mathbf{A}|$ and A are alternate notations for the magnitude of vector $\mathbf{A}$.)

$$\mathbf{A} \cdot \mathbf{A} = |\mathbf{A}|^2 = A^2. \quad (2.7.6)$$

The proof is trivial. By the definition of the dot product (2.7.1) and the Pythagorean theorem:

$$\begin{aligned} \mathbf{A} \cdot \mathbf{A} &= A_x A_x + A_y A_y \\ &= A_x^2 + A_y^2 = A^2 \\ &= |\mathbf{A}|^2 \end{aligned}$$

Taking the square root of each side gives the magnitude of $\mathbf{A}$

$$\sqrt{\mathbf{A} \cdot \mathbf{A}} = A = |\mathbf{A}|. \quad (2.7.7)$$

The result is similar for three-dimensional vectors.

Example 2.7.1 Find Vector Magnitude using the Dot Product.

Find the magnitude of vector $\mathbf{F}$ with components $F_x = 30$ N, $F_y = -40$ N and $F_z = 50$ N

Solution.

$$\mathbf{F} = \langle 30 \text{ N}, -40 \text{ N}, 50 \text{ N} \rangle$$

$$\begin{aligned}\mathbf{F} \cdot \mathbf{F} &= F_x^2 + F_y^2 + F_z^2 \\ &= (30 \text{ N})^2 + (-40 \text{ N})^2 + (50 \text{ N})^2 \\ &= 5000 \text{ N}^2\end{aligned}$$

$$\begin{aligned}F &= |\mathbf{F}| = \sqrt{\mathbf{F} \cdot \mathbf{F}} \\ &= \sqrt{5000 \text{ N}^2} \\ &= 70.7 \text{ N}\end{aligned}$$

2.7.3 Angle between Two Vectors

A second application of the dot product is to find the angle between two vectors. Equation (2.7.2) provides the procedure.

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= |\mathbf{A}||\mathbf{B}| \cos \theta \\ \cos \theta &= \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|}\end{aligned}\tag{2.7.8}$$

Example 2.7.2 Angle between Orthogonal Unit Vectors.

Find the angle between $\mathbf{i} = \langle 1, 0, 0 \rangle$ and $\mathbf{j} = \langle 0, 1, 0 \rangle$.

Solution.

$$\begin{aligned}\cos \theta &= \frac{\mathbf{i} \cdot \mathbf{j}}{|\mathbf{i}||\mathbf{j}|} \\ &= \frac{(1)(0) + (0)(1) + (0)(0)}{(1)(1)} \\ &= 0\end{aligned}$$

$$\begin{aligned}\theta &= \cos^{-1}(0) \\ &= 90^\circ\end{aligned}$$

This shows that $\mathbf{i}$ and $\mathbf{j}$ are perpendicular to each other.

Example 2.7.3 Angle between Two Vectors.

Find the angle between $\mathbf{F} = \langle 100 \text{ N}, 200 \text{ N}, -50 \text{ N} \rangle$ and $\mathbf{G} = \langle -75 \text{ N}, 150 \text{ N}, -40 \text{ N} \rangle$.

Solution.

$$\begin{aligned}
 \cos \theta &= \frac{\mathbf{F} \cdot \mathbf{G}}{|\mathbf{F}||\mathbf{G}|} \\
 &= \frac{F_x G_x + F_y G_y + F_z G_z}{\sqrt{F_x^2 + F_y^2 + F_z^2} \sqrt{G_x^2 + G_y^2 + G_z^2}} \\
 &= \frac{(100)(-75) + (200)(150) + (-50)(-40)}{\sqrt{100^2 + 200^2 + (-50)^2} \sqrt{(-75)^2 + 150^2 + (-40)^2}} \\
 &= \frac{24500}{(229.1)(172.4)} \\
 &= 0.620 \\
 \theta &= \cos^{-1}(0.620) \\
 &= 51.7^\circ
 \end{aligned}$$

2.7.4 Vector Projection

The dot product is used to find the **projection** of one vector onto another. You can think of a projection of $\mathbf{B}$ on $\mathbf{A}$ as a vector the length of the shadow of $\mathbf{B}$ on the line of action of $\mathbf{A}$ when the sun is directly above $\mathbf{A}$. More precisely, the projection of $\mathbf{B}$ onto $\mathbf{A}$ produces the rectangular component of $\mathbf{B}$ in the direction *parallel* to $\mathbf{A}$. This is one side of a rectangle aligned with $\mathbf{A}$, having $\mathbf{B}$ as its diagonal.

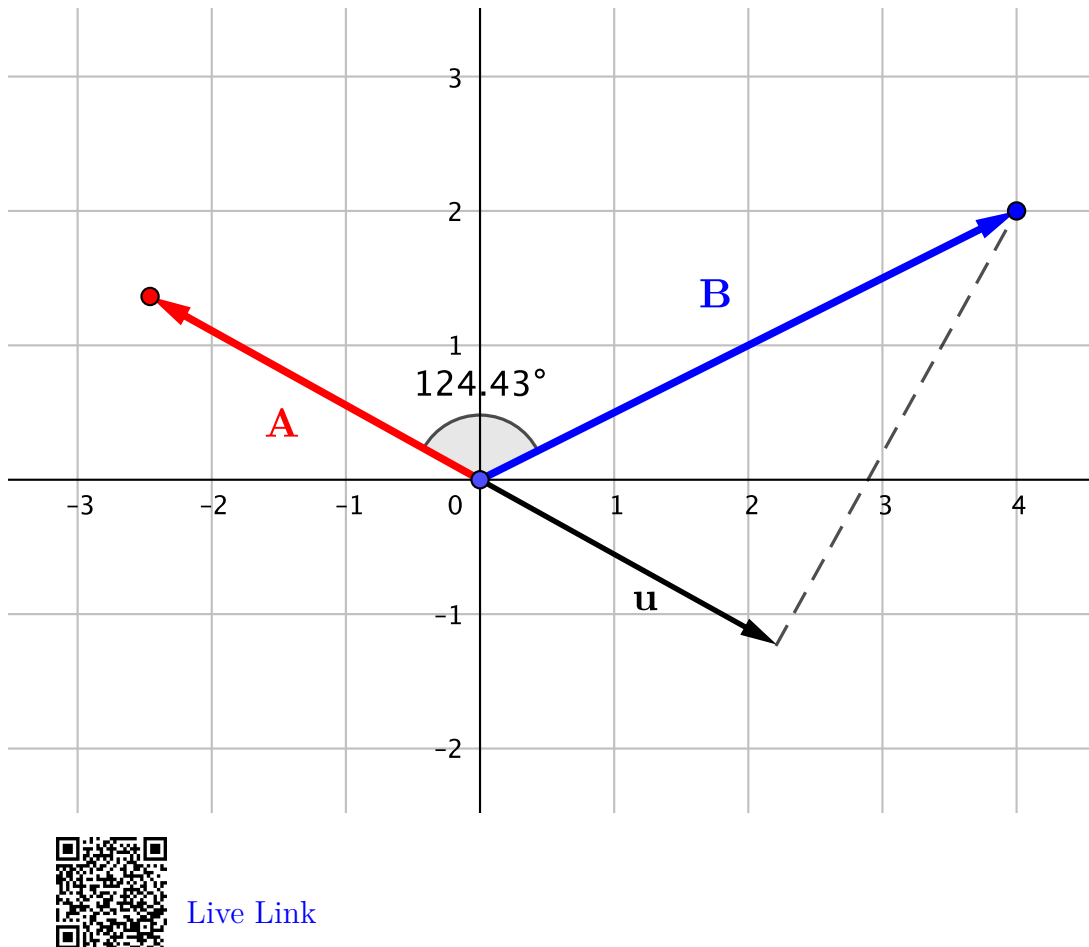
This is illustrated in Figure 2.7.4, where $\mathbf{u}$ is the projection of $\mathbf{B}$ onto $\mathbf{A}$, or alternately $\mathbf{u}$ is the rectangular component of $\mathbf{B}$ in the direction of $\mathbf{A}$.

In this text we will use the symbols

- $\text{proj}_{\mathbf{A}} \mathbf{B}$ to mean the **vector projection** of $\mathbf{B}$ on $\mathbf{A}$
- $|\text{proj}_{\mathbf{A}} \mathbf{B}|$ to mean the *magnitude* of the vector projection, a positive or zero-valued scalar, and
- $\|\text{proj}_{\mathbf{A}} \mathbf{B}\|$ to mean the **scalar projection**. This value represents the component of $\mathbf{B}$ in the $\mathbf{A}$ direction, and can have a positive, zero, or negative value.

As we have mentioned before, the magnitude of a vector is its length and is always positive or zero, while a scalar component is a signed value that can be positive or negative. When a scalar component is multiplied by a unit vector

the result is a vector in that direction when the scalar component is positive, or 180° opposite when the scalar component is negative.



[Live Link](#)

Figure 2.7.4 Vector projection in two dimensions.

The interactive shows that the projection is the adjacent side of a right triangle with $\mathbf{B}$ as the hypotenuse. From the definition of the dot product (2.7.2) we find that

$$\mathbf{A} \cdot \mathbf{B} = A(B \cos \theta) = A \|\text{proj}_{\mathbf{A}} \mathbf{B}\|, \quad (2.7.9)$$

where $B \cos \theta$ is the scalar component of the projection. So, the dot product of $\mathbf{A}$ and $\mathbf{B}$ gives us the projection of $\mathbf{B}$ onto $\mathbf{A}$ times the magnitude of $\mathbf{A}$. This value will be positive when $\theta < 90^\circ$, negative when $\theta > 90^\circ$, and zero when the vectors are perpendicular because of the properties of the cosine function.

So, to find the scalar value of the projection of $\mathbf{B}$ onto $\mathbf{A}$ we divide by the magnitude of $\mathbf{A}$.

$$\|\text{proj}_{\mathbf{A}} \mathbf{B}\| = \frac{\mathbf{A} \cdot \mathbf{B}}{A} = \frac{\mathbf{A}}{A} \cdot \mathbf{B} = \hat{\mathbf{A}} \cdot \mathbf{B} \quad (2.7.10)$$

where $\hat{\mathbf{A}} = \frac{\mathbf{A}}{A}$ is the unit vector in the direction of $\mathbf{A}$.

If you want the *vector* projection of $\mathbf{B}$ onto $\mathbf{A}$, as opposed to the *scalar* projection we just found, multiply the *scalar* projection by the unit vector $\hat{\mathbf{A}}$.

$$\text{proj}_{\mathbf{A}} \mathbf{B} = \|\text{proj}_{\mathbf{A}} \mathbf{B}\| \hat{\mathbf{A}} = (\hat{\mathbf{A}} \cdot \mathbf{B}) \hat{\mathbf{A}} \quad (2.7.11)$$

Similarly, the vector projection of $\mathbf{A}$ onto $\mathbf{B}$ is

$$\text{proj}_{\mathbf{B}} \mathbf{A} = \|\text{proj}_{\mathbf{B}} \mathbf{A}\| \hat{\mathbf{B}} = (\hat{\mathbf{B}} \cdot \mathbf{A}) \hat{\mathbf{B}} \quad (2.7.12)$$

The spatial interpretation of the results the scalar projection $\|\text{proj}_{\mathbf{A}} \mathbf{B}\|$ is

- *Positive values* mean that $\mathbf{A}$ and $\mathbf{B}$ are generally in the same direction.
- *Negative values* mean that $\mathbf{A}$ and $\mathbf{B}$ are generally in opposite directions.
- *Zero* means that $\mathbf{A}$ and $\mathbf{B}$ are perpendicular.
- *Magnitude smaller than $\mathbf{B}$* This is the most common answer. This means that the vectors are neither parallel nor perpendicular.
- *Magnitude equal to $\mathbf{B}$* means that the vectors point in the same direction, and all of $\mathbf{B}$ acts in the direction of $\mathbf{A}$.
- *Magnitude larger than $\mathbf{B}$* This answer is impossible. Check your algebra; you might have forgotten to divide by the magnitude of $\mathbf{A}$.

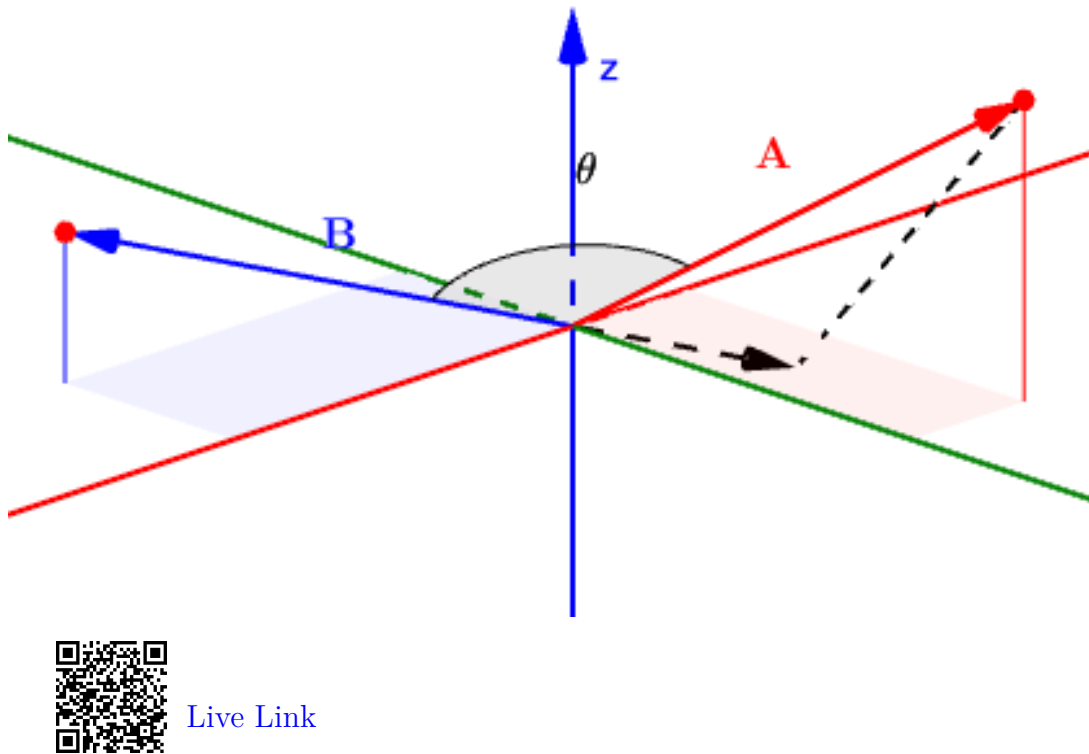


Figure 2.7.5 Vector projections in three dimensions.

2.7.5 Perpendicular Components

The final application of dot products is to find the component of one vector perpendicular to another.

To find the component of $\mathbf{B}$ perpendicular to $\mathbf{A}$, first find the vector projection of $\mathbf{B}$ on $\mathbf{A}$, then subtract that from $\mathbf{B}$. What remains is the perpendicular component.

$$\mathbf{B}_\perp = \mathbf{B} - \text{proj}_\mathbf{A} \mathbf{B} \quad (2.7.13)$$

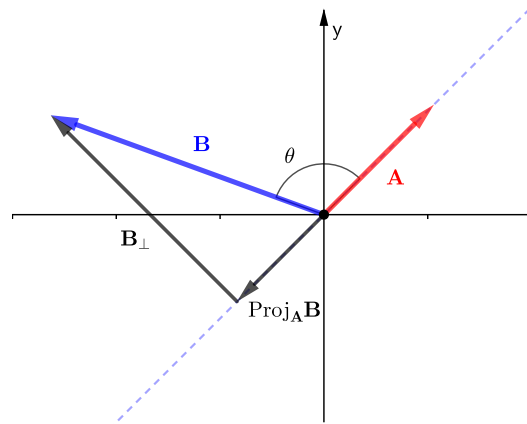
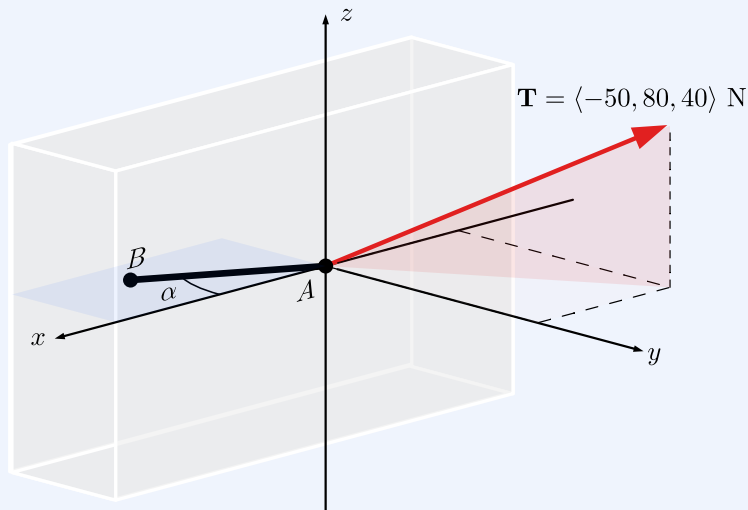


Figure 2.7.6 Perpendicular and parallel components of $\mathbf{B}$.

Example 2.7.7 Dot Products.

A cable pulls with tension $\mathbf{T} = \langle -50, 80, 40 \rangle$ N on a 0.4 m long anchor AB embedded in a concrete wall. The anchor lies in the xy plane at an angle $\alpha = 30^\circ$ from the x axis.



For the system above, compute the following:

- (a) Find the dot product of the cable tension $\mathbf{T}$ and the anchor $\mathbf{AB}$

Answer.

$$\mathbf{T} \cdot \mathbf{AB} = -33.32 \text{ N}\cdot\text{m}$$

Solution. When you know the magnitudes and angle between two vectors, it is easiest to use the second dot product equation (2.7.2), but in this case it will be easier to find the components of $\mathbf{AB}$ and use (2.7.1).

$$\begin{aligned}\mathbf{T} &= \langle -50, 80, 40 \rangle \text{ N} \\ \mathbf{AB} &= \langle 0.4 \cos 30^\circ, -0.4 \sin 30^\circ, 0 \rangle \text{ m} \\ &= \langle 0.3464, -0.2, 0 \rangle \text{ m}\end{aligned}$$

$$\begin{aligned}\mathbf{T} \cdot \mathbf{AB} &= T_x(AB_x) + T_y(AB_y) + T_z(AB_z) \\ &= (-50 \text{ N})(0.3464 \text{ m}) + (80 \text{ N})(-0.2 \text{ m}) + (40 \text{ N})(0 \text{ m}) \\ &= -33.32 \text{ N}\cdot\text{m}\end{aligned}$$

The units of a dot product are the product of the units of the two vectors.

(b) Find the angle θ between the cable tension $\mathbf{T}$ and the anchor $\mathbf{AB}$.

Answer.

$$\theta = 144.38^\circ$$

Solution. With the dot product $\mathbf{T} \cdot \mathbf{AB}$ known from the previous step, we can use the (2.7.2) to find the angle between force $\mathbf{T}$ and anchor.

$$\mathbf{T} \cdot \mathbf{AB} = |\mathbf{T}||\mathbf{AB}| \cos \theta = -33.32 \text{ N}\cdot\text{m}$$

The magnitude of $|\mathbf{AB}| = 0.4 \text{ m}$ was given but we need to calculate the magnitude of $|\mathbf{T}|$,

$$|\mathbf{T}| = \sqrt{(-50)^2 + 80^2 + 40^2} = 102.47 \text{ N}$$

$$\begin{aligned}\theta &= \cos^{-1} \frac{\mathbf{T} \cdot \mathbf{AB}}{|\mathbf{T}||\mathbf{AB}|} \\ &= \cos^{-1} \frac{-33.32 \text{ N}\cdot\text{m}}{(102.47 \text{ N})(0.4 \text{ m})} \\ &= 144.38^\circ\end{aligned}$$

Note that $\theta > 90^\circ$ correctly corresponds to the negative dot product and indicates that the two vectors generally oppose each other.

- (c) Find the scalar projection of the cable tension $\mathbf{T}$ onto the anchor $\mathbf{AB}$.

Answer.

$$\|\text{proj}_{\mathbf{AB}} \mathbf{T}\| = -83.30 \text{ N}$$

Solution 1. Recall from [Subsection 2.7.4](#) that the scalar projection is the scalar component of one vector in the direction of another, in other words, how much of one vector is parallel to another. This is one of the most direct and practical applications of the dot product. The dot product of $\mathbf{T}$ with $\mathbf{AB}$ gives the product of the length of the anchor AB and the scalar projection of the tension in the direction of the anchor.

$$\mathbf{T} \cdot \mathbf{AB} = AB (T \cos \theta) = AB \|\text{proj}_{\mathbf{AB}} \mathbf{T}\|$$

and has units of N-m.

To find the projection, divide the dot product by the magnitude of $\mathbf{AB}$

$$\begin{aligned} \|\text{proj}_{\mathbf{AB}} \mathbf{T}\| &= \frac{\mathbf{T} \cdot \mathbf{AB}}{AB} = \frac{-33.32 \text{ N}\cancel{\text{m}}}{0.4 \cancel{\text{m}}} \\ &= -83.30 \text{ N} \end{aligned}$$

Solution 2. Alternately, you can apply [\(2.7.10\)](#) and calculate the dot product of force vector $\mathbf{T}$ with the unit vector $\widehat{\mathbf{AB}}$. First, find the unit vector

$$\widehat{\mathbf{AB}} = \frac{\mathbf{AB}}{AB} = \frac{\langle 0.3464, -0.2, 0 \rangle \text{ m}}{0.4 \text{ m}} = \langle 0.866, -0.5, 0 \rangle$$

Or, since $\mathbf{AB}$ is in the xy plane with its direction defined by the $\alpha = 30^\circ$, the unit vector $\widehat{\mathbf{AB}}$ is found

$$\widehat{\mathbf{AB}} = \langle \cos 30^\circ, -\sin 30^\circ, 0 \rangle = \langle 0.866, -0.5, 0 \rangle.$$

Then find the projection

$$\begin{aligned} \|\text{proj}_{\mathbf{AB}} \mathbf{T}\| &= \mathbf{T} \cdot \widehat{\mathbf{AB}} \\ &= (-50 \text{ N})(0.866) + (80 \text{ N})(-0.5) + (40 \text{ N})(0) \\ &= -83.30 \text{ N} \end{aligned}$$

- (d) Find the vector projection of the cable tension $\mathbf{T}$ onto the anchor $\mathbf{AB}$.

Answer.

$$\text{proj}_{\mathbf{AB}} \mathbf{T} = \langle -72.14, 41.65, 0 \rangle \text{ N}$$

Solution. The vector projection is the scalar projection value multiplied by a unit direction vector to turn it a vector. So we multiply the scalar projection with the unit vector of $\widehat{\mathbf{AB}}$ to compute the vector projection of T onto AB .

$$\begin{aligned} \text{proj}_{\mathbf{AB}} \mathbf{T} &= \|\text{proj}_{\mathbf{AB}} \mathbf{T}\| \cdot \widehat{\mathbf{AB}} \\ &= 83.301 \text{ N} (\langle 0.866, -0.5, 0 \rangle) \\ &= \langle -72.14, 41.65, 0 \rangle \text{ N} \end{aligned}$$

- (e) Find the vector portion of cable tension $\mathbf{T}$ perpendicular to the anchor $\mathbf{AB}$.

Answer.

$$\mathbf{T}_{\perp \mathbf{AB}} = \langle 22.14, 38.35, 40 \rangle \text{ N}$$

Solution. Recall that a two-dimensional vector can be represented by the sum of two perpendicular components. In the same way, a right triangle can be represented by a vector along the hypotenuse equal to the sum of the two right-triangle sides.

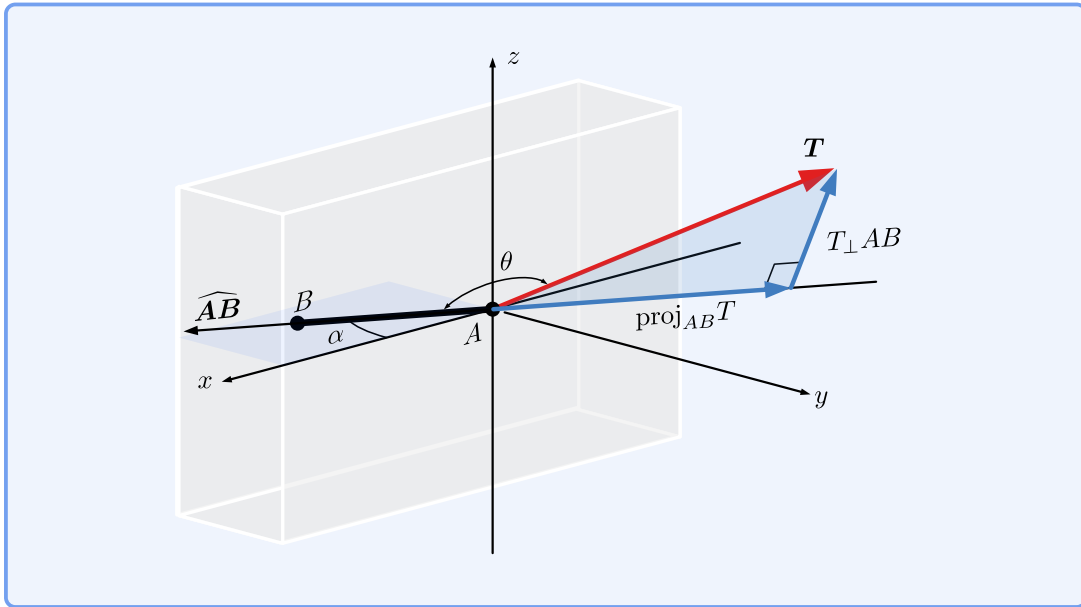
Thus, any vector can be divided into two vectors parallel and perpendicular to another line. The vector projection $\text{proj}_{\mathbf{AB}} \mathbf{T}$, from Part (d), is the portion of $\mathbf{T}$ parallel to $\mathbf{AB}$. So the sum of $\mathbf{T}$ can be expressed as the parallel and perpendicular terms:

$$\mathbf{T} = \text{proj}_{\mathbf{AB}} \mathbf{T} + (\mathbf{T} \perp \mathbf{AB})$$

We want to find the part of $\mathbf{T}$ perpendicular to $\mathbf{AB}$, so we can rearrange the equation to find:

$$\begin{aligned} \mathbf{T} \perp \mathbf{AB} &= \mathbf{T} - \text{proj}_{\mathbf{AB}} \mathbf{T} \\ &= \langle -50, 80, 40 \rangle - \langle -72.14, 41.65, 0 \rangle \\ &= \langle 22.14, 38.35, 40 \rangle \text{ N} \end{aligned}$$

Nice effort if you worked through all the parts of this example. Graphically the results for parts (b), (d), and (e) are shown in this diagram.



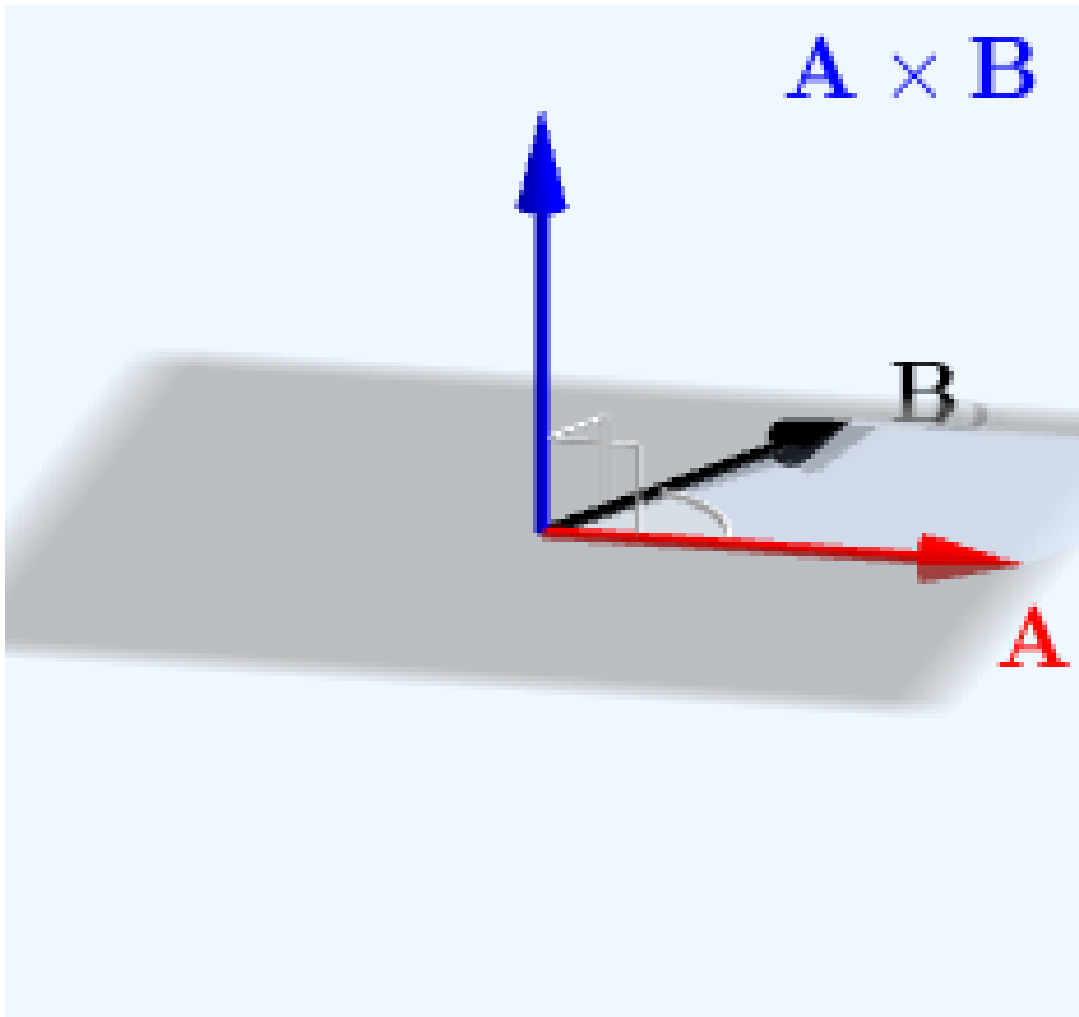
2.8 Cross Products

Key Questions

- How is a cross product different than a dot product?
- What is a determinant?
- What defines a right-handed Cartesian coordinate system?
- How do you use the cross-product circle to find the cross product of two unit vectors?

The vector **cross product** is a multiplication operation applied to two vectors which produces a third mutually perpendicular vector as a result. It's sometimes called the **vector product** to emphasize this and to distinguish it from the dot product which produces a scalar result. The $\times$ symbol is used to indicate this operation.

Cross products are used in mechanics to find the moment of a force about a point.



[Live Link](#)

Figure 2.8.1 Direction of a cross product.

The cross product is a vector multiplication process defined by

$$\mathbf{A} \times \mathbf{B} = A B \sin \theta \hat{\mathbf{u}}. \quad (2.8.1)$$

The result is a vector mutually perpendicular to both with a sense determined by the right-hand rule. If $\mathbf{A}$ and $\mathbf{B}$ are in the xy plane, this is

$$\mathbf{A} \times \mathbf{B} = (A_y B_x - A_x B_y) \mathbf{k}. \quad (2.8.2)$$

The operation is not commutative, in fact reversing the order introduces a negative sign.

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}.$$

The magnitude of the cross product is the product of the perpendicular component of $\mathbf{A}$ with the magnitude of $\mathbf{B}$,. This is the area of the parallelogram

formed by vectors $\mathbf{A}$ and $\mathbf{B}$. The magnitude of the cross product is zero if $\mathbf{A}$ and $\mathbf{B}$ are parallel, and it is maximum when they are perpendicular. The magnitude of the cross product of two perpendicular *unit* vectors is one.

Notice that the cross product equation are similar to the dot product, except that $\sin$ is used rather than $\cos$ and the product includes a unit vector $\hat{\mathbf{u}}$ making the result a vector. This unit vector $\hat{\mathbf{u}}$ is simple to find in a two-dimensional problem as it will always be perpendicular to the page, but for three-dimensional cross products a vector determinant is used, as discussed in [Subsection 2.8.3](#).

2.8.1 Direction of the Vector Cross Product

The direction of a cross product is determined by the **right-hand rule**. There are two ways to apply the right-hand rule, the three-finger method, and the point-and-curl method. You don't need both, but you will need to master at least one to find the direction of cross products.

The three-finger method uses the fact that your extended index finger, middle finger, and thumb are all roughly mutually perpendicular. If you align your index finger with the first vector and your middle finger with the second, then your thumb will point in the direction of the cross product. Alternately, if you align your thumb with the first vector and your index finger with the second, your middle finger will point in the direction of the cross-product.

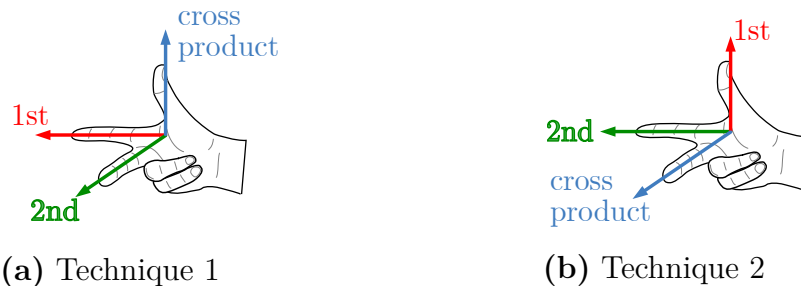


Figure 2.8.2 Right-hand rule using three fingers.

The point-and-curl method involves placing your right hand flat with your fingertips pointing in the direction of the first vector. Then rotate your hand until the second vector is can curl your fingers around your thumb. In this position, your thumb defines the direction of the cross product.

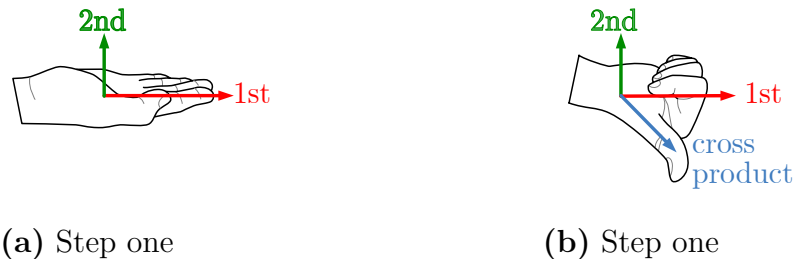
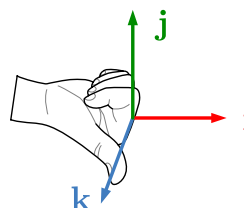
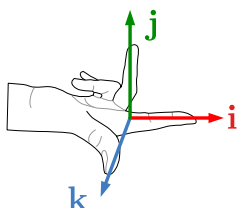


Figure 2.8.3 Right-hand rule using the point-and-curl technique.

2.8.2 Cross Product of Unit Vectors

The Figure 2.8.4(a) demonstrates how you apply these techniques to find the cross product of $\mathbf{i} \times \mathbf{j}$. Assuming the x axis points right and the y axis points up, the cross product points in the positive z direction. Recalling that the magnitude of the cross product of two perpendicular unit vectors is one, we conclude that

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}.$$



(a) Using the three-finger method. (b) Using the point-and-curl method.

Figure 2.8.4 Crossing $\mathbf{i}$ into $\mathbf{j}$ to get $\mathbf{k}$.

Similarly, the cross products of the other pairs of vectors are:

$$\mathbf{i} \times \mathbf{i} = 0$$

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}$$

$$\mathbf{i} \times \mathbf{k} = -\mathbf{j}$$

$$\mathbf{j} \times \mathbf{i} = -\mathbf{k}$$

$$\mathbf{j} \times \mathbf{j} = 0$$

$$\mathbf{j} \times \mathbf{k} = \mathbf{i}$$

$$\mathbf{k} \times \mathbf{i} = \mathbf{j}$$

$$\mathbf{k} \times \mathbf{j} = -\mathbf{i}$$

$$\mathbf{k} \times \mathbf{k} = 0$$

An alternate way to remember this is to use the cross-product circle shown. For example when you cross $\mathbf{i}$ with $\mathbf{j}$ you are going in the positive (counter-clockwise) direction around the blue inner circle and thus the answer is $+\mathbf{k}$. But when you cross $\mathbf{j}$ into $\mathbf{i}$ you go in the negative (clockwise) direction around the circle and thus get a $-\mathbf{k}$. Remember that the order of cross products matter. If you put the vectors in the wrong order you will introduce a sign error.

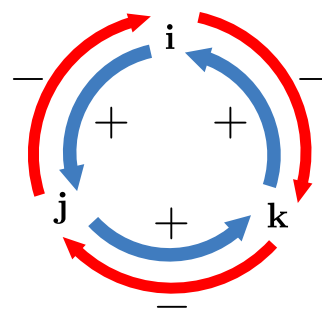


Figure 2.8.5 Unit vector cross product circle.

If you have any negative unit vectors it is easiest to pull out the negative signs before you take the cross product, like the following.

$$-\mathbf{j} \times \mathbf{i} = (-1)(\mathbf{j} \times \mathbf{i}) = (-1)(-\mathbf{k}) = +\mathbf{k}$$

2.8.3 Cross Product of Arbitrary Vectors

The cross product of two arbitrary three-dimensional vectors can be calculated by evaluating the **determinant** of this 3×3 matrix.

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \quad (2.8.3)$$

Here, the first row contains the unit vectors, the second row contains the components of $\mathbf{A}$, and the third row, the components of $\mathbf{B}$. The determinant of this 3×3 matrix is evaluated using the method of cofactors, as follows

$$\mathbf{A} \times \mathbf{B} = + \begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} \mathbf{i} - \begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix} \mathbf{j} + \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix} \mathbf{k}. \quad (2.8.4)$$

Each term contains a 2×2 determinant which is evaluated with the formula

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc. \quad (2.8.5)$$

After simplifying, the resulting formula for a three-dimensional cross product is

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y) \mathbf{i} - (A_x B_z - A_z B_x) \mathbf{j} + (A_x B_y - A_y B_x) \mathbf{k}. \quad (2.8.6)$$

In practice, the easiest way to remember this equation is to use the augmented determinant below, where the first two columns have been copied and placed after the determinant. The cross product is then calculated by adding the product of the red diagonals and subtracting the product of blue diagonals. The result is identical to (2.8.6).

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \begin{vmatrix} \mathbf{i} & \mathbf{j} \\ A_x & A_y \\ B_x & B_y \end{vmatrix}$$

Figure 2.8.6 Augmented determinant

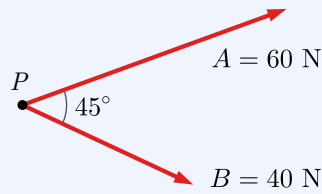
In two dimensions, vectors $\mathbf{A}$ and $\mathbf{B}$ have no z components, so (2.8.6) reduces to

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & 0 \\ B_x & B_y & 0 \end{vmatrix} = (A_x B_y - A_y B_x) \mathbf{k}. \quad (2.8.7)$$

This equation produces the same result as equation (2.8.1) and you may use it if it is more convenient.

Example 2.8.7 2D Cross Product.

The two vectors $\mathbf{A}$ and $\mathbf{B}$ shown lie in the xy plane. Determine the cross product $\mathbf{A} \times \mathbf{B}$.



Solution 1. In this solution we will apply equation (2.8.1).

$$\mathbf{A} \times \mathbf{B} = A B \sin \theta \hat{\mathbf{u}}$$

The direction of the cross product is determined by applying the right-hand rule. With the right hand, rotating $\mathbf{A}$ towards $\mathbf{B}$ we find that our thumb points into the xy plane, so the direction of $\hat{\mathbf{u}}$ is $-\mathbf{k}$.

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (60 \text{ N})(40 \text{ N}) \sin 45^\circ (-\mathbf{k}) \\ &= 1,697 \text{ N}^2 (-\mathbf{k}) \\ &= -1,697 \text{ N}^2 \mathbf{k} \end{aligned}$$

Solution 2. In this solution we will use (2.8.7).

First, establish a coordinate system with the origin P and with the x axis aligned with $\mathbf{A}$, then find the rectangular components and apply the cross product equation.

$$\begin{array}{ll} A_x = 60 \text{ N} & A_y = 0 \text{ N} \\ B_x = 40 \text{ N} \cos 45^\circ & B_y = -40 \text{ N} \sin 45^\circ \\ = 28.28 \text{ N} & = -28.28 \text{ N} \end{array}$$

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (A_x B_y - A_y B_x) \mathbf{k} \\ &= (60)(-28.28) - (0)(28.28) \text{ N}^2 \mathbf{k} \\ &= -1697 \text{ N}^2 \mathbf{k} \end{aligned}$$

Example 2.8.8 3D Cross Product.

Find the cross product of $\mathbf{A} = \langle 2, 4, -1 \rangle \text{ m}$ and $\mathbf{B} = \langle 10, 25, 20 \rangle \text{ N}$. Here, we are crossing a *distance* $\mathbf{A}$ and with a *force* $\mathbf{B}$. This calculation is equivalent to finding the moment about a point P caused by force $\mathbf{B}$

acting distance $\mathbf{A}$ from P . You will learn about moments in [Chapter 4](#).

Solution 1. To solve, set up the augmented determinant and evaluate it by adding the left-to-right diagonals and subtracting the right-to-left diagonals using equation (2.8.6).

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 4 & -1 \\ 10 & 25 & 20 \end{vmatrix} \begin{vmatrix} \mathbf{i} & \mathbf{j} \\ 2 & 4 \\ 10 & 25 \end{vmatrix} \\ &= (4 \cdot 20) \mathbf{i} + (-1 \cdot 10) \mathbf{j} + (2 \cdot 25) \mathbf{k} - (4 \cdot 10) \mathbf{k} - (-1 \cdot 25) \mathbf{i} - (2 \cdot 20) \mathbf{j} \\ &= (80 + 25) \mathbf{i} + (-10 - 40) \mathbf{j} + (50 - 40) \mathbf{k} \\ &= \langle 105, -50, 10 \rangle \text{ N}\cdot\text{m}\end{aligned}$$

Thus, the force $\mathbf{B}$ creates a three-dimensional rotational moment equal to $\langle 105, -50, 10 \rangle \text{ N}\cdot\text{m}$.

Solution 2. Calculating three-dimensional cross products by hand is tedious and error-prone. Whenever you can, you should use technology to do the grunt work for you and focus on the meaning of the results. In this solution, we will use an embedded Sage calculator to calculate the cross product. This same calculator can be used to do other problems.

Given:

$$\begin{aligned}\mathbf{A} &= \langle 2, 4, -1 \rangle \text{ m} \\ \mathbf{B} &= \langle 10, 25, 20 \rangle \text{ N}\end{aligned}$$

$\mathbf{A}$ and $\mathbf{B}$ are defined in the first two lines, and `A.cross_product(B)` is the expression to be evaluated. Click **Evaluate** to see the result. You'll have to work out the correct units for yourself.

```
A = vector([2, 4, -1]);
B = vector([10, 25, 20]);
A.cross_product(B)
```

```
(-105, -50, 10)
```

Try changing the third line to `B.cross_product(A)`. What changes?

2.9 Exercises (Ch. 2)

Chapter 2 Exercises

Vectors	0/140
Rectangular components: P to R	0/20 Not attempted
Rectangular components: R to P	0/20 Not attempted
Find magnitude and direction of three vectors	0/60 Not attempted
Parallelogram Rule	0/20 Not attempted
Arbitrary components	0/20 Not attempted
Vector Addition	0/200
Tip-to-tail method	0/20 Not attempted
Parallelogram rule	0/30 Not attempted
Summing scalar components	0/60 Not attempted
Vector addition of three forces	0/20 Not attempted
Vector addition by summing scalar components	0/40 Not attempted
Description only	0/30 Not attempted

Display options

Done

Created using Numbas v10.0, developed by Newcastle University.



Standalone
In Context

Chapter 3

Equilibrium of Particles

3.1 Equilibrium

Engineering statics is the study of rigid bodies in equilibrium so it's appropriate to begin by defining what we mean by *rigid bodies* and what we mean by *equilibrium*.

A **body** is an object, possibly made up of many parts, which may be examined as a unit. In statics, we consider the forces acting on the object as a whole and also examine it in greater detail by studying each of its parts, which are bodies in their own right. The choice of the body is an engineering decision based on what we are interested in finding out. We might, for example, consider an entire high-rise building as a body for the purpose of designing the building's foundation, and later consider each column and beam of the structure to ensure that they are strong enough to perform their individual roles.

A **rigid body** is a body that doesn't deform under load, that is to say, an object which doesn't bend, stretch, or twist when forces are applied to it. It is an idealization or approximation because no objects in the real world behave this way; however, this simplification still produces valuable information. You will drop the rigid body assumption and study deformation, stress, and strain in a later course called Strength of Materials or Mechanics of Materials. In that course, you will perform analysis of non-rigid bodies, but each problem you do *there* will begin with the rigid body analysis you will learn to do *here*.

A body in **equilibrium** is not accelerating. As you learned in physics, acceleration is velocity's time rate of change and is a vector quantity. For linear motion,

$$\mathbf{a} = \frac{d\mathbf{v}}{dt}.$$

For an object in equilibrium $\mathbf{a} = 0$ which implies that the body is either stationary or moving with a constant velocity

$$\mathbf{a} = 0 \implies \begin{cases} \mathbf{v} = 0 \\ \mathbf{v} = C. \end{cases}$$

The acceleration of an object is related to the net force acting on it by **Newton's Second Law**

$$\Sigma \mathbf{F} = m\mathbf{a}.$$

So for the special case of static equilibrium Newton's Law becomes

$$\Sigma \mathbf{F} = 0. \quad (3.1.1)$$

This simple equation is one of the two foundations of engineering statics.

There are several ways to think about this equation. Reading it from left to right it says that *if all the forces acting on a body sum to zero, then the body will be in equilibrium*. If you read it from right to left it says that *if a body is in equilibrium, then all the forces acting on the body must sum to zero*. Both interpretations are equally valid but we will be using the second one more often. In a typical problem equilibrium of a body implies that the forces sum to zero, and we use that fact to find the unknown forces which make it so. Remember that we are talking about vector addition here, so the sums of the forces must be calculated using the rules of vector addition; you won't get correct answers if you can't add vectors!

We'll be using all of the different vector addition techniques introduced in [Section 2.6](#), which may lead to some confusion. It doesn't matter, mathematically, which technique you use but part of the challenge and reward of statics is learning to select the best tool for the job at hand; to select the simplest, easiest, fastest, or clearest way to get to the solution. You'll do best in this course if you can use multiple approaches to solve the same problem.

In [Chapter 5](#) we will add another requirement for equilibrium, namely equilibrium equation (5.3.2) which says the forces which cause rotational motion and angular acceleration α also must sum to zero, but for the problems of this chapter the only condition we'll need for equilibrium is $\Sigma \mathbf{F} = 0$.

3.2 Particles

We'll begin our study of Equilibrium with the simplest possible object in the simplest possible situation — a **particle** in a **one-dimensional coordinate system**. Also, in this chapter and the next all forces will be represented as **concentrated forces**. In later sections, we will address more complicated situations, higher dimensions, and distributed forces, but beginning with very simple situations will help you to develop engineering sense and problem-solving skills which will be useful later.

The defining characteristic of a particle is that all forces that act on it are coincident¹ or concurrent², not that it is small. Forces are coincident if they have the same line of action, and concurrent if they intersect at a point. The moon, earth and sun can all be treated as particles, but we probably won't encounter

¹Two lines are coincident when one lies on top of the other.

²Two or more lines are concurrent if they intersect at a single point.

them in statics since they're not in equilibrium. Forces are coincident/concurrent if their lines of action all intersect at a single, common point. Two or more forces are also considered concurrent if they share the same line of action. One practical consequence of this is that particles are never subjected to forces that cause rotation. So a see-saw, for example, is not a particle because the weights of the children tend ³ to cause rotation.

Another consequence of concurrent forces is that Equation (3.1.1) is the only equilibrium equation that applies. This vector equation can be used to solve for a maximum of one unknown per dimension. If you find yourself trying to solve a two-dimensional particle equilibrium problem and you are seeking more than two unknowns, it's likely that you have missed something and need to re-read the question.

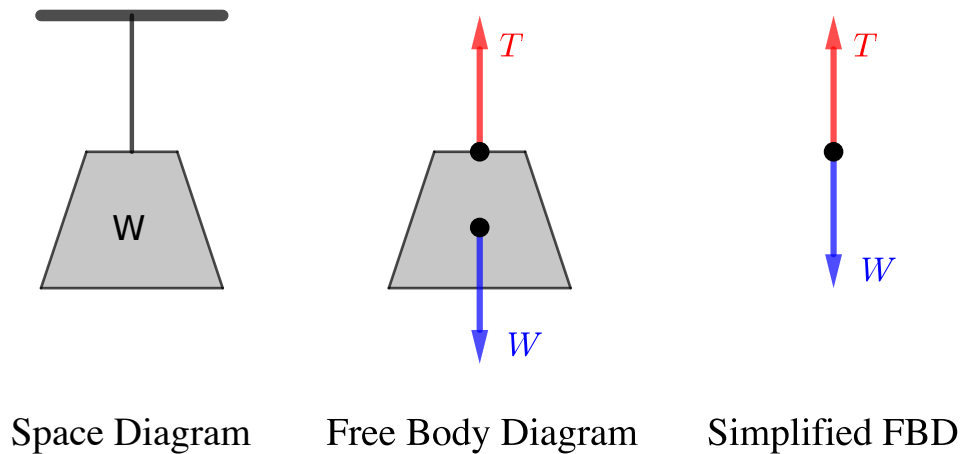
Another simplification we will be making is to treat all forces as **concentrated**. Concentrated forces act at a single point, have a well-defined line of action, and can be represented with an arrow — in other words, they are vectors. Real forces don't actually act at a single mathematical point but concentrating them is intuitive and will be justified in a [later chapter 7.8](#). You're already familiar with the concept if you have ever placed all the weight of an object at its center of gravity.

3.3 1D Particle Equilibrium

3.3.1 A simple case

Consider the weight suspended by a rope shown in [Figure 3.3.1](#) when the slider is all the way to the left. Diagrams of this type are called **space diagrams**; they show the objects as they exist in space.

³We say “tend to cause rotation” because in a static's context, all objects are static — so no actual rotation occurs.



[Live Link](#)

Figure 3.3.1 A suspended weight

In mechanics we are interested in studying the forces acting on objects and in this course, the objects will be in equilibrium. The best way to do this is to draw a diagram that focuses on the forces acting on the object, not the mechanisms that hold it in place.

Free-body diagrams are used in engineering for this purpose. They are called *free-body* diagrams because objects are shown disconnected or freed from their supporting mechanisms and corresponding forces are shown instead. You can see the free-body diagram for this situation by moving the slider in the interactive to the middle position. This emphasizes that there are two forces acting on the object; the tension of the rope holding it up, and the weight of the object trying to pull it down. We treat the weight as a single force acting at the object's center of gravity.

The actual shape of the weight is not important to us, so it can simply be represented with a dot, as shown when the view control is in position three. The forces have been slid along their common line of action until they both act on the dot, which is an example of an [equivalent transformation](#) called the “Principle of Transmissibility.” The **simplified free-body diagram** shown when the slider is moved to the right is completely sufficient for this situation.

Drawing free-body diagrams can be surprisingly tricky. The reason for this is that you must identify all the forces acting on the object and correctly represent them on the free-body diagram. If you fail to account for all the forces, include additional ones, or represent them incorrectly, your analysis will surely be wrong.

So what kind of analysis can we do here? Admittedly not much. We can find the tension in the rope caused by a particular weight and use it to select an

appropriately strong rope, or we can determine the maximum weight a particular rope can safely support.

The actual analysis is so trivial that you've probably already done it in your head, nevertheless several ways to approach it will be shown next.

In the vector approach we will use the [equation of equilibrium](#).

Example 3.3.2 1D Vector Addition.

Find the relationship between the tension in the rope and the suspended weight for the system of [Figure 3.3.1](#).

Solution.

The free-body diagram shows two forces acting on the particle, and since the particle is in equilibrium they must add to zero.

$$\begin{aligned}\Sigma \mathbf{F} &= 0 \\ \mathbf{T} + \mathbf{W} &= 0 \\ \mathbf{T} &= -\mathbf{W}\end{aligned}$$

We conclude that force $\mathbf{T}$ is equal and opposite to $\mathbf{W}$, that is, since the weight is acting down, the rope acts with the same magnitude but up.

Tension is the magnitude of the rope's force. Recall that the magnitude of a vector is always a positive scalar. We use normal (non-bold) typefaces or absolute value bars surrounding a vector to indicate its magnitude. For any force $\mathbf{F}$,

$$F = |\mathbf{F}|.$$

To find how the tension is related to $\mathbf{W}$, take the absolute value of both sides

$$\begin{aligned}|\mathbf{T}| &= |-\mathbf{W}| \\ T &= W\end{aligned}$$

We can also formulate this example in terms of unit vectors. Recall that $\mathbf{j}$ is the unit vector that points up. It has a magnitude of one with no units associated. So in terms of unit vector $\mathbf{j}$, $\mathbf{T} = T\mathbf{j}$ and $\mathbf{W} = -W\mathbf{j}$.

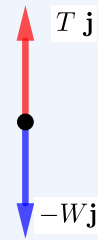
Example 3.3.3 1D Vector Addition using unit vectors.

Find the relation between the tension T and weight W for the system of [Figure 3.3.1](#) using unit vectors.

Solution.

Express the forces in terms of their magnitudes and the unit vector $\mathbf{j}$ then proceed as before,

$$\begin{aligned}\Sigma \mathbf{F} &= 0 \\ \mathbf{T} + \mathbf{W} &= 0 \\ T\mathbf{j} + W(-\mathbf{j}) &= 0 \\ T\mathbf{j} &= W\mathbf{j} \\ T &= W\end{aligned}$$



In the previous example, the unit vector $\mathbf{j}$ completely dropped out of the equation leaving only the coefficients of $\mathbf{j}$. This will be the case whenever you add vectors which all act along the same line of action.

The coefficients of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are known as the **scalar components**. A scalar component times the associated unit vector is a force vector.

When you use scalar components, the forces are represented by scalar values and the equilibrium equations are solved using normal algebraic addition rather than vector addition. This leads to a slight simplification of the solution as shown in the next example.

Example 3.3.4 1D Vector Addition using scalar components.

Find the relation between the tension T and weight W for the system of Figure 3.3.1 using scalar components.

Solution.

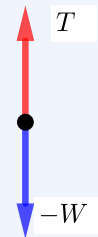
The forces in this problem are $\mathbf{W} = -W \mathbf{j}$ and $\mathbf{T} = T \mathbf{j}$, so the corresponding scalar components are

$$W_y = -W \qquad T_y = T$$

Adding scalar components gives,

$$\begin{aligned}\Sigma F_y &= 0 \\ W_y + T_y &= 0 \\ -W + T &= 0 \\ T &= W\end{aligned}$$

Unsurprisingly, we get the same result.



3.3.2 Scalar Components

The scalar component of a vector is a *signed* number which indicates the vector's **magnitude** and **sense**, and is usually identified by a symbol with a subscript

which indicates the line of action of the vector.

So for example, $F_x = 10$ N is a scalar component. We can tell it's not a vector because it F_x is not bold. 10 N is the magnitude of the associated vector; the subscript x indicates that the force acts “in the x direction,” in other words it acts on a line of action which is parallel to the x axis; and the (implied) positive sign means that the vector points towards the positive end of the x axis — towards positive infinity. So a scalar component, while not a vector, contains all the information necessary to completely describe and draw the corresponding vector. Be careful not to confuse scalar components with vector magnitudes. A force with a magnitude of 10 N can point in any direction, but can never have a negative magnitude.

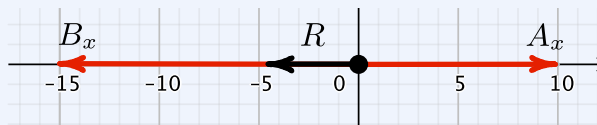
Scalar components can be added together algebraically, but only if they act “in the same direction.” It makes no sense to add F_x to F_y . If that's what you want to do, first you must convert the scalar components to vectors, then add them according to the rules of vector addition.

Example 3.3.5 1D Scalar Addition.

If $A_x = 10$ lb and $B_x = -15$ lb, find the magnitude and direction of their resultant $\mathbf{R}$.

Solution. Start by sketching the two forces. The subscripts indicate the line of action of the force, and the sign indicates the direction along the line of action. A negative B_x points towards the negative end of the x axis.

$$\begin{aligned} R &= A_x + B_x \\ &= 10 \text{ lb} + -15 \text{ lb} \\ &= -5 \text{ lb} \end{aligned}$$



R is the scalar component of the resultant $\mathbf{R}$.

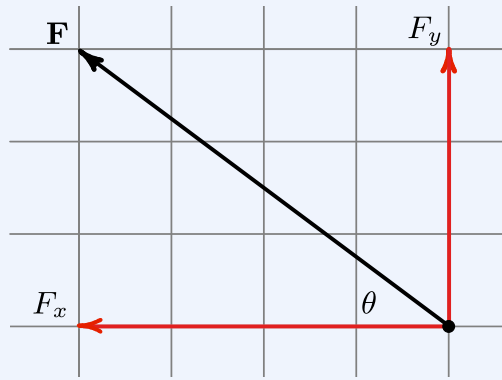
The negative sign on the result indicates that the resultant force acts to the left.

Example 3.3.6 2D Scalar Addition.

If $F_x = -40$ N and $F_y = 30$ N, find the magnitude and direction of their resultant $\mathbf{F}$.

Solution. In this example the scalar components have different subscripts indicating that they act along different lines of action, and this must be accounted for when they are added together.

Make a sketch of the two vectors and add them using the parallelogram rule. In this case, the parallelogram is a rectangle, so right-triangle trig is appropriate.



$$\begin{aligned}
 F &= \sqrt{F_x^2 + F_y^2} \\
 &= \sqrt{(-40 \text{ N})^2 + (30 \text{ N})^2} \\
 &= 50 \text{ N} \\
 \theta &= \tan^{-1} \left| \frac{F_y}{F_x} \right| \\
 &= \tan^{-1} \left| \frac{30 \text{ N}}{-40 \text{ N}} \right| \\
 &= 36.9^\circ
 \end{aligned}$$

θ is measured from the negative x axis. The direction of $\mathbf{F}$ from the positive x axis is $(180^\circ - \theta) = 143.1^\circ$, so

$$\mathbf{F} = 50 \text{ N at } 143.1^\circ \angle$$

3.3.3 Two-force Bodies

As you might expect from the name, a **two-force body** is a body with two forces acting on it, like the weight just discussed. As we just saw, in order for a two-force body to be in equilibrium the two forces must add to zero. There are only three possible ways that this can happen:

The two forces must either

- share the same line of action, have the same magnitude, and point away from each other, or
- share the same line of action, have the same magnitude, and point towards each other, or
- both forces have zero magnitude.

When two forces have the same magnitude but act in diametrically opposite directions, we say that they are **equal-and-opposite**. When equal and opposite

forces act on an object and they point towards each other we say that the object is **in compression**, when they point away from each other the object is **in tension**. Tension and compression describe the internal state of the object.

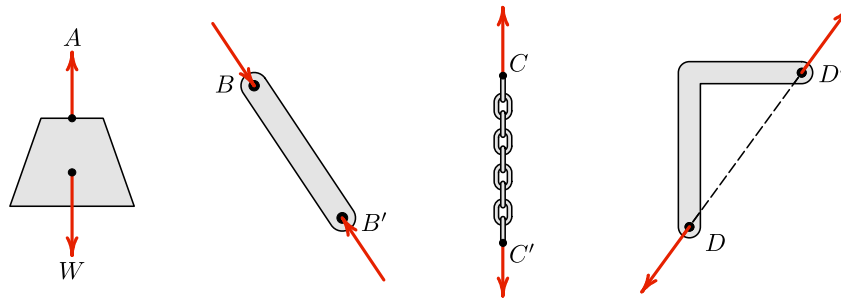


Figure 3.3.7 Examples of two-force bodies

Did you notice that last three examples in [Figure 3.3.7](#) did not include the object's weight? These are simplifications that ignore the object's weight to make them two-force bodies. If the object's weight was included, it would be a *three-force* body. This approximation is justifiable when the object's weight is small in comparison with the tensile or compressive forces. In this case, we say that the weight is **negligible**, i.e. small enough to neglect. Also note that all these examples show single forces acting at each point. If several forces act at a point, they should be combined into a single resultant force acting there.

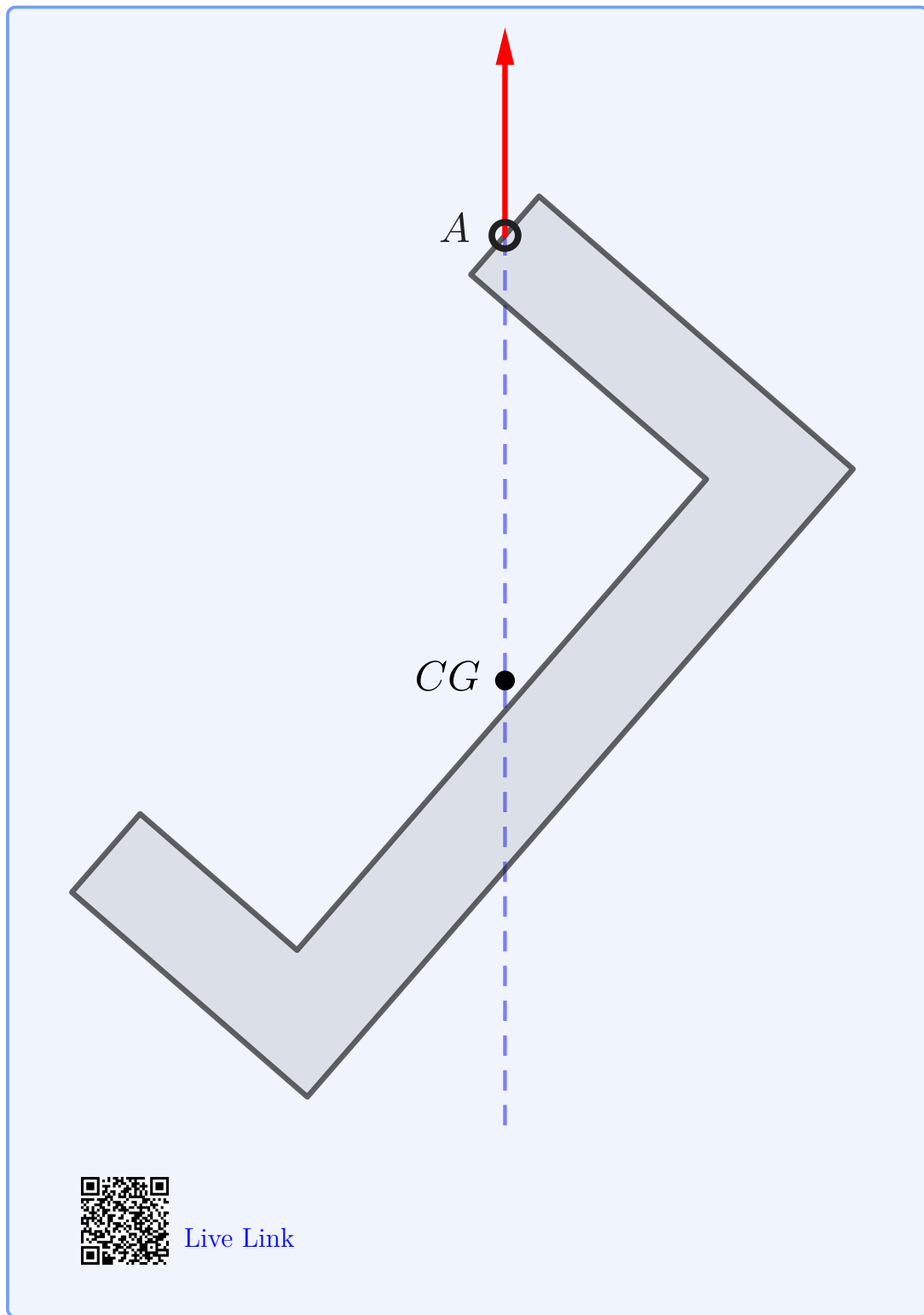
Two force bodies appear frequently in multipart structures and machines which will be covered in [Chapter 6](#). Some examples of two force bodies are struts and linkages, ropes, cables and guy wires, and springs.

Thinking Deeper 3.3.8 Locating the Center of Gravity.

As we will see in [Chapter 7](#), the center of gravity is the point where all the weight of an object can be considered to be concentrated.

An object suspended by a cable or a frictionless pin is a two-force body. When hung freely it will naturally rotate until its center of gravity lies directly beneath the support point to ensure that the lifting force and the weight share the same line of action. This means that the center of gravity of an object can be found by suspending it from several different points, and noting intersection of lines drawn straight down from the hook (like a plumb bob).

In practical terms, to safely lift a heavy object with a chain fall or crane, you must always ensure that the hook is directly above the center of gravity before hoisting the load. The load will be unstable if lifted from any other point.



Example 3.3.9 Tug of War.

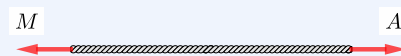
Marines and Airmen at Goodfellow Air Force Base are competing in a tug of war and have reached a stalemate. The Marines are pulling with a force of 1500 lb. How hard are the Airmen pulling? What is the tension in the rope?



This is a simple question, but students often get it wrong at first.

Solution.1. *Assumptions.*

A free-body diagram of the rope is shown.

**Figure 3.3.10**

We'll solve this with scalar components because there's no need for the additional complexity of the vector approaches in this simple situation.

We'll align the x axis with the rope with positive to the right as usual to establish a coordinate system.

Assume that the pull of each team can be represented by a single force. Let force M be supplied by the Marines and force A by the Airmen; call the tension in the rope T .

Assume that the weight of the rope is negligible; then the rope can be considered a particle because both forces lie along same line of action.

2. *Givens.*

$$M = 1500 \text{ lb.}$$

3. *Procedure.*

Since they're stalemated we know that the rope is in equilibrium.

Applying the equation of equilibrium gives:

$$\begin{aligned}
 \Sigma F_x &= 0 \\
 -M + A &= 0 \\
 A &= M \\
 &= 1500 \text{ lb}
 \end{aligned}$$

We find out that both teams pull with the same force. This was probably obvious without drawing the free-body diagram or solving the equilibrium equation.

It may seem equally obvious that if both teams are pulling with 1500 lb in opposite directions that the tension in the rope must be 3000 lb. This is wrong however.

The tension in the rope T is an example of an internal force and in order to learn its magnitude we need a free-body diagram which includes force T . To expose the internal force we take an imaginary cut through the rope and draw (or imagine) a free-body diagram of either half of the rope.

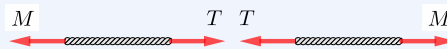


Figure 3.3.11

The correct answer is easily seen to be $T = A = M = 1500 \text{ lb}$.

Example 3.3.12 Hanging Weight.

The wire spool being lifted into the truck consists of 750 m of three strand medium voltage (5 kV) 1/0 AWG electrical power cable with a 195 amp capacity at 90°C, weighing 927 kg/km, on a 350 kg steel reel.

How much weight is supported by the hook and high tension polymer lifting sling?



Solution. The entire weight of the wire and the spool is supported by the hook and sling.

Remember that weight is not mass and mass is not force. The total weight is found by multiplying the total mass by the gravitational constant g .

$$W = mg$$

$$\begin{aligned}
 &= (m_w + m_s) g \\
 &= ((0.75 \text{ km})(927 \text{ kg/km}) + 350 \text{ kg}) g \\
 &= (1045 \text{ kg})(9.81 \text{ m/s}^2) \\
 &= 10300 \text{ N}
 \end{aligned}$$

Question 3.3.13

How can we apply the principles of mechanics in the two previous examples if the rope and the sling are clearly not “rigid bodies?”

Answer. They are not rigid, but they are inextensible and in tension. Under these conditions they don’t change shape, so we can treat them as rigid. If the force were to change direction and put either into compression, our assumptions and analysis would fail. That is why “tug of war” involves pulling and not pushing.

3.3.4 General Procedure

The general procedure for solving one, two, or three-dimensional particle equilibrium problems is essentially the same. Start with the 5-step method for creating a free-body diagram and solve for your unknowns using your equilibrium equations.

Draw a Free-Body Diagram

1. *Select and isolate the particle.* The “free-body” in free-body diagram means that a concurrent force particle or connection must be isolated from the supports physically holding it in place. This means creating a separate free-body diagram from your problem sketch.
2. *Establish a coordinate system.* This step is simple for one-dimensional problems: just label a positive direction for the forces.
3. *Identify all loads.* Include force vectors on your free-body diagram representing each applied load pushing or pulling the body, in addition to the body’s weight, if it is non-negligible. Every vector should have a descriptive variable name and a clear arrowhead indicating its direction.
4. *Identify all reactions.* Reactions represent the resistance of the physical supports you cut away by isolating the body in step 1. All particle supports are two-force members that result in tension or compression forces. Label each reaction with a descriptive variable name and a clear arrowhead.
5. *Label the diagram.* Verify that every force is labeled with either a value or a symbolic name if the value is unknown. Your final free-body diagram

should be a stand-alone presentation and is the basis of your equilibrium equations.

Create and Solve Equilibrium Equations

1. *Write the equilibrium equation.* Now represent your free-body diagram as an equilibrium equation. Your computation should start with the governing equation, like $\Sigma \mathbf{F} = 0$.
2. *Solve for unknown.* Use algebra to simplify the equilibrium equation and solve for the unknown value. Write the unit of your answer. All answers in engineering have units unless you *prove* that they don't. Finally, underline or box your answers.
3. *Check your work.* Do the results seem reasonable given the situation? Have you included appropriate units?

3.4 2D Particle Equilibrium

3.4.1 Introduction

In this section we will study situations where everything of importance occurs in a two-dimensional plane and the third dimension is not involved. Studying two-dimensional problems is worthwhile because they illustrate all the important principles of engineering statics while being easier to visualize and less mathematically complex.

We will normally work in the “plane of the page,” that is, a two-dimensional Cartesian plane with a horizontal x axis and a vertical y axis discussed in [Section 2.3](#) previously. This coordinate system can represent either the front, side, or top view of a system as appropriate. In some problems it may be worthwhile to **rotate the coordinate** system, that is, to establish a coordinate system where the x and y axes are not horizontal and vertical. This is usually done to simplify the mathematics by avoiding simultaneous equations.

3.4.2 General Procedure

The general procedure for solving two-dimensional particle equilibrium is a step up from solving [Subsection 3.3.1](#), as you now need to find equilibrium in two independent directions. The major difference is that you must carefully find each independent vector component and then solve for the equilibrium in each component direction. The process follows the same five-step method for creating a free-body diagram, followed by steps to solve your equilibrium equations.

Draw a Free-Body Diagram:

1. *Select and isolate the particle.* The “free-body” in free-body diagram means that a concurrent force particle or connection must be isolated from the supports that are physically holding it in place. This means creating a separate free-body diagram from your problem sketch.
2. *Establish a coordinate system.* Draw a right-handed coordinate system to use as a reference for your equilibrium equations. Look ahead and select a coordinate system that minimizes the number of force components. This will simplify your vector algebra. The choice is technically arbitrary, but a good choice will simplify your calculations and reduce your effort.
3. *Identify all loads.* Add force vectors to your free-body diagram representing each applied load pushing or pulling the body, in addition to the body’s weight, if it is non-negligible. If a force vector has a known direction, draw it. If its direction is unknown, assume one, and your later algebra will check your assumption. Every vector should have a descriptive variable name and a clear arrowhead indicating its direction.
4. *Identify all reactions.* Reactions represent the resistance of the physical supports you cut away by isolating the body in step 1. All particle supports are some type of two-force members with tension or compression reaction forces. These reactions will all be concurrent with the body loads from Step 2. Label each reaction with a descriptive variable name and a clear arrowhead. Again, if a vector’s direction is unknown, just assume one.
5. *Label the diagram.* Verify that every dimension, angle, force, and moment is labeled with either a value or a symbolic name if the value is unknown. In our eyes, dimensioning is optional. Having the information needed for your calculations is helpful, but don’t clutter the diagram up with unneeded details. Your final free-body diagram should be a stand-alone presentation and is the basis of your equilibrium equations.

Create and Solve Equilibrium Equations

1. *Break vectors into components.* Compute each force’s x and y components using right-triangle trigonometry.
2. *Write equilibrium equations.* Now represent your free-body diagram as two equilibrium equations, $\Sigma \mathbf{F}_x = 0$ and $\Sigma \mathbf{F}_y = 0$.
3. *Count knowns and unknowns.* At this point, you should have at most two unknown values. If you have more than two, reread the problem and look for overlooked information.
4. *Solve for unknowns.* Use algebra to simplify the equilibrium equations and solve for unknowns. All answers in Statics will have units - unless you have solved for a dimensionless value, like a friction coefficient. Finally, underline or box your answers.

5. *Check your work.* If you add the components of the forces, do they add to zero? Do the results seem reasonable given the situation? Have you included appropriate units?

3.4.3 Force Triangle Method

The force triangle method applies to situations where there are (exactly) three forces acting on a particle, and no more than two unknown magnitudes or directions.

If such a particle is in equilibrium then the three forces must add to zero. Graphically, if you arrange the force vectors tip-to-tail, they will form a closed, three-sided polygon, i.e. a triangle. This is illustrated in [Figure 3.4.1](#).

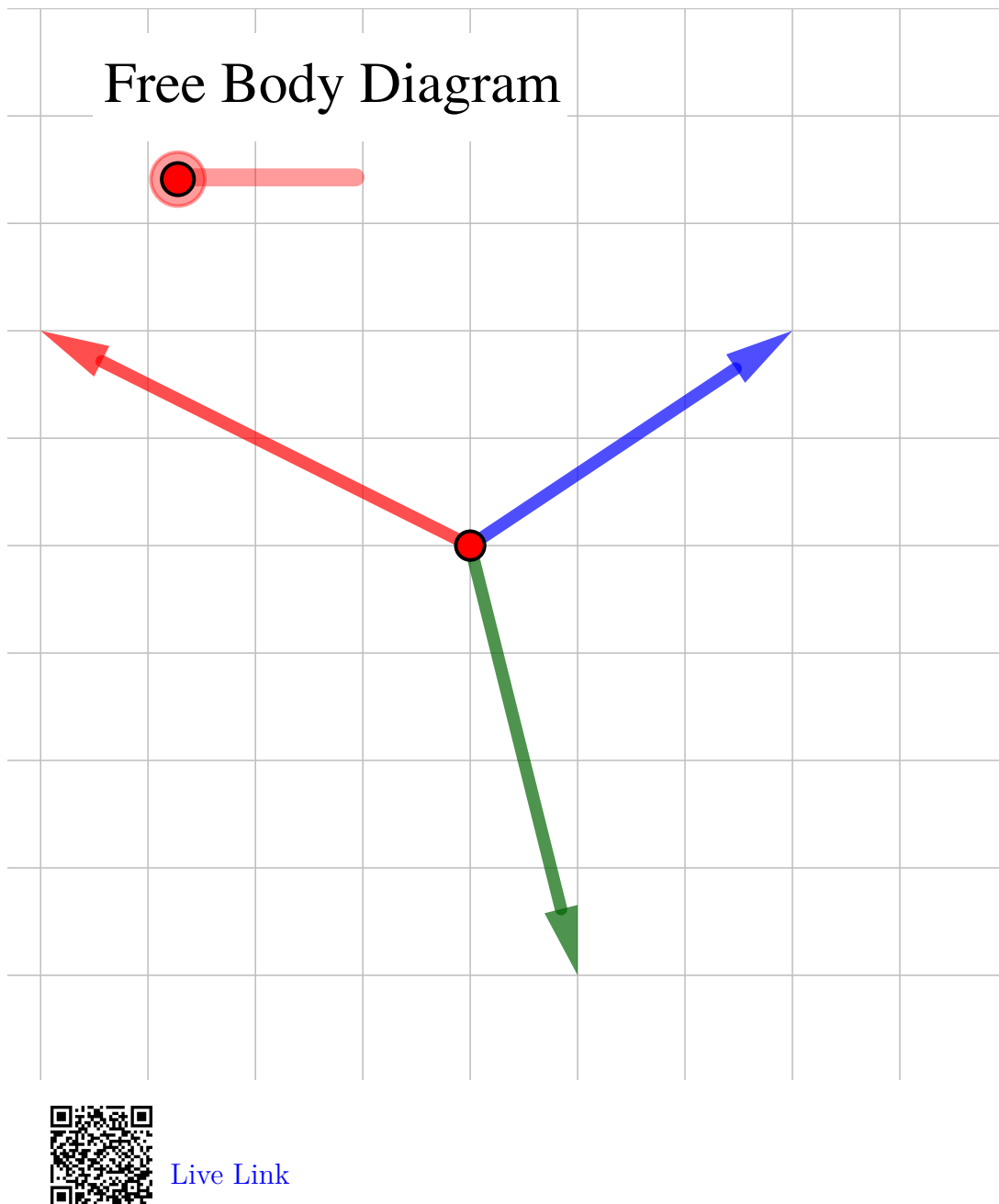


Figure 3.4.1 Free-Body Diagram and Force Triangle

Question 3.4.2

Why do the forces always form a closed polygon?

Answer. Because their resultant is zero.

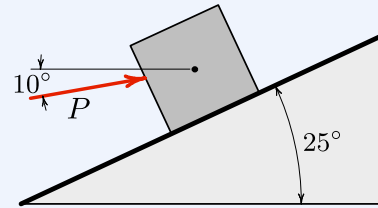
The force triangle is a graphical representation of the vector [equilibrium equation \(3.1.1\)](#). It can be used to solve for unknown values in multiple different ways, which will be illustrated in the next two examples. In [Example 3.4.3](#) We will use

a graphical approach to find the forces causing equilibrium, and in [Example 3.4.4](#) we will use trigonometry to solve for the unknown forces mathematically.

In the next example we will use technology to draw a scaled diagram of the force triangle representing the equilibrium situation. We are using [Geogebra](#) to make the drawing, but you could use CAD, another drawing program, or even a ruler and protractor as you prefer. Since the diagram is accurately drawn, the lengths and angles represent the magnitudes and directions of the forces which hold the particle in equilibrium.

Example 3.4.3 Frictionless Incline.

A force P is being applied to a 100 lb block resting on a frictionless incline as shown. Determine the magnitude and direction of force P and of the contact force on the bottom of the block.



Solution.

1. Assumptions.

We must assume that the block is in equilibrium, that is, either motionless or moving at a constant velocity in order to use the equilibrium equations. We will represent the block's weight and the force between the incline and the block as concentrated forces. The force of the inclined surface on the block must act in a direction that is normal to the surface since it is frictionless and can't prevent motion along the surface.

2. Givens.

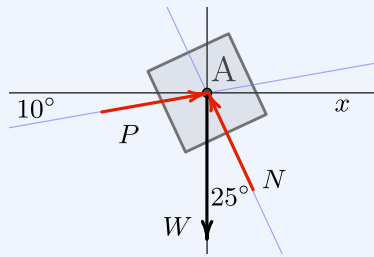
The knowns here are the weight of the block, the direction of the applied force, and the slope of the incline. The slope of the incline provides the direction of the normal force.

The unknown values are the magnitudes of forces P and N .

3. Free-Body Diagram.

You should always begin a statics problem by drawing a free-body diagram. It allows you to think about the situation, identify knowns and unknowns, and define symbols.

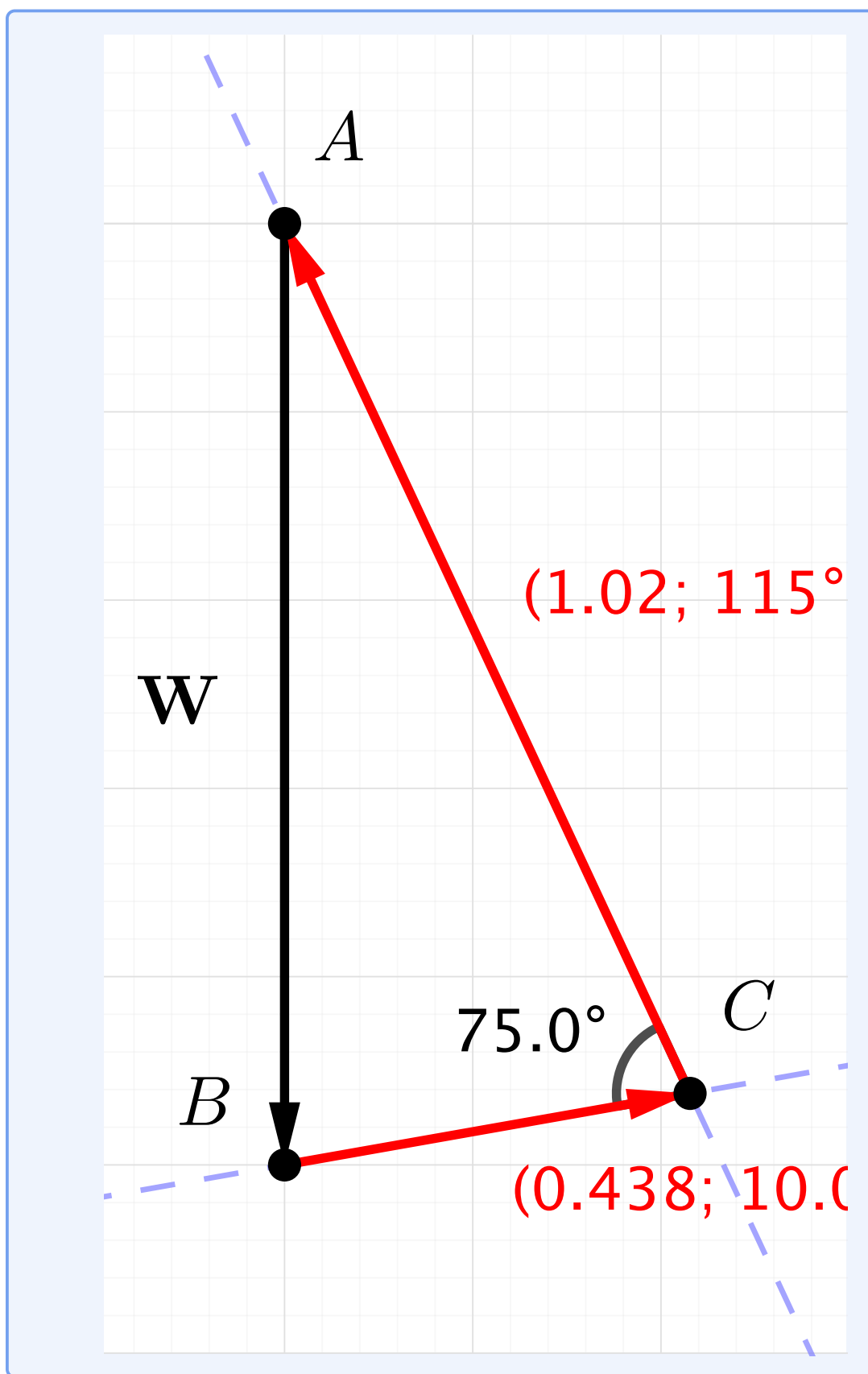
We define three symbols, W , N , and P , representing the weight, normal force, and applied force respectively. The angles could be given symbols too, but since we know their values it isn't necessary.



The free-body can be a quick sketch or an accurate drawing but it must show *all* the forces acting on the particle and define the symbols. In most cases, you won't know the magnitudes of all the forces, so the lengths of the vectors are just approximate.

Notice that the force N is represented as acting 25° from the y axis, which is 90° away from the direction of the surface.

4. *Force Triangle.*





[Live Link](#)

Use the known information to carefully and accurately construct the force triangle.

- (a) Start by placing point A at the origin.
- (b) Draw force $\mathbf{W}$ straight down from A with a length of 1, and place point B at its tip. The length of this vector represents the weight.
- (c) We know the direction of force $\mathbf{P}$ but not its magnitude. For now, just draw line BC passing through point B with an angle of 10° from the horizontal.
- (d) Similarly we know force $\mathbf{N}$ acts at 25° from vertical because it is perpendicular to the inclined surface, and it will close the triangle. So draw line CA passing through point A and at a 25° angle from the y axis.
- (e) Call the point where lines BC and CA intersect point C . Points A , B , and C define the force triangle.
- (f) Now draw force $\mathbf{P}$ from point B to point C , and
- (g) Draw force $\mathbf{N}$ from point C back to point A .

Can you prove from the geometry of the triangle that angle BCA is 75° ?

5. Results.

In steps 6 and 7, Geogebra tells us that $\mathbf{p} = (0.438; 10.0^\circ)$ which means force P is 0.438 units long with a direction of 10° , similarly $\mathbf{n} = (1.02; 115^\circ)$ means N is 1.02 units long at 115° . These angles are measured counter-clockwise from the positive x axis.

These are not the answers we are looking for, but we're close. Remember that for this diagram, our scale is

$$1 \text{ unit} = 100 \text{ lbs},$$

so scaling the lengths of $\mathbf{p}$ and $\mathbf{n}$ by this factor gives

$$\begin{aligned} P &= (0.438 \text{ unit})(100 \text{ lb/ unit}) \\ &= 43.8 \text{ lb at } 10^\circ \angle \\ N &= (1.02 \text{ unit})(100 \text{ lb/ unit}) \\ &= 102 \text{ lb at } 115^\circ \angle \end{aligned}$$

If you use technology such as Geogebra, as we did here, or CAD software to draw the force triangle, it will accurately produce the solution.

If technology isn't available to you, such as during an exam, you can still use a ruler and protractor to draw the force triangle, but your results will only be as accurate as your diagram. In the best case, using a sharp pencil and carefully measuring lengths and angles, you can only expect about two significant digits of accuracy from a hand-drawn triangle. Nevertheless, even a roughly drawn triangle can give you an idea of the correct answers and be used to check your work after you use another method to solve the problem.

3.4.4 Trigonometric Method

The general approach for solving particle equilibrium problems using the trigonometric method is to:

1. Draw and label a free-body diagram.
2. Rearrange the forces into a force triangle and label it.
3. Identify the knowns and unknowns.
4. Use trigonometry to find the unknown sides or angles of the triangle.

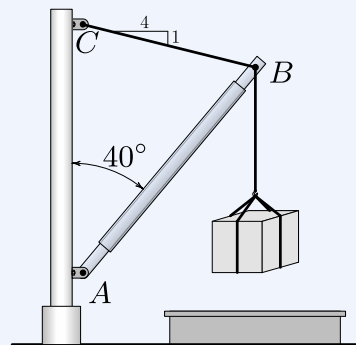
There must be no more than two unknowns to use this method, which may be either magnitudes or directions. During the problem setup, you will probably need to use the geometry of the situation to find one or more angles.

If the force triangle has a right angle you can use [Section B.2](#) to find the unknown values, but in most cases, the triangle will be oblique and you will need to use either or both of the [Law of Sines](#) or the [Law of Cosines](#) to find the sides or angles.

Example 3.4.4 Cargo Boom.

A 24 kN crate is being lowered into the cargo hold of a ship. Boom AB is 20 m long and acts at a 40° angle from kingpost AC . The boom is held in this position by topping lift BC which has a 1:4 slope.

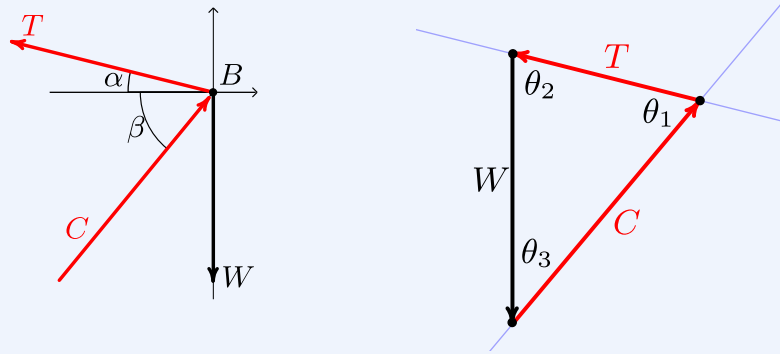
Determine the forces in the boom and in the topping lift.



Solution.1. *Draw diagrams.*

Start by identifying the particle and drawing a free-body diagram. The particle in this case is point B at the end of the boom because it is the point where all three forces intersect. Let T be the tension of the topping lift, C be the force in the boom, and W be the weight of the load. Let α and β be the angles that forces T and C make with the horizontal.

Rearrange the forces acting on point B to form a force triangle as was done in the previous example.

2. *Find angles.*

Angle α can be found from the slope of the topping lift.

$$\alpha = \tan^{-1} \left(\frac{1}{4} \right) = 14.0^\circ.$$

Angle β is the complement of the 40° angle the boom makes with the vertical kingpost.

$$\beta = 90^\circ - 40^\circ = 50^\circ$$

Use these values to find the three angles in the force triangle.

$$\theta_1 = \alpha + \beta = 64.0^\circ$$

$$\theta_2 = 90^\circ - \alpha = 76.0^\circ$$

$$\theta_3 = 90^\circ - \beta = 40.0^\circ$$

3. *Solve force triangle.*

With the angles and one side of the force triangle known, apply the [Law of Sines](#) to find the two unknown sides.

$$\frac{\sin \theta_1}{W} = \frac{\sin \theta_2}{C} = \frac{\sin \theta_3}{T}$$

$$\begin{aligned} T &= W \left(\frac{\sin \theta_3}{\sin \theta_1} \right) & C &= W \left(\frac{\sin \theta_2}{\sin \theta_1} \right) \\ T &= 24 \text{ kN} \left(\frac{\sin 40.0^\circ}{\sin 64.0^\circ} \right) & C &= 24 \text{ kN} \left(\frac{\sin 76.0^\circ}{\sin 64.0^\circ} \right) \\ T &= 17.16 \text{ kN} & C &= 25.9 \text{ kN} \end{aligned}$$

3.4.5 Scalar Components Method

The general statement of equilibrium of forces, (3.1.1), can be expressed as the sum of forces in the **i**, **j** and **k** directions

$$\Sigma \mathbf{F} = \Sigma F_x \mathbf{i} + \Sigma F_y \mathbf{j} + \Sigma F_z \mathbf{k} = 0. \quad (3.4.1)$$

This statement will only be true if all three coefficients of the unit vectors are themselves equal to zero, leading to this scalar interpretation of the equilibrium equation

$$\Sigma \mathbf{F} = 0 \implies \begin{cases} \Sigma F_x = 0 \\ \Sigma F_y = 0 \\ \Sigma F_z = 0 \end{cases} \quad (\text{three dimensions}). \quad (3.4.2)$$

In other words, the single vector equilibrium equation is equivalent to three independent scalar equations, one for each coordinate direction.

In two-dimensional situations, no forces act in the **k** direction leaving just these two equilibrium equations to be satisfied

$$\Sigma \mathbf{F} = 0 \implies \begin{cases} \Sigma F_x = 0 \\ \Sigma F_y = 0 \end{cases} \quad (\text{two-dimensions}). \quad (3.4.3)$$

We will use this equation as the basis for solving two-dimensional particle equilibrium problems in this section and equation (3.4.2) for three-dimensional problems in Section 3.5.

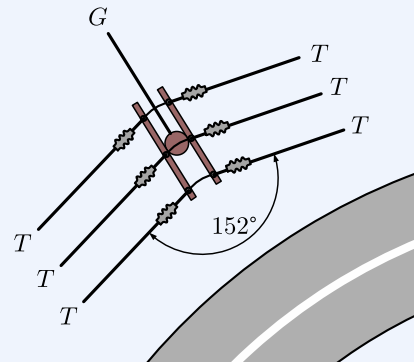
You are undoubtedly familiar with utility poles, which carry electric, cable and telephone lines, but have you ever noticed as you drive down a winding road that the poles will switch from one side of the road to the other and back again? Why is this?

If you consider the forces acting on the top of a pole beside a curving section of road you'll observe that the tensions of the cables produce a net force towards the road. This force is typically opposed by a "guy wire" pulling in the opposite direction which prevents the pole from tipping over due to unbalanced

forces. The power company tries to keep poles beside road segments with convex curvature. If they didn't switch sides, the guy wire for poles at concave curves would extend into the road... which is a poor design.

Example 3.4.5 Utility Pole.

Consider the utility pole next to the road shown below. A top view is shown in the right-hand diagram. If each of the six power lines pulls with a force of 10.0 kN, determine the magnitude of the tension in the guy wire.



Solution.

1. Assumptions.

A utility pole isn't two-dimensional, but we can solve this problem as if it was by first considering the force components acting in a horizontal plane, and then considering the components in a vertical plane.

It also isn't a concurrent force problem because the lines of action of the forces don't all intersect at a single point. However, we can make it into one by replacing the forces of the three power lines in each direction with a single force three times larger. This is an example of an *equivalent transformation*, a trick engineers use frequently to turn complex situations into simpler ones. It works here because all the tensions are equal, and the outside wires are equidistant from the center wire. You must be careful to justify all equivalent transformations because they will lead to errors if they are not applied correctly. Equivalent transformations will be discussed in greater detail in [Section 4.6](#) later.

2. *Givens.*

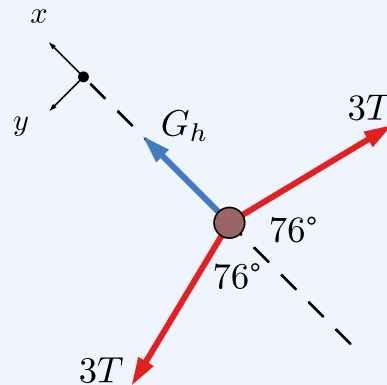
$T = 10.0$ kN and 38° and 152° angles.

3. *Free-Body Diagram.*

Begin by drawing a neat, labeled, free-body diagram of the top view of the pole, establishing a coordinate system and indicating the directions of the forces.

Call the tension in one power line T and the tension in the guy wire G . Resolve the the tension of the guy wire into a horizontal component G_h , and a vertical component G_v . Only the horizontal component of G is visible in the top view.

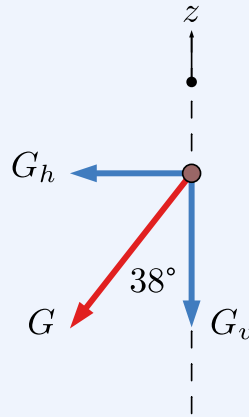
Although it is not necessary, it simplifies this problem considerably to note the symmetry and establish the x axis along the axis of symmetry.

4. *Solution.*

Solve for G_h by applying the equations of equilibrium. The symmetry of this problem means that the ΣF_x equation is sufficient.

$$\begin{aligned}\Sigma F_x &= 0 \\ G_h - 6 T_x &= 0 \\ G_h &= 6 (T \cos 76^\circ) \\ &= 14.5 \text{ kN}\end{aligned}$$

Once G_h is determined, the tension of the guy wire G is easily found by considering the components of G in the side view. Note that the vertical component G_v tends to compress the pole.



$$\begin{aligned} G_h/G &= \sin 38^\circ \\ G &= G_h / \sin 38^\circ \\ G &= 23.6 \text{ kN} \end{aligned}$$

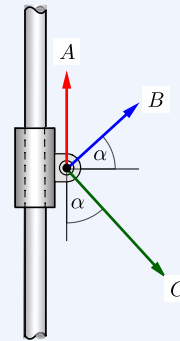
This problem could have also been solved using the force triangle method. See [Subsection 3.4.3](#).

In the next example we look at the conditions of equilibrium by considering the load and the constraints, rather than taking a global equilibrium approach which considers both the load and reaction forces.

Example 3.4.6 Slider.

Three forces act on a machine part that is free to slide along a vertical, frictionless rod. Forces A and B have a magnitude of 20 N and force C has a magnitude of 30 N. Force B acts α degrees from the horizontal, and force C acts at the same angle from the vertical.

Determine the angle α required for equilibrium, and the magnitude and direction of the reaction force acting on the slider.

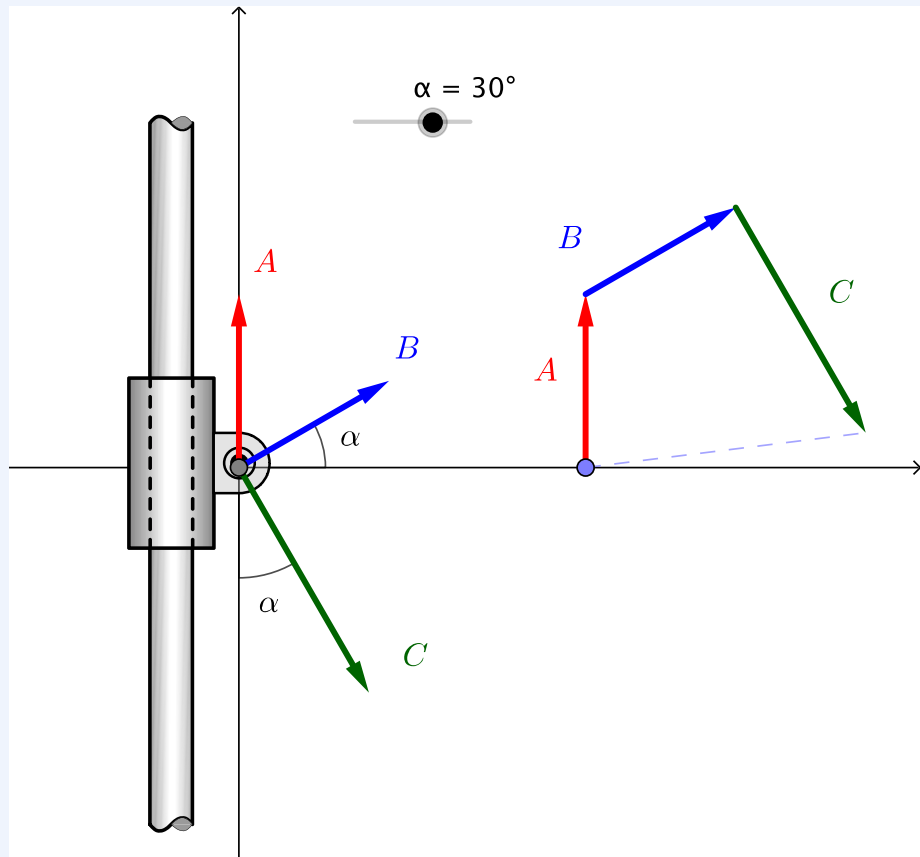


Solution.

1. Givens.

We are given magnitudes of forces $A = 20 \text{ N}$, $B = 20 \text{ N}$, and $C = 30 \text{ N}$. The unknowns are angle α and resultant force R .

2. Procedure.


[Live Link](#)

Since the rod is frictionless, it cannot prevent the slider from moving vertically. Consequently, the slider will only be in equilibrium if the resultant of the three load forces is horizontal. Since a horizontal force has no y component, we can establish this equilibrium condition:

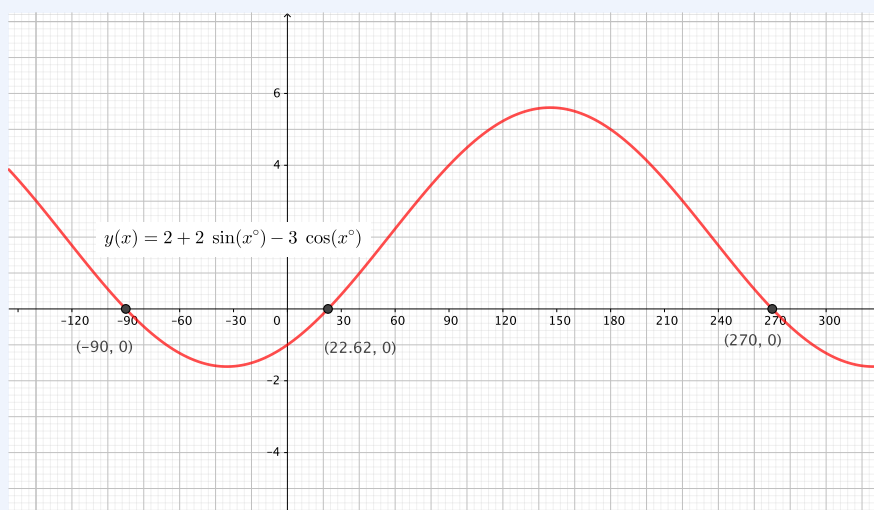
$$R_y = \Sigma F_y = A_y + B_y + C_y = 0$$

Inserting the known values into the equilibrium relation and simplifying gives an equation in terms of unknown angle α .

$$\begin{aligned} R_y &= A_y + B_y + C_y = 0 \\ A + B \sin \alpha - C \cos \alpha &= 0 \\ 20 + 20 \sin \alpha - 30 \cos \alpha &= 0 \end{aligned}$$

$$2 + 2 \sin \alpha - 3 \cos \alpha = 0$$

This is a single equation with a single unknown, although it is not particularly easy to solve with algebra. One approach is described at socratic.org. An alternate approach is to use technology to graph the function $y(x) = 2 + 2 \sin x - 3 \cos x$. The roots of this equation correspond to values of α which satisfy the equilibrium condition above. The root occurring closest to $x = 0$ will be the answer corresponding to our problem, in this case $\alpha = 22.62^\circ$ which you can verify by plugging it back into the equilibrium equation. Note that -90° also satisfies this equation, but it is not the solution we are looking for.



Once α is known, we can find the reaction force by adding the x components of A , B , and C .

$$\begin{aligned} R_x &= A_x + B_x + C_x \\ &= A + B \cos \alpha + C \sin \alpha \\ &= 0 + 20 \cos(22.62^\circ) + 30 \sin(22.62^\circ) \\ &= 30.00 \text{ N} \end{aligned}$$

The resultant force $\mathbf{R}$ is the vector sum of R_x and R_y , but in this situation R_y is zero, so the resultant acts purely to the right with a magnitude of R_x .

$$\mathbf{R} = 30.00 \text{ N} \rightarrow .$$

Note that this value is the **resultant** force, i.e. the net force applied to the slider by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$. However, the question asks for the

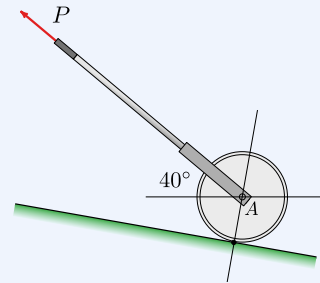
reaction force, which is the force required for equilibrium. The reaction is equal and opposite to the resultant.

$$\mathbf{R}' = -\mathbf{R} = 30.00 \text{ N} \leftarrow$$

The next example demonstrates how rotating the coordinate system can simplify the solution. In the first solution, the standard orientation of the x and y axes is chosen, and in the second the coordinate system is rotated to align with one of the unknowns, which enables the solution to be found without solving simultaneous equations.

Example 3.4.7 Roller.

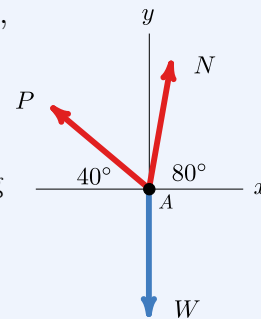
A lawn roller which weighs 160 lb is being pulled up a 10° slope at a constant velocity. Determine the required pulling force P .



Solution 1.

1. Strategy.

- Select a coordinate system, in this case, horizontal and vertical.
- Draw a free-body diagram
- Solve the equations of equilibrium using the scalar approach.



2. Procedure.

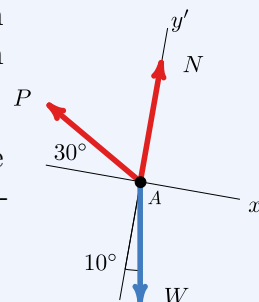
$$\begin{aligned} \Sigma F_x &= 0 & \Sigma F_y &= 0 \\ -P_x + N_x &= 0 & P_y + N_y &= 0 \\ N \cos 80^\circ &= P \cos 40^\circ & P \sin 40^\circ + N \sin 80^\circ &= W \\ N &= P \left(\frac{0.766}{0.174} \right) & 0.643P + 0.985N &= 160 \text{ lb} \end{aligned}$$

Solving simultaneously for P

$$\begin{aligned}
 0.643P + 0.985(4.40P) &= 160 \text{ lb} \\
 4.98P &= 160 \text{ lb} \\
 P &= 32.1 \text{ lb}
 \end{aligned}$$

Solution 2.1. *Strategy.*

- (a) Rotate the standard coordinate system 10° clockwise to align the new y' axis with force N .
- (b) Draw a free-body diagram and calculate the angles between the forces and the rotated coordinate system.
- (c) Solve for force P directly.

2. *Procedure.*

$$\begin{aligned}
 \Sigma F_{x'} &= 0 \\
 -P_{x'} + W_{x'} &= 0 \\
 P \cos 30^\circ &= W \sin 10^\circ \\
 P &= 160 \text{ lb} \left(\frac{0.1736}{0.866} \right) \\
 P &= 32.1 \text{ lb}
 \end{aligned}$$

3.4.6 Multi-Particle Equilibrium

When two or more particles interact with each other there will always be common forces between them as a result of Newton's Third Law, the action-reaction principle.

Consider the two boxes with weights W_1 and W_2 connected to each other and the ceiling shown in the interactive diagram.

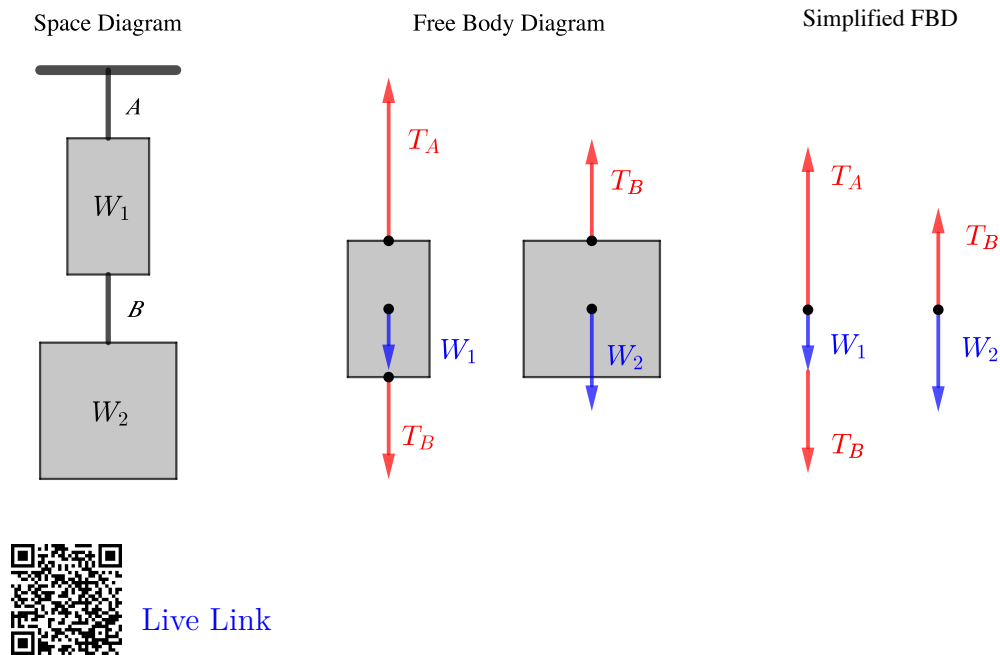


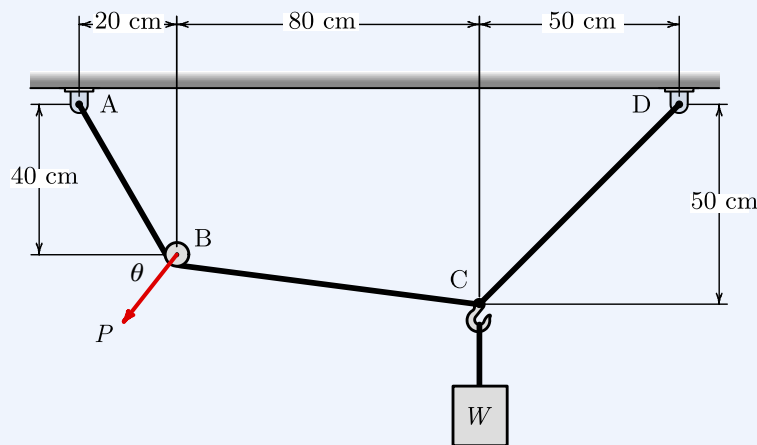
Figure 3.4.8 Two suspended weights

Position one shows the physical arrangement of the objects, position two shows their free-body diagrams, and position three shows simplified free-body diagrams where the objects are represented by points. The boxes were freed by replacing the cables with tension forces T_A and T_B .

From the free-body diagrams you can see that cable B only supports the weight of the bottom box, while cable A and the ceiling support the combined weight. The tension T_B is common to both diagrams. Recognizing the common force is the key to solving multi-particle equilibrium problems.

Example 3.4.9 Two hanging weights.

A 100 N weight W is supported by cable $ABCD$. There is a frictionless pulley at B and the hook is firmly attached to the cable at point C .



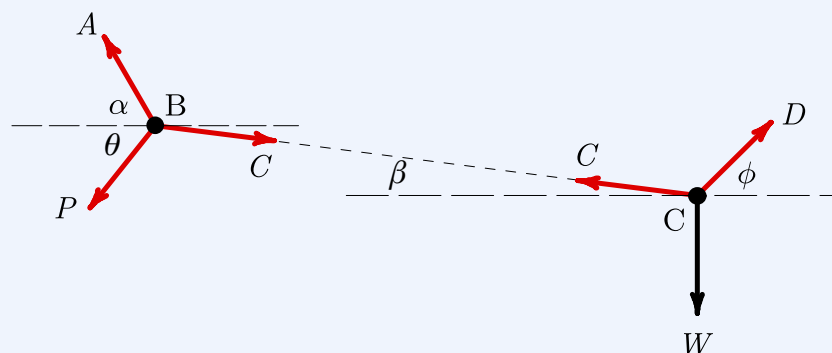
What is the magnitude and direction of force $\mathbf{P}$ required to hold the system in the position shown?

Hint. The particles are points B and C . The common force is the tension in rope segment BC .

Solution.

1. *Strategy.*

Following the [General Procedure](#) we identify the particles as points A and B , and draw free-body diagrams of each. We label the rope tensions A , C , and D for the endpoints of the rope segments, and label the angles of the forces α , β , and ϕ . We will use the standard Cartesian coordinate system and use the [scalar components method](#).



Weight W was given, and we can easily find angles α , β , and ϕ so the knowns are:

$$W = 100 \text{ N}$$

$$\alpha = \tan^{-1} \left(\frac{40}{20} \right) = 63.4^\circ$$

$$\beta = \tan^{-1} \left(\frac{10}{80} \right) = 7.13^\circ$$

$$\phi = \tan^{-1} \left(\frac{50}{50} \right) = 45^\circ$$

Counting unknowns we find that there are two on the free-body diagram of particle C (C and D), but four on particle B , (AC , P and θ).

Two unknowns on particle C means it is solvable since there are two [equilibrium equations](#) available, so we begin there.

2. Solve Particle C .

$$\begin{array}{ll} \Sigma F_x = 0 & \Sigma F_y = 0 \\ -C_x + D_x = 0 & C_y + D_y - W = 0 \\ C \cos \beta = D \cos \phi & C \sin \beta + D \sin \phi = W \\ C = D \left(\frac{\cos 45^\circ}{\cos 7.13^\circ} \right) & C \sin 7.13^\circ + D \sin 45^\circ = 100 \text{ N} \\ C = 0.713D & 0.124C + 0.707D = 100 \text{ N} \end{array}$$

Solving these two equations simultaneously gives

$$C = 89.6 \text{ N} \qquad D = 125.7 \text{ N}$$

With particle C solved, we can use the results to solve particle B . There are three unknowns remaining, tension A , magnitude P , and direction θ . Unfortunately, we still only have two available equilibrium equations. When you find yourself in this situation with more unknowns than equations, it generally means that you are missing something. In this case, it is the pulley. When a cable wraps around a frictionless pulley the tension doesn't change. The missing information is that $A = C$. Knowing this, the magnitude and direction of force $\mathbf{P}$ can be determined.

Because $A = C$, the free-body diagram of particle B is symmetric, and the technique used in [Example 3.4.5](#) to rotate the coordinate system could be applied here.

3. Solve Particle B.

Referring to the FBD for particle B we can write these equations.

$$\begin{aligned}\Sigma F_x &= 0 & \Sigma F_y &= 0 \\ -A_x - P_x + C_x &= 0 & A_y - P_y - C_y &= 0 \\ P \cos \theta &= C \cos \beta - A \cos \alpha & P \sin \theta &= A \sin \alpha - C \sin \beta\end{aligned}$$

Since $A = C = 89.6 \text{ N}$, substituting and solving simultaneously gives

$$\begin{aligned}P \cos \theta &= 48.8 \text{ N} & P \sin \theta &= 69.0 \text{ N} \\ P &= 84.5 \text{ N} & \theta &= 54.7^\circ\end{aligned}$$

These are the magnitude and direction of vector $\mathbf{P}$. If you wish, you can express $\mathbf{P}$ in terms of its scalar components. The negative signs on the components have been applied by hand since $\mathbf{P}$ points down and to the left.

$$\begin{aligned}\mathbf{P} &= \langle -P \cos \theta, -P \sin \theta \rangle \\ &= \langle -48.8 \text{ N}, -69.0 \text{ N} \rangle\end{aligned}$$

3.5 3D Particle Equilibrium

The world we live in has three dimensions. One and two-dimensional textbook problems have been useful for learning the principles of engineering mechanics, but to model real-world problems we will have to consider all three.

Fortunately, all the principles you have learned so far still apply, but many students have difficulty visualizing three-dimensional problems drawn on two-dimensional paper and the mathematics becomes a bit harder. It is especially important to have good diagrams and keep your work neat and organized to avoid errors.

3.5.1 Three-Dimensional Coordinate Frame

We need a coordinate frame for three dimensions, just as we did in two dimensions, so we add a third orthogonal axis z to our existing two-dimensional frame.

For equilibrium of a particle, usually the origin of the coordinate frame is at the particle, the x axis is horizontal, and the y axis is vertical just as in a two-dimensional situation. The orientation of the z axis is determined by the **right-hand rule**. Using your *right hand*, put your palm at the origin and point your fingers along the positive x axis. Then curl your fingers towards the positive y axis. Your thumb will point in the direction of the positive z axis. For example, in the plane of the page with the positive x axis horizontal and to the right and the positive y axis vertical and upwards, the positive z axis will point *towards* you out of the page. Remember that the three axes are mutually perpendicular, i.e. each axis is perpendicular to both of the others. The right-hand rule is important in many aspects of engineering, so make sure that you understand how it works. Mistakes will lead to sign errors.

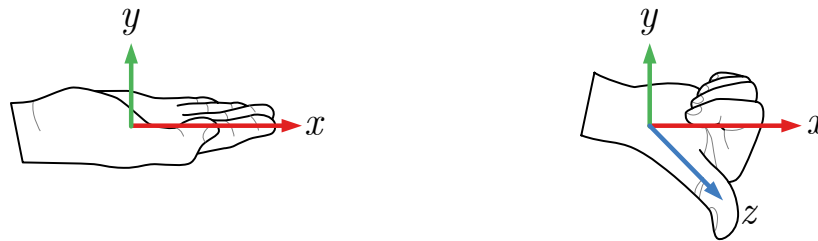


Figure 3.5.1 Point-and-curl right-hand rule technique.

3.5.2 Free-Body Diagrams

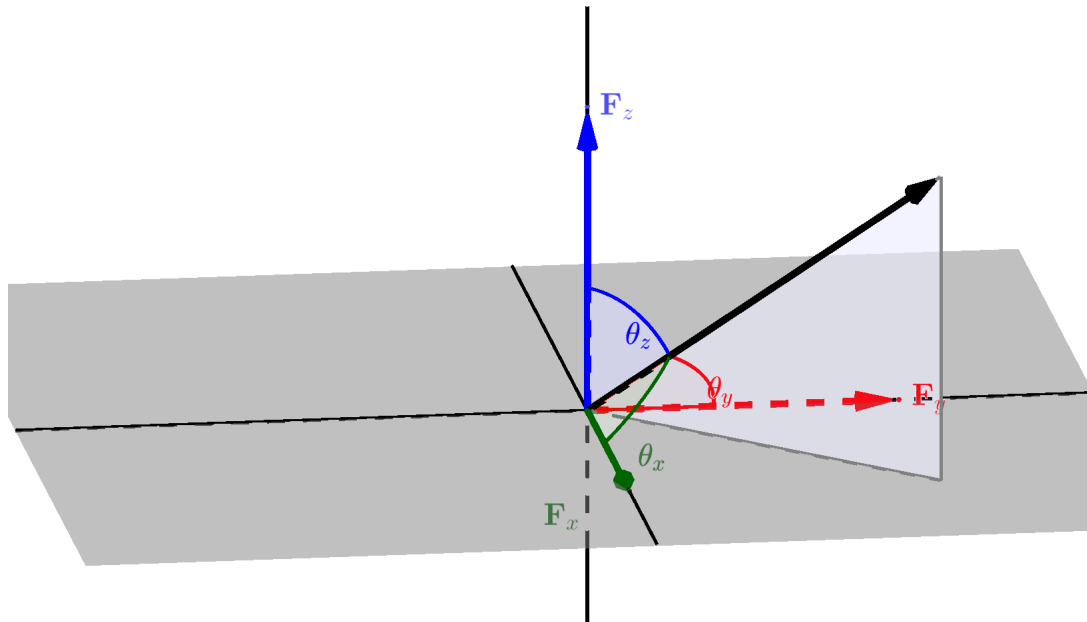
As we did before, we begin our analysis by drawing a free-body diagram that shows all forces and moments acting on the object of interest. Drawing a FBD in three dimensions can be difficult. It is sometimes hard to see things in three dimensions when they are drawn on a two-dimensional sheet. Consequently, it is important to carefully label vectors and angles, but not to clutter up the diagram with too much and/or unnecessary information. When working in two dimensions, you only need one angle to determine the direction of the vector, but when working in three dimensions you need two or three angles.

3.5.3 Angles

As stated above, when working in three dimensions you need three angles to determine the direction of the vector, namely, the angle with respect to the x axis, the angle with respect to the y axis and the angle with respect to the z axis. The three angles mentioned above are not necessarily located in any of the coordinate planes. Think of it this way — three points determine a plane, and in this case, the three points are: the origin, the tip of the vector, and a point on an axis. The plane made by those three points is not necessarily the xy , yz , or xz plane. It is most likely a “tilted” plane.

As you learned in [Subsection 2.4.2](#), one way to quantify the direction of a vector is with direction cosine angles. These direction cosine angles are measured from the positive x , y , and z axes and are often labeled θ_x , θ_y , and θ_z , respectively.

As with two dimensions, angles can be determined from geometry — a distance vector going in the same direction as the force vector. This is the three-dimensional equivalent of similar triangles that you used in the two-dimensional problems.



[Live Link](#)

Figure 3.5.2 Direction Cosine Angles

If you know that the line of action of a force vector goes between two points, then you can use the distance vector that goes from one point to the other to determine the angles.

Let's suppose that the line of action goes through two points A and B , and the direction of the force is from A towards B . The first step in determining the three angles is to write the distance vector from point A towards point B . Let's call this vector $\mathbf{r}_{AB}$. Starting at point A , you need to determine how to get to point B by moving in each of the three directions. Ask yourself: to get from point A to point B do I have to move in the x direction? If so, how far do I have to travel? This becomes the x component of the vector $\mathbf{r}_{AB}$ namely r_{AB_x} . Next, to get from point A to point B how far do I move in y direction? This distance is r_{AB_y} . Finally, to get from point A to point B how far do I move in the z -direction? This distance is r_{AB_z} .

When writing these scalar components pay attention to which way you move along the axes. If you travel toward the positive end of an axis, the corresponding scalar component gets a positive sign. Travel towards the negative end results in a negative sign. The sign is important.

Once you have determined the components of the distance vector r_{AB} , you can determine the total distance from point A to B using the [three-dimensional Pythagorean Theorem](#)

$$r_{AB} = \sqrt{(r_{ABx})^2 + (r_{ABy})^2 + (r_{ABz})^2} \quad (3.5.1)$$

Lastly, the angles are determined by the direction cosines, namely

$$\cos \theta_x = \frac{r_{ABx}}{r_{AB}} \quad \cos \theta_y = \frac{r_{ABy}}{r_{AB}} \quad \cos \theta_z = \frac{r_{ABz}}{r_{AB}}$$

Since the force vector has the same line of action as the distance vector, by the three-dimensional version of similar triangles,

$$\frac{r_{ABx}}{r_{AB}} = \frac{F_x}{F} \quad \frac{r_{ABy}}{r_{AB}} = \frac{F_y}{F} \quad \frac{r_{ABz}}{r_{AB}} = \frac{F_z}{F}$$

. So,

$$F_x = \left(\frac{r_{ABx}}{r_{AB}} \right) F \quad F_y = \left(\frac{r_{ABy}}{r_{AB}} \right) F \quad F_z = \left(\frac{r_{ABz}}{r_{AB}} \right) F$$

Now, that is a bit of math there, but the important things to remember are:

- You can use three angles to determine the direction of a force in three dimensions.
- You can use the geometry to get them from a distance vector that lies along the line of action of the force.

The three direction cosine angles are not mutually independent. From [\(3.5.1\)](#) you can easily show that

$$\cos^2 \theta_x + \cos^2 \theta_y + \cos^2 \theta_z = 1, \quad (3.5.2)$$

so if you know two direction cosine angles you can find the third from this relationship.

3.5.4 General Procedure

The general procedure for solving three-dimensional particle equilibrium is essentially the same as for two-dimensional particle equilibrium using the components method. The major differences are that you must carefully find each vector component using the techniques from [Section 2.4](#). The process follows the same five-step method for creating a free-body diagram, followed by steps to solve your equilibrium equations.

Draw a Free-body Diagram:

1. *Select and isolate the particle.* The “free-body” in free-body diagram means that a concurrent force particle or connection must be isolated from the supports that are physically holding it in place. This means creating a separate free-body diagram from your problem sketch.
2. *Establish a coordinate system.* Draw a right-handed coordinate system to use as a reference for your equilibrium equations. Look ahead and select a coordinate system that minimizes the number of force components. This will simplify your vector algebra. The choice is technically arbitrary, but a good choice will simplify your calculations and reduce your effort.
3. *Identify all loads.* Add force vectors to your free-body diagram representing each applied load pushing or pulling the body, in addition to the body’s weight, if it is non-negligible. If a force vector has a known direction, draw it. If its direction is unknown, assume one, and your later algebra will check your assumption. Every vector should have a descriptive variable name and a clear arrowhead indicating its direction.
4. *Identify all reactions.* Reactions represent the resistance of the physical supports you cut away by isolating the body in step 1. All particle supports are some type of two-force members with tension or compression reaction forces. These reactions will all be concurrent with the body loads from Step 2. Label each reaction with a descriptive variable name and a clear arrowhead. Again, if a vector’s direction is unknown, just assume one.
5. *Label the diagram.* Verify that every dimension, angle, force, and moment is labeled with either a value or a symbolic name if the value is unknown. In our eyes, dimensioning is optional. Having the information needed for your calculations is helpful, but don’t clutter the diagram up with unneeded details. Your final free-body diagram should be a stand-alone presentation and is the basis of your equilibrium equations.

Create and Solve Equilibrium Equations

1. *Break vectors into components.* Compute each force’s x , y , and z components using the tools outlined in [Section 2.4](#). While the components in two-dimensional problems can often be found with right triangle trigonometry, three-dimensional problems often use unit vectors.
2. *Write equilibrium equations.* Now represent your free-body diagram as equilibrium equations. For a three-dimensional particle equilibrium problem, you can have up to three force equilibrium equations corresponding to a force balance in the three independent x , y , and z directions. Each equation should start with the governing equation, like $\Sigma F_x = 0$.
3. *Count knowns and unknowns.* At this point, you should have at most three unknowns remaining. If you have over three, reread the problem and look for overlooked information.

4. *Solve for unknowns.* Use algebra to simplify the equilibrium equations and solve for unknowns. With multiple unknowns scattered across multiple equations, linear algebra may be more efficient than substitution. Assume that all answers have units - unless you prove that they don't. Finally, underline or box your answers.
5. *Check your work.* If you add the components of the forces, do they add to zero? Do the results seem reasonable given the situation? Have you included appropriate units?

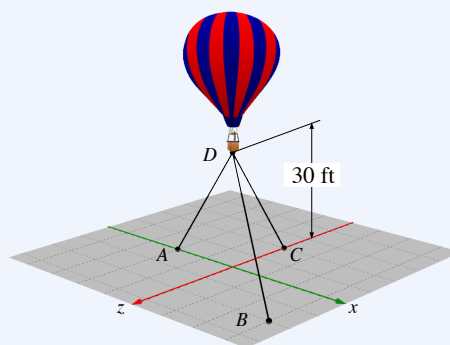
Now let's see how that process looks on an example problem.

Example 3.5.3 Balloon.

A hot air balloon 30 ft above the ground is tethered by three cables as shown in the diagram.

If the balloon is pulling upwards with a force of 900 lb, what is the tension in each of the three cables?

The grid lines on the ground plane are spaced 10 ft apart.



Solution.

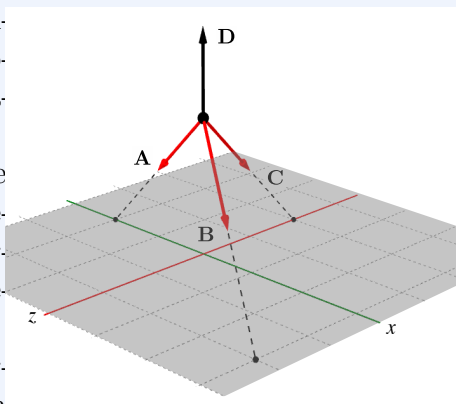
1. Strategy.

The three tensions are the unknowns which we can find by applying the three equilibrium equations.

We'll establish a coordinate system with the origin directly below the balloon and the y axis vertical, then draw and label a free-body diagram.

Next we'll use the given information to find two points on each line of action to find the components of each force in terms of the unknowns.

When the x , y and z components of all forces can be expressed in terms of known values, the equilibrium equations can be solved.



2. Geometry.

From the diagram, the coordinates of the points are

$$A = (-20, 0, 0) \quad B = (30, 0, 20) \quad C = (0, 0, -20) \quad D = (0, 30, 0)$$

Use the point coordinates to find the x , y and z components of the forces.

$$\begin{aligned} A_x &= \frac{-20}{L_A} A & A_y &= \frac{-30}{L_A} A & A_z &= \frac{0}{L_A} A \\ B_x &= \frac{30}{L_B} B & B_y &= \frac{-30}{L_B} B & B_z &= \frac{20}{L_B} B \\ C_x &= \frac{0}{L_C} C & C_y &= \frac{-30}{L_C} C & C_z &= \frac{-20}{L_C} C \end{aligned}$$

Where L_A , L_B and L_C are the lengths of the three cables found with the [distance formula](#).

$$\begin{aligned} L_A &= \sqrt{(-20)^2 + (-30)^2 + 0^2} & &= 36.1 \text{ ft} \\ L_B &= \sqrt{30^2 + (-30)^2 + 20^2} & &= 46.9 \text{ ft} \\ L_C &= \sqrt{0^2 + (-30)^2 + (-20)^2} & &= 36.1 \text{ ft} \end{aligned}$$

3. Equilibrium Equations.

Applying the three equations of equilibrium yields three equations in terms of the three unknown tensions.

$$\begin{aligned} \Sigma F_x &= 0 \\ A_x + B_x + C_x &= 0 \\ -\frac{20}{36.1} A + \frac{30}{46.9} B + 0 C &= 0 \\ A &= 1.153 B \end{aligned} \tag{1}$$

$$\begin{aligned} \Sigma F_z &= 0 \\ A_z + B_z + C_z &= 0 \\ 0 A + \frac{20}{46.9} B - \frac{20}{36.1} C &= 0 \\ C &= 0.769 B \end{aligned} \tag{2}$$

$$\begin{aligned}
\Sigma F_y &= 0 \\
A_y + B_y + C_y + D &= 0 \\
-\frac{30}{36.1}A - \frac{30}{46.9}B - \frac{30}{36.1}C + 900 &= 0 \\
0.832A + 0.640B + 0.832C &= 900 \text{ lb} \quad (3)
\end{aligned}$$

Solving these equations simultaneously yields the answers we are seeking. One way to do this is to substitute equations (1) and (2) into (3) to eliminate A and C and solve the resulting equation for B .

$$\begin{aligned}
0.832(1.153B) + 0.640B + 0.832(0.769B) &= 900 \text{ lb} \\
2.24B &= 900 \text{ lb} \\
B &= 402 \text{ lb}
\end{aligned}$$

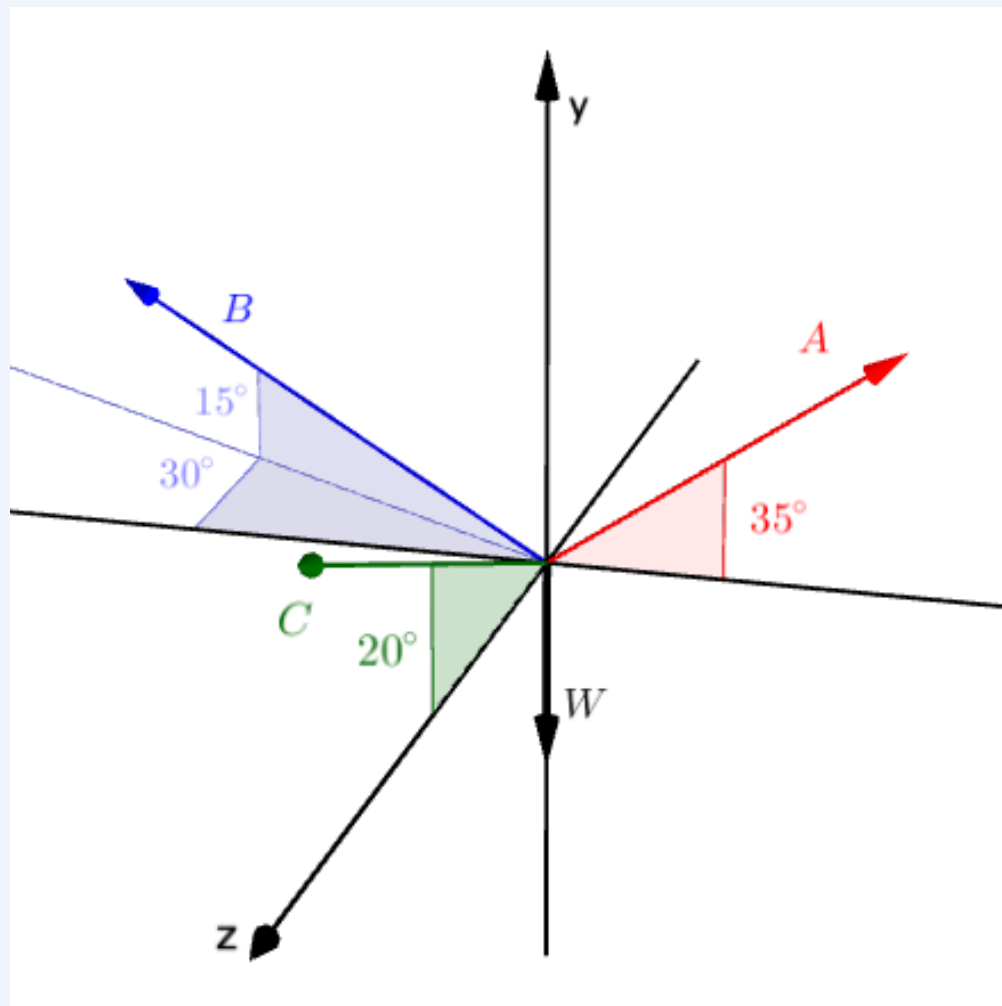
With B known, substitute it into equations (1) and (2) to find A and C .

$$\begin{aligned}
A &= 1.153B & C &= 0.769B \\
&= 464 \text{ lb} & &= 309 \text{ lb}
\end{aligned}$$

Example 3.5.4 Skycam.

The skycam at Stanford University Stadium has a mass of 20 kg and is supported by three cables as shown. Assuming that it is currently in equilibrium, find the tension in each of the three supporting cables.





[Live Link](#)

Solution. In this situation, the directions of all four forces are specified by the angles in the free-body diagram, and the magnitude of the weight is known. The three unknowns are the magnitudes of forces **A**, **B**, and **C**.

$$W = mg = 20 \text{ kg } 9.81 \text{ m/s}^2 = 196.2 \text{ N}$$

We will first find unit vectors in the directions of the four forces by inspection of the free-body diagram. This step requires visualizing the component's unit vectors and determining the angles each makes with the coordinate axis.

$$\hat{\mathbf{W}} = \langle 0, -1, 0 \rangle$$

$$\hat{\mathbf{A}} = \langle \cos 35^\circ, \cos 55^\circ, 0 \rangle$$

$$\hat{\mathbf{B}} = \langle -\cos 15^\circ \cos 30^\circ, \cos 75^\circ, -\cos 15^\circ \cos 60^\circ \rangle$$

$$\hat{\mathbf{C}} = \langle 0, \cos 70^\circ, \cos 20^\circ \rangle$$

Particle equilibrium requires that $\sum \mathbf{F} = 0$.

$$A \hat{\mathbf{A}} + B \hat{\mathbf{B}} + C \hat{\mathbf{C}} = -W \hat{\mathbf{W}}$$

This is a 3×3 system of three simultaneous equations, one for each coordinate direction, which needs to be solved for A , B , and C .

$$A \cos 35^\circ - B \cos 15^\circ \cos 30^\circ + 0 = 0 \quad (\Sigma F_x = 0)$$

$$A \cos 55^\circ + B \cos 75^\circ + C \cos 70^\circ = 196.2 \text{ N} \quad (\Sigma F_y = 0)$$

$$0 - B \cos 15^\circ \cos 60^\circ + C \cos 20^\circ = 0 \quad (\Sigma F_z = 0)$$

These can be solved by any method you choose. Here we will use Sage. Evaluating the coefficients and expressing the equations in matrix form gives

$$\begin{bmatrix} 0.819 & -0.837 & 0 \\ 0.574 & 0.259 & 0.342 \\ 0 & -0.482 & 0.940 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ 196.2 \text{ N} \\ 0 \end{bmatrix}.$$

This is an equation in the form

$$[A][x] = [B].$$

Entering the coefficient matrices into Sage.

```
A = Matrix([[0.819, -0.837, 0], [0.574, 0.259, 0.342], [0, -0.482, 0.940]])
B = vector([0, 196.2, 0])
x = A.solve_right(B)
x
```

```
(196.391530042156, 192.168056277808, 98.5372373679827)
```

After evaluating, we learn that

$$A = 196.4 \text{ N}$$

$$B = 192.2 \text{ N}$$

$$C = 98.5 \text{ N}$$

.

3.6 Exercises (Ch. 3)

Chapter 3 Exercises

Particle Equilibrium	0/192
Equilibrium equation method	0/20 Not attempted
Particle Equilibrium: Use slopes to find components	0/20 Not attempted
Four forces	0/20 Not attempted
Block on Incline	0/10 Not attempted
Hanging weight	0/22 Not attempted
Ball in a trough	0/20 Not attempted
Rope and pulley	0/20 Not attempted
3D Equilibrium: Hanging Plate	0/60 Not attempted
Multi-particle Equilibrium	0/60
Two suspended loads	0/20 Not attempted
Blocks on Inclines	0/10 Not attempted
Cylinders in a trough	0/30 Not attempted

Display options

Done

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